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Multi-view support vector machine classifier via $L_{0/1}$ soft-margin loss with structural information

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ABSTRACT

Multi-view learning seeks to leverage the advantages of various views to complement each other and make full use of the latent information in the data. Nevertheless, effectively exploring and utilizing common and complementary information across diverse views remains challenging. In this paper, we propose two multi-view classifiers: multi-view support vector machine via $L_{0/1}$ soft-margin loss (Mv$L_{0/1}$-SVM) and structural Mv$L_{0/1}$-SVM (Mv$SL_{0/1}$-SVM). The key difference between them is that Mv$SL_{0/1}$-SVM additionally fuses structural information, which simultaneously satisfies the consensus and complementarity principles. Despite the discrete nature inherent in the $L_{0/1}$ soft-margin loss, we successfully establish the optimality theory for Mv$SL_{0/1}$-SVM. This includes demonstrating the existence of optimal solutions and elucidating their relationships with P-stationary points. Drawing inspiration from the P-stationary point optimality condition, we design and integrate a working set strategy into the proximal alternating direction method of multipliers. This integration significantly enhances the overall computational speed and diminishes the number of support vectors. Last but not least, numerical experiments show that our suggested models perform exceptionally well and have faster computational speed, affirming the rationality and effectiveness of our methods.

1. Introduction

In comparison to other soft-margin support vector machine (SVM) models, $L_{0/1}$-SVM [1] is the ideal soft-margin SVM model because it efficiently captures the core of binary classification and ensures fewer support vectors (SV). Fig. 1 shows the classification of common single-view loss functions on convexity and smoothness. Despite the loss function being non-convex, Wang et al. [1] successfully established the optimality theory related to the P-stationary point (PSP), which renders $L_{0/1}$-SVM to be numerically tractable. Leveraging the concept of $L_{0/1}$ SVs inspired by the PSP, they integrated a working set into the proximal alternating direction method of multipliers (ADMM). This integration resulted in a notable reduction in SVs and a considerable computation acceleration. As a result, $L_{0/1}$-SVM demonstrated exceptional performance and fast computation speed with fewer SVs, particularly for large-scale datasets. Hence, employing $L_{0/1}$ soft-margin loss holds potential benefits for fulfilling multi-view classification tasks.

Although $L_{0/1}$-SVM is highly effective in handling single-view data, it overlooks the fact that most real-world data today is in a multi-view format. Recently, multi-view learning (MVL) has emerged as a highly promising approach, drawing on data from various distinct feature sets to achieve both theoretical soundness and notable practical achievements [2]. Several studies have shown that integrating information and jointly learning from multiple views outperform the simplistic approach of learning from each view individually [3–5].

By considering the complementarity principle, MVL models can fully extract complementary discriminant information. Thus, combining multiple relational views has demonstrated the potential to enhance performance in machine learning tasks [6,7]. In the realm of machine learning, an immediate question is how to effectively utilize multi-perspective data for learning, which is entailed by ensuring consistency among the data while also leveraging the complementary advantages of different perspectives [8,9].

Consensus and complementarity (2C) are two essential principles determining how effective MVL algorithms are. By maximizing the agreement among multiple views, the consensus principle strives for precise classification on every view. In contrast, the complementarity principle highlights the significance of utilizing complementary information shared by views to give a comprehensive description of the

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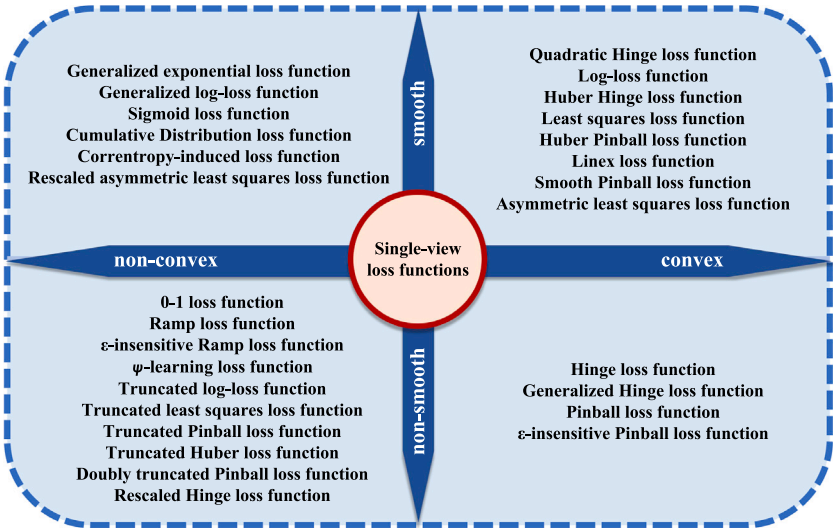


Fig. 1. Classification of common single-view loss functions on convexity and smoothness.

Table 1
A summary of multi-view classification models.

Theme	Time	Consensus	Complementarity	Views for classification
SVM-2K [10]	2005	✓	×	2
MvTSVMs [11]	2015	✓	×	2
MvCVM [12]	2016	✓	×	>3
MvMED-2C [13]	2016	✓	✓	2
MVNPSVM [14]	2018	✓	×	2
MvLSSVM [15]	2018	×	✓	>3
RMvLSTSVMs [16]	2018	✓	×	2
PSVM-2V [17]	2018	✓	✓	2
MvIGSVMs [18]	2019	✓	×	2
MCPK [19]	2019	✓	✓	2
GAMvMED [20]	2019	✓	✓	>3
Mvs-SVM [21]	2020	✓	×	>3
MvSVM-2C [22]	2020	✓	✓	>3
ρTMV-RKM [23]	2021	×	✓	>3
MV-Linex [24]	2021	✓	✓	>3
MVLSSVM-2C [25]	2021	✓	✓	>3
GMGSVMs [26]	2022	✓	✓	>3
MvTHSVM [27]	2022	✓	×	2
MvTHSVM-2C [27]	2022	✓	✓	>3
MPWTSVM [28]	2022	✓	✓	2
MVAST-BX [29]	2023	✓	✓	2
MvSLMC [6]	2023	✓	✓	>3
DMvTSVM [30]	2023	✓	×	>3
PIFMvSVM [31]	2023	✓	✓	>3
MVLDM [32]	2024	✓	✓	2

data. 2C principles offer reliable guidance for constructing models in MVL algorithms. In Table 1, we present a summary of various multi-view classification models along with their key characteristics, such as the year of publication, consensus and/or complementarity among views, and the number of views that can be utilized for classification at the same time. Moreover, a summary of the representative multi-view models and key techniques for implementing 2C principles is shown in Fig. 2.

Meanwhile, the majority of these models, such as SVM-2K and MvTSVMs, primarily address problems in only two views. Despite being general multi-view models, multi-view SVMs with the consensus and complementary information (MvSVM-2C) [22] and improved multi-view PSVM (IPSVM) [33] still use a pairwise method to examine the relationships between each pair of views. These models inevitably lead to a minimum of C_V^2 constraints, where V represents the number of views.

Despite the fact that many excellent support vector machine [34] models have been expanded to the field of MVL with different approaches, achieving good performance [10,11,28], most multi-view classifiers based on SVM tend to solve a large quadratic programming problem, which is time consuming, especially for large MVL problems. Besides, they ignore the prior information within classes, which may lead to poor performance. These models are still subject to the limitations of SVM, emphasizing class separation over structural information (SI) within classes, thereby imposing restrictions on their effectiveness.

In this work, we focus on multi-view classification problems within the SVM framework. The first challenge we address is the inability of $L_{0/1}$-SVM to be directly applied to the multi-view domain. To tackle this, a novel multi-view support vector machine classifier via the $L_{0/1}$ soft-margin loss, called Mv$L_{0/1}$-SVM, is introduced. Then, we aim to address the prevalent issue of failing to utilize SI by enhancing Mv$L_{0/1}$-SVM. This enhancement results in the development of Mv$SL_{0/1}$-SVM, which incorporates clustering techniques alongside a structural regularization term to capture and leverage SI effectively. This approach adheres to the 2C principles, achieving robust performance without relying on pairwise comparisons. The overall flow of Mv$SL_{0/1}$-SVM is shown in Fig. 3. More specifically, the $L_{0/1}$ soft-margin loss is embedded within the MVL framework to derive the optimal hyperplane. This loss function enables our model to achieve robust performance by effectively distinguishing support vectors from noise, enhancing its resilience to noise in multi-view data. Furthermore, by incorporating the hierarchical agglomerative clustering (HAC) technique into the SVM framework, we improve the model's ability to capture and leverage the SI inherent in multi-view data. SI from different views is integrated into a structural regularization term through cross-view information exchange, further improving classification performance. These enhancements collectively result in superior performance of our models compared to several state-of-the-art approaches, particularly in handling large-scale multi-view datasets, as demonstrated in our experimental results.

To the best of our knowledge, this study is the first to incorporate the $\ell_{0/1}(\cdot)$ loss function into MVL. Our key contributions are detailed below:

(1) To increase the speed and improve the robustness to noise of MVL algorithm, we propose two MVL methods, termed as

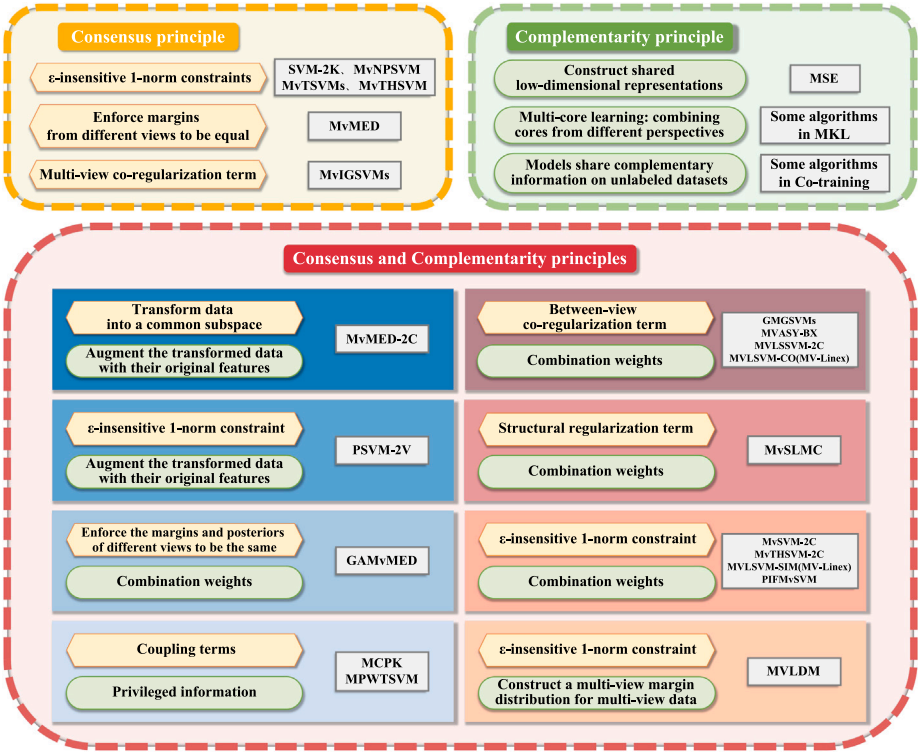


Fig. 2. Representative multi-view learning models and key techniques for implementing consensus and complementarity principles.

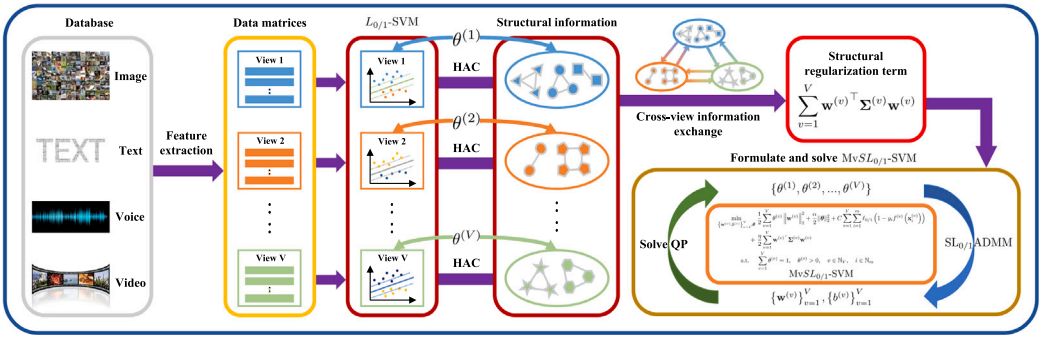


Fig. 3. The framework of Mv$SL_{0/1}$-SVM. By integrating $L_{0/1}$ soft-margin loss and the HAC technique into the multi-view SVM framework, the model's ability is improved to capture and utilize the structural information inherent in multi-view data. Additionally, structural information from different views is combined into a structural regularization term through cross-view information exchange.

Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM. By incorporating view weights, both Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM adhere to the complementarity principle and avoid reliance on pairwise comparisons.

(2) Considering that Mv$L_{0/1}$-SVM fails to follow the consensus principle, we introduce Mv$SL_{0/1}$-SVM, which integrates SI through HAC and cross-view information exchange to enhance classification performance. Mv$L_{0/1}$-SVM can be seen as a special case of Mv$SL_{0/1}$-SVM. Although Mv$SL_{0/1}$-SVM is NP-hard, we demonstrate that globally optimal solutions exist and derive the optimality conditions for discovering these solutions. The connection between these conditions and the PSP is pivotal, offering a novel approach for solving both Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM effectively.

(3) To tackle the computational challenges associated with Mv$SL_{0/1}$-SVM, we design a fast iterative approach to solve them by using ADMM. Additionally, we integrate a working set into the proximal ADMM framework, which can significantly accelerate computation and reduce the number of SVs. These computational improvements make our models more practical for large-scale datasets.

The rest of the paper is organized as follows. Research works associated with our model are introduced in Section 2. Section 3 presents the proposed Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM. In Section 4, we aim to establish the optimality condition of our Mv$SL_{0/1}$-SVM. In Section 5, we develop an iterative algorithm based on SL$_{0/1}$ADMM to solve Mv$SL_{0/1}$-SVM. Section 6 reports the comparative results across different datasets. Finally, in Section 7, we present the conclusion and future directions.

Table 2
List of main notations.

Symbol	Denotation	Scope
m	Number of training samples	$\mathbb{N}$
n	Total dimension of the sample	$\mathbb{N}$
$n^{(v)}$	Dimension of the sample in v-th view	$\mathbb{N}$
V	Number of the views	$\mathbb{N}$
$\mathbf{w}$	Weight vector of the hyperplane	$\mathbb{R}^n$
$\mathbf{w}^{(v)}$	Weight vector of the hyperplane in v-th view	$\mathbb{R}^{n^{(v)}}$
b	Bias of the hyperplane	$\mathbb{R}$
$b^{(v)}$	Bias of the hyperplane in v-th view	$\mathbb{R}^n$
$\boldsymbol{\theta}$	View weights	$(0,1)^V$
$\theta^{(v)}$	Weight of the v-th view	$(0,1)$
$\mathbb{N}_m$	$\{1,2,\ldots,m\}$	–
$\mathcal{I},\mathcal{T}$	A subset of $\mathbb{N}_m$	–
$\mathbf{1}$	$(1,1,\ldots,1)^\top$	$\mathbb{R}^m$
$\mathcal{S}^{(v)}$	Dataset containing data matrix from v views	–
$\mathbf{X}^{(v)}$	Data matrix of the v-th view	$\mathbb{R}^{m\times n^{(v)}}$
$\mathbf{y}$	$(y_1,y_2,\ldots,y_m)^\top$	$\{-1,1\}^m$
$\mathbf{Y}$	Diag$\{y_1,y_2,\ldots,y_m\}$	$\mathbb{R}^{m\times m}$
$\mathbf{M}$	$[y_1\mathbf{x}_1 y_2\mathbf{x}_2\cdots y_m\mathbf{x}_m]^\top$	$\mathbb{R}^{m\times n}$
$\mathbf{M}^{(v)}$	$\left[y_1\mathbf{x}_1^{(v)}\ y_2\mathbf{x}_2^{(v)}\ \cdots\ y_m\mathbf{x}_m^{(v)}\right]^\top$	$\mathbb{R}^{m\times n^{(v)}}$
$\mathbf{M}_{\mathcal{I}}^{(v)}$	Submatrix containing rows of $\mathbf{M}^{(v)}$ indexed on $\mathcal{I}$	$\mathbb{R}^{\lvert\mathcal{I}\rvert\times n^{(v)}}$
$\widetilde{\mathbf{M}}$	$[\mathbf{M}\ \mathbf{y}]$	$\mathbb{R}^{m\times(n+1)}$
$\widetilde{\mathbf{M}}^\dagger$	Generalized inverse of $\widetilde{\mathbf{M}}$	$\mathbb{R}^{(n+1)\times m}$
$\mathbf{L}$	$\begin{bmatrix}\mathbf{I}_{n\times n} & \mathbf{0}\\ \mathbf{0} & 0\end{bmatrix}\widetilde{\mathbf{M}}^\dagger$, here $\mathbf{0}\in\mathbb{R}^n$	$\mathbb{R}^{(n+1)\times m}$
$\lambda_{\mathbf{L}}$	Maximum eigenvalue of $\mathbf{L}^\top\mathbf{L}$	$\mathbb{R}$
$\mathbf{t}$	$\mathbf{1}-\mathbf{M}\mathbf{w}-b\mathbf{y}$	$\mathbb{R}^m$
$\mathbf{t}^{(v)}$	$\mathbf{1}-\mathbf{M}^{(v)}\mathbf{w}^{(v)}-b^{(v)}\mathbf{y}$	$\mathbb{R}^m$
$\mathbf{t}_+$	$\left((t_1)_+,\ldots,(t_m)_+\right)^\top$	$\mathbb{R}^m$
$\mathbf{t}_+^{(v)}$	$\left(\left(t_1^{(v)}\right)_+,\ldots,\left(t_m^{(v)}\right)_+\right)^\top$	$\mathbb{R}^m$
t_+	$\max\{t,0\}$	$\mathbb{R}$
$t_+^{(v)}$	$\max\{t^{(v)},0\}$	$\mathbb{R}$
$\lVert\cdot\rVert_0$	The number of non-zero elements	$\mathbb{N}$
$\mathcal{O}$	Time complexity of the model	–
$\boldsymbol{\xi}$, $\boldsymbol{\eta}$, $\boldsymbol{\xi}^{(v)}$, $\boldsymbol{\eta}^{(ij)}$,	Slack variables	$\mathbb{R}^m$
C, C_1, C_2, D, ϵ, ρ, σ, γ, α, η	Nonnegative parameters	$\mathbb{R}$
l	Counter for the outer loop of the algorithm	$\mathbb{N}$
k	Counter for the inner loop of the algorithm	$\mathbb{N}$

2. Related work

This section briefly overviews $L_{0/1}$-SVM and some relevant MVL models. Let $\mathcal{S}^{(V)}$ be a V-view dataset $\{\mathbf{X}^{(1)},\ldots,\mathbf{X}^{(V)},\mathbf{y}\} = \{(\mathbf{x}_i^{(1)},\ldots,\mathbf{x}_i^{(V)},y_i)\}_{i=1}^m \subseteq \mathcal{X}_1\times\cdots\times\mathcal{X}_V\times\mathcal{Y}$, where $\mathcal{X}_v\,(v\in\mathbb{N}_V)$ are feature spaces and $\mathcal{Y}=\{-1,+1\}$ is the label space. For simplification, we have the main notations presented in Table 2.

2.1. $L_{0/1}$-SVM

Given a one-view data set $\mathcal{S}^{(1)}$, the ideal soft-margin SVM, known as $L_{0/1}$-SVM, can be expressed as:

$$\min_{\mathbf{w}\in\mathbb{R}^n,b\in\mathbb{R}} \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^m \ell_{0/1}\left(1-y_i f_1\left(\mathbf{x}_i\right)\right),\quad i\in\mathbb{N}_m. \tag{1}$$

where $f_1\left(\mathbf{x}_i\right) := \langle\mathbf{w},\mathbf{x}_i\rangle + b$. The soft-margin loss function $\ell_{0/1}(\cdot)$ of (1) is

$$\ell_{0/1}\left(t_i\right) = \begin{cases} 1, & t_i>0,\\ 0, & t_i\le 0,\end{cases}\quad i\in\mathbb{N}_m,$$

and $t_i = 1-y_i\langle\mathbf{w},\mathbf{x}_i\rangle - y_i b = 1-y_i f_1\left(\mathbf{x}_i\right), i\in\mathbb{N}_m$.

Wang et al. [1] rewrote $L_{0/1}$-SVM (1) into the matrix form:

$$\begin{aligned}\min_{\mathbf{w}\in\mathbb{R}^n,b\in\mathbb{R},\mathbf{t}\in\mathbb{R}^m}\ & \frac{1}{2}\|\mathbf{w}\|_2^2 + C\left\|\mathbf{t}_+\right\|_0\\ \text{s.t.}\ & \mathbf{t}+\mathbf{M}\mathbf{w}+b\mathbf{y}=\mathbf{1}.\end{aligned} \tag{2}$$

Moreover, they defined the PSP of (2) in Definition 2.1 and proved the optimality conditions of (2) in Theorem 2.1.

Definition 2.1 ([1]). P-stationary point of (2). For a given $C>0$, we call $(\mathbf{w}^*;b^*;\mathbf{t}^*)$ a proximal stationary (P-stationary) point of (2) if there is a Lagrangian multiplier $\lambda^*\in\mathbb{R}^m$ and a constant $\gamma>0$ such that

$$\begin{cases}\mathbf{w}^*+\mathbf{M}^\top\lambda^* = \mathbf{0},\\ \langle\mathbf{y},\lambda^*\rangle = 0,\\ \mathbf{t}^*+\mathbf{M}\mathbf{w}^*+b^*\mathbf{y}=\mathbf{1},\\ \text{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{t}^*-\gamma\lambda^*\right)=\mathbf{t}^*,\end{cases}$$

where

$$\left[\text{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{z}^*\right)\right]_i = \begin{cases}0, & 0<z_i^*\le\sqrt{2\gamma C},\\ z_i^*, & z_i^*>\sqrt{2\gamma C}\ \text{or}\ z_i^*\le 0,\end{cases}$$

and $\mathbf{z}^* := \mathbf{t}^*-\gamma\lambda^*$.

Theorem 2.1. For problem (2), the following relations hold.

(i) Assume that $\widetilde{\mathbf{M}}$ is full column rank. When $C>0$ is given, if $(\mathbf{w}^*;b^*;\mathbf{t}^*)$ is a globally optimal solution of (2), then it is a P-stationary point of (2) with $0<\gamma<1/\lambda_{\mathbf{L}}$.

(ii) When $C>0$ is given, if $(\mathbf{w}^*;b^*;\mathbf{t}^*)$ is a P-stationary point of (2) with $\gamma>0$, then it is a locally optimal solution of (2).

The theorem elucidates the relationship between local/global minimizers and a PSP of the $L_{0/1}$-SVM. In [1], the PSP serves as a termination criterion for the iterative points generated by $\text{L}_{0/1}$ADMM, an algorithm designed explicitly for numerically solving (2). Additionally, they introduced $L_{0/1}$ SVs based on the PSP of $L_{0/1}$-SVM and working set strategy based on Theorem 2.1. The working set strategy reduces the number of SVs and speeds up the computation. In their work, $L_{0/1}$-SVM exhibits significant advantages both theoretically and experimentally.

2.2. MVL models

SVM-2K adheres to the consistency principle of MVL through the use of kernel canonical correlation analysis (KCCA) theory but fails to follow the principle of complementarity. Given a two-view data set $\mathcal{S}^{(2)}$, the formulation of SVM-2K can be outlined as follows:

$$\begin{aligned}\min_{\mathbf{w}^{(1)/(2)},\boldsymbol{\xi}^{(1)/(2)},\boldsymbol{\eta}}\ & \frac{1}{2}\left\|\mathbf{w}^{(1)}\right\|_2^2+\frac{1}{2}\left\|\mathbf{w}^{(2)}\right\|_2^2 + C_1\mathbf{1}^\top\boldsymbol{\xi}^{(1)}+C_2\mathbf{1}^\top\boldsymbol{\xi}^{(2)}+D\mathbf{1}^\top\boldsymbol{\eta}\\ \text{s.t.}\ & \left|\mathbf{X}^{(1)}\mathbf{w}^{(1)}-\mathbf{X}^{(2)}\mathbf{w}^{(2)}\right|\le\boldsymbol{\eta}+\epsilon\mathbf{1},\\ & \mathbf{Y}\mathbf{X}^{(1)}\mathbf{w}^{(1)}\ge\mathbf{1}-\boldsymbol{\xi}^{(1)},\\ & \mathbf{Y}\mathbf{X}^{(2)}\mathbf{w}^{(2)}\ge\mathbf{1}-\boldsymbol{\xi}^{(2)},\\ & \boldsymbol{\xi}^{(1)}\ge\mathbf{0},\boldsymbol{\xi}^{(2)}\ge\mathbf{0},\boldsymbol{\eta}\ge\mathbf{0}.\end{aligned} \tag{3}$$

SVM-2K solely incorporates the consensus constraint and does not adhere to the complementarity principle. Therefore, based on SVM-2K, subsequent SVM-based MVL classifiers primarily explore complementarity between views. Examples include MvSVM-2C, which utilizes view weights, and its formulation is:

$$\begin{aligned}\min_{\mathbf{w}^{(v)},\boldsymbol{\xi}^{(v)},\boldsymbol{\eta}^{(ij)},\boldsymbol{\theta}}\ & \frac{1}{2}\sum_{v=1}^V\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|_2^2 + D\sum_{i=1}^V\sum_{i<j}^V\mathbf{1}^\top\boldsymbol{\eta}^{(ij)}+C\sum_{v=1}^V\mathbf{1}^\top\boldsymbol{\xi}^{(v)}+\frac{\rho}{2}\|\boldsymbol{\theta}\|_2^2\\ \text{s.t.}\ & \left|\mathbf{X}^{(i)}\mathbf{w}^{(i)}-\mathbf{X}^{(j)}\mathbf{w}^{(j)}\right|\le\boldsymbol{\eta}^{(ij)}+\epsilon\mathbf{1},\\ & \mathbf{Y}\mathbf{X}^{(v)}\mathbf{w}^{(v)}\ge\mathbf{1}-\boldsymbol{\xi}^{(v)},\\ & \boldsymbol{\xi}^{(v)}\ge\mathbf{0},\boldsymbol{\eta}^{(ij)}\ge\mathbf{0},\\ & \sum_{i=1}^V\theta^{(v)}=1,\ \theta^{(v)}>0,\ i,j,v\in\mathbb{N}_V,\ 1\le i\le j\le V.\end{aligned}$$

MvSVM-2C incorporates C_V^2 constraints in a pairwise way to explore consensus information across diverse views. The combination weight $\boldsymbol{\theta}$ maintains balance in relationships and uncovers complementary information among different views.

2.3. Structural information

Extracting and preserving SI in data often provides advantages for machine learning [6,35]. Generally speaking, some researchers have categorized SI into two types.

(1) The inherent structure within a single sample, including block, sparse, and low-rank structures [36] that might be presented in matrix data, as well as the sequential arrangement observed in biological data, and so forth.
(2) SI among samples, like the prevalent topological details.

In contrast to the inherent structure, the SI among samples is more widespread in real-world datasets. Researchers have explored and utilized SI in various methods to enhance learning outcomes and deepen their comprehension of data, such as manifold regularization terms [37,38], structural granularity [39], and others. Multi-view data, originating from diverse sources, inherently contains richer SI.

Moreover, a framework was introduced to adeptly integrate Same-Subject-Modalities-Interactions to exploit the SI present in Magnetic Resonance Imaging and Positron Emission Tomography datasets for Alzheimer's disease classification [40]. This approach exemplifies the application of SI in enhancing learning outcomes, as it captures the nuanced interactions between modalities that are often characteristic of rich multi-view data. However, extracting and preserving SI may require substantial computational resources to process and analyze effectively.

3. Foundation of Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM

In this section, in order to better capture the essence of binary classification in MVL, we extend the $L_{0/1}$-SVM framework to Mv$L_{0/1}$-SVM to tackle multi-view classification tasks. Moreover, we develop Mv$SL_{0/1}$-SVM to leverage SI among instances effectively.

3.1. Mv$L_{0/1}$-SVM

Given a multi-view data set $S^{(V)}$, our goal is to obtain the prediction function $f(\mathbf{x})$. Adhering to the principles of SVM, Mv$L_{0/1}$-SVM can be written like:

$$\begin{aligned}\min_{\{\mathbf{w}^{(v)},b^{(v)}\}_{v=1}^{V},\theta}\ & \frac{1}{2}\sum_{v=1}^{V}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|_2^2+\frac{\alpha}{2}\|\theta\|_2^2+C\sum_{v=1}^{V}\sum_{i=1}^{m}\ell_{0/1}\left(1-y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right)\right)\\ \text{s.t.}\quad & \sum_{v=1}^{V}\theta^{(v)}=1,\quad \theta^{(v)}>0,\quad v\in\mathbb{N}_V,\quad i\in\mathbb{N}_m,\end{aligned}\tag{4}$$

where α and C are penalty coefficients and $f^{(v)}\left(\mathbf{x}_i^{(v)}\right):=\left\langle\mathbf{w}^{(v)},\mathbf{x}_i^{(v)}\right\rangle+b^{(v)}$. Here, $\ell_{0/1}(\cdot)$ is the ideal loss function given by

$$\ell_{0/1}\left(y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right)\right)=\begin{cases}1, & 1-y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right)>0,\\ 0, & 1-y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right)\le 0.\end{cases}$$

We refer to (4) as Mv$L_{0/1}$-SVM, which minimizes the number of samples in each view that are misclassified by a soft margin. For explanatory purposes, we present the following illustrations of the Mv$L_{0/1}$-SVM mechanism.

- The regularization term $(1/2)\sum_{v=1}^{V}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|_2^2$ is utilized to prevent overfitting. The view weight $\theta^{(v)}$ indicates the significance of the v-th view and facilitates exploring complementary information among multiple views.
- θ represents the collection of trainable view weights, and the term $(\alpha/2)\|\theta\|_2^2$ is employed to prevent the trivial solution.
- The predictor relies on $\left\{\mathbf{w}^{(v)},b^{(v)}\right\}_{v=1}^{V}$ and θ. For a testing instance $\mathbf{x}'=\left(\mathbf{x}^{(1)},\ldots,\mathbf{x}^{(V)}\right)$, its prediction is given by

$$f(\mathbf{x}')=\operatorname{sign}\left(\sum_{v=1}^{V}\theta^{(v)}\left(\mathbf{w}^{(v)\top}\mathbf{x}^{(v)}+b^{(v)}\right)\right).$$

3.2. Mv$SL_{0/1}$-SVM

3.2.1. Problem formulation

Given a multi-view data set $S^{(V)}$, our objective is to obtain the prediction function $f(\mathbf{x})$. Adhering to the principles of SVM, Mv$SL_{0/1}$-SVM can be written like:

$$\begin{aligned}\min_{\{\mathbf{w}^{(v)},b^{(v)}\}_{v=1}^{V},\theta}\ & \frac{1}{2}\sum_{v=1}^{V}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|_2^2+\frac{\alpha}{2}\|\theta\|_2^2+C\sum_{v=1}^{V}\sum_{i=1}^{m}\ell_{0/1}\left(1-y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right)\right)\\ & +\frac{\eta}{2}\sum_{v=1}^{V}\mathbf{w}^{(v)\top}\boldsymbol{\Sigma}^{(v)}\mathbf{w}^{(v)}\\ \text{s.t.}\quad & \sum_{v=1}^{V}\theta^{(v)}=1,\quad \theta^{(v)}>0,\quad v\in\mathbb{N}_V,\quad i\in\mathbb{N}_m,\end{aligned}\tag{5}$$

where α, C and η are penalty coefficients. $f^{(v)}\left(\mathbf{x}_i^{(v)}\right)$ and $\ell_{0/1}(\cdot)$ are defined identically as in Mv$L_{0/1}$-SVM. We refer to (5) as Mv$SL_{0/1}$-SVM, which minimizes the number of samples in each view that are misclassified by a soft margin. For explanatory purposes, we present the following illustrations of the mechanism of Mv$SL_{0/1}$-SVM. Mv$SL_{0/1}$-SVM builds upon the foundation of Mv$L_{0/1}$-SVM by introducing the following additional SI. The depiction is as follows.

- $\boldsymbol{\Sigma}^{(v)}$ $(v\in\mathbb{N}_V)$, which is composed of the weighted sum of many covariance matrices, displays the SI of each view while satisfies the properties of the covariance matrix at the same time, such as symmetric and positive semidefinite.
- The structural regularization term $(\eta/2)\sum_{v=1}^{V}\mathbf{w}^{(v)\top}\boldsymbol{\Sigma}^{(v)}\mathbf{w}^{(v)}$ enhances the compactness within classes, thereby promoting accurate classification of samples.

3.2.2. Fusing via structural information

In SVMs, the primary goal is to maximize the margin between classes during the construction of the optimal classification hyperplane. However, this emphasis on maximizing inter-class distance might neglect crucial structural details present in different classes, particularly in real-world problems. This challenge is further compounded in multi-view scenarios, where the complexities of capturing and representing diverse SI across multiple views arise.

In order to tackle this difficulty, we draw inspiration from [39] and use the HAC technique to fuse SI. Furthermore, Ward's linkage is used to calculate the separation between two clusters.

In essence, by integrating HAC and Ward's linkage into the SVM framework, we aim to enhance its capability to capture and leverage SI inherent in multi-view data, thus addressing the limitations of traditional SVM in handling diverse class structures. Specifically, when given two clusters C_1 and C_2 with their respective means $\boldsymbol{\mu}_{C_1}$ and $\boldsymbol{\mu}_{C_2}$, Ward's linkage can be calculated as follows:

$$W(C_1,C_2)=\frac{|C_1|\cdot|C_2|\cdot\left\|\boldsymbol{\mu}_{C_1}-\boldsymbol{\mu}_{C_2}\right\|_2^2}{(|C_1|+|C_2|)}.$$

HAC is carried out independently for every class in every view in Mv$SL_{0/1}$-SVM. At each iteration of HAC, the algorithm merges the two clusters with the shortest Ward's linkage. We assume that the positive and negative classes form a total of $c^{(v)}$ clusters together in the v-th view and denote the j-th cluster in the v-th view as $c_j^{(v)}$. To represent the information of $c_j^{(v)}$ in the u-th view $(u\in\mathbb{N}_V, u\neq v)$, the corresponding covariance matrix can be defined as:

$$\boldsymbol{\Sigma}_{c_j^{(v)}}^{(u)}=\frac{1}{n_j^{(v)}}\sum_{k\in c_j^{(v)}}\left(\mathbf{x}_k^{(u)}-\boldsymbol{\mu}_{c_j^{(v)}}^{(u)}\right)\left(\mathbf{x}_k^{(u)}-\boldsymbol{\mu}_{c_j^{(v)}}^{(u)}\right)^{\top},$$

where $\boldsymbol{\mu}^{(u)}_{c^{(v)}_j}$ and $n^{(v)}_j$ denote the corresponding mean and the number of samples in cluster $c^{(v)}_j$. Crucially, sample indices are stored in the cluster. We can calculate the covariance matrix of these samples in different views by using these indexes. This approach enables the conveyance of SI from one view to another, thereby enhancing cross-view information exchange.

As views may have varying degrees of contribution, we additionally incorporate view weights into the covariance matrix $\boldsymbol{\Sigma}^{(u)}$, which can be calculated with the following formula:

$$\boldsymbol{\Sigma}^{(u)} = \sum_{v=1}^{V} \theta^{(v)} \sum_{j=1}^{c^{(v)}} \boldsymbol{\Sigma}^{(u)}_{c^{(v)}_j}.$$

Moreover, our Mv$SL_{0/1}$-SVM model adopts an adaptive method proposed by Salvador and Chan [41] to determine cluster numbers in each view. The number of clusters is usually chosen based on the knee point, as the point usually depicts the maximum curvature.

4. Optimality theory and analysis of Mv$SL_{0/1}$-SVM

Since Mv$L_{0/1}$-SVM is a particular case of Mv$SL_{0/1}$-SVM, we begin by analyzing the latter and subsequently draw direct conclusions about Mv$L_{0/1}$-SVM.

In this section, we aim to establish the optimality condition of our Mv$SL_{0/1}$-SVM, which has a significant relationship with the PSP and turns out to be highly practical in addressing the problem. Before that, we first have to demonstrate the existence of globally optimal solutions.

As stated in (5), there are two sets of variables that need to be solved: $\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^{V}$ and $\boldsymbol{\theta}$. However, optimizing them simultaneously presents a challenge. To address this, we fix one set of variables while solving for another. So we solve the following optimization problem after fixing $\boldsymbol{\theta}$ in order to obtain $\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^{V}$:

$$\begin{aligned}\min_{\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^{V}} \frac{1}{2}\sum_{v=1}^{V}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|_2^2 + C\sum_{v=1}^{V}\sum_{i=1}^{m}\ell_{0/1}\left(1-y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right)\right)\\ +\frac{\eta}{2}\sum_{v=1}^{V}\mathbf{w}^{(v)\top}\boldsymbol{\Sigma}^{(v)}\mathbf{w}^{(v)}.\end{aligned} \tag{6}$$

It is clear that $\mathbf{w}$ and b are solved independently for each view here. Hence, without sacrificing generality, we can focus our research solely on the following sub-problem in the v-th view:

$$\begin{aligned}\min_{\mathbf{w}^{(v)}\in\mathbb{R}^{n^{(v)}}, b^{(v)}\in\mathbb{R}} \frac{1}{2}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|_2^2 + C\sum_{i=1}^{m}\ell_{0/1}\left(1-y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right)\right)\\ +\frac{\eta}{2}\mathbf{w}^{(v)\top}\boldsymbol{\Sigma}^{(v)}\mathbf{w}^{(v)},\end{aligned} \tag{7}$$

where $\theta^{(v)} \in (0,1)$ represents the weight of the v-th view, C and η are nonnegative tradeoff parameters given in advance. We name (7) as $SL_{0/1}$-SVM, which facilitates the correct classification of instances by the structural regularization.

It can be easily checked that $t_i^{(v)} = 1 - y_i\left\langle \mathbf{w}^{(v)}, \mathbf{x}_i^{(v)}\right\rangle - y_i b^{(v)} = 1 - y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right), i \in \mathbb{N}_m$. Thus, we can rewrite the soft-margin loss function $\ell_{0/1}(\cdot)$ of $SL_{0/1}$-SVM as

$$\ell_{0/1}\left(t_i^{(v)}\right) = \begin{cases} 1, & t_i^{(v)} > 0,\\ 0, & t_i^{(v)} \le 0,\end{cases} \quad i \in \mathbb{N}_m.$$

This indicates that we can calculate the total count of all positive elements in $\mathbf{t}^{(v)}$ by the function $L_{0/1}\left(\mathbf{t}^{(v)}\right) = \left\|\mathbf{t}_+^{(v)}\right\|_0$, which was introduced as the $L_{0/1}$ soft-margin loss function [1].

$$\begin{aligned}\sum_{i=1}^{m}\ell_{0/1}\left(1-y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right)\right) &= \sum_{i=1}^{m}\ell_{0/1}\left(t_i^{(v)}\right)\\ &= \left\|\mathbf{t}_+^{(v)}\right\|_0 =: L_{0/1}\left(\mathbf{t}^{(v)}\right).\end{aligned}$$

Consequently, we can get the equivalent optimization problem of the $SL_{0/1}$-SVM (7),

$$\begin{aligned}\min_{\mathbf{w}^{(v)}\in\mathbb{R}^{n^{(v)}}, b^{(v)}\in\mathbb{R}} f(\mathbf{w}^{(v)}; b^{(v)}) := \frac{1}{2}\theta^{(v)}\|\mathbf{w}^{(v)}\|_2^2 + \frac{\eta}{2}\mathbf{w}^{(v)\top}\boldsymbol{\Sigma}^{(v)}\mathbf{w}^{(v)}\\ + C\left\|(\mathbf{1}-\mathbf{M}^{(v)}\mathbf{w}^{(v)} - b^{(v)}\mathbf{y})_+\right\|_0,\end{aligned} \tag{8}$$

or the subsequent problem with an additional variable $\mathbf{t}^{(v)}$,

$$\begin{aligned}\min_{\mathbf{w}^{(v)}\in\mathbb{R}^{n^{(v)}}, b^{(v)}\in\mathbb{R}, \mathbf{t}^{(v)}\in\mathbb{R}^m} \frac{1}{2}\theta^{(v)}\|\mathbf{w}^{(v)}\|_2^2 + C\left\|\mathbf{t}_+^{(v)}\right\|_0 + \frac{\eta}{2}\mathbf{w}^{(v)\top}\boldsymbol{\Sigma}^{(v)}\mathbf{w}^{(v)}\\ \text{s.t.} \quad \mathbf{t}^{(v)} + \mathbf{M}^{(v)}\mathbf{w}^{(v)} + b^{(v)}\mathbf{y} = \mathbf{1}.\end{aligned} \tag{9}$$

Observing the structure of the $SL_{0/1}$-SVM model (8) or (9), we find that they have a similar structure as $\min_{\mathbf{x}\in\mathbb{R}^m} h(\mathbf{x}) + \gamma\|\mathbf{x}\|_0$, where $\gamma > 0$ is a given penalty parameter and $h : \mathbb{R}^m \to \mathbb{R}$ is smooth or nonsmooth function, just with difference between $\|(\cdot)_+\|_0$ and $\|(\cdot)\|_0$. Similarly, we analyze the optimality of (8) or (9) in this section by using continuous optimization theory.

4.1. Existence of $SL_{0/1}$-SVM optimal solution

In order to meet the optimality condition of the $SL_{0/1}$-SVM, we first show that there is a global minimizer to (8).

Theorem 4.1. *Given $b^{(v)} \in \mathcal{G} := [-N, N]$ and $0 < N < +\infty$. Then, the solution set of (8) is bounded, and there exists a globally optimal solution.*

Proof. From (8), we can check that

$$\min_{\mathbf{w}^{(v)}\in\mathbb{R}^{n^{(v)}}, b^{(v)}\in\mathbb{R}} f(\mathbf{w}^{(v)}; b^{(v)}) \le f(\mathbf{1}; b^{(v)}) < \frac{1}{2}\theta^{(v)}n^{(v)} + Cm + \frac{\eta}{2}\|\boldsymbol{\Sigma}^{(v)}\|_1 < +\infty,$$

where $\|\mathbf{A}\|_1 = \sum_{i=1}^{m}\sum_{j=1}^{n}\left|a_{ij}\right|$. Denote

$$S := \left\{(\mathbf{w}^{(v)}; b^{(v)}) \in \mathbb{R}^{n^{(v)}} \times \mathcal{G} : f(\mathbf{w}^{(v)}; b^{(v)}) < \frac{1}{2}\theta^{(v)}n^{(v)} + Cm + \frac{\eta}{2}\|\boldsymbol{\Sigma}^{(v)}\|_1\right\}.$$

It is clear that $S \neq \emptyset$ because of $(\mathbf{1}; b^{(v)}) \in S$. Given any $(\mathbf{w}^{(v)}; b^{(v)}) \in S$, due to the semi-positive nature of the covariance matrix, we can derive

$$\frac{1}{2}\theta^{(v)}\|\mathbf{w}^{(v)}\|_2^2 \le f(\mathbf{w}^{(v)}; b^{(v)}) < \frac{1}{2}\theta^{(v)}n^{(v)} + Cm + \frac{\eta}{2}\|\boldsymbol{\Sigma}^{(v)}\|_1.$$

Therefore, we can derive that $\mathbf{w}^{(v)}$ and $b^{(v)} \in \mathcal{G}$ are bounded. By applying [42, Theorem 4.10] and leveraging the lower semicontinuity of f, we can deduce that the global optimal solution to (8) exists and the solution set is bounded, which completes the proof. □

For any $b^{(v)} \in \mathcal{G}$, and considering the fact that $y_i \in \{-1, 1\}$, we can derive that

$$f\left(\mathbf{0}; b^{(v)}\right) = C\left\|\left(\mathbf{1} - b^{(v)}\mathbf{y}\right)_+\right\|_0 = \begin{cases} Cm_-, & b^{(v)} \ge 1,\\ Cm_+, & b^{(v)} \le -1,\\ Cm, & |b^{(v)}| < 1,\end{cases}$$

where m_+ and m_- denote the number of positive and negative samples respectively. Therefore, if $\left(\mathbf{w}^{(v)*}; b^{(v)*}\right)$ represents an optimal solution to (8) (which exists according to Theorem 4.1), then

$$f\left(\mathbf{w}^{(v)*}; b^{(v)*}\right) \le C\min\left\{m_+, m_-\right\}.$$

In numerical experiments, the above observation provides us with a hint to set the starting point $\left(\mathbf{w}^{(v)0}; b^{(v)0}\right)$ satisfying

$$f\left(\mathbf{w}^{(v)0}; b^{(v)0}\right) \le C\min\left\{m_+, m_-\right\}. \tag{10}$$

4.2. First-order optimality condition

One crucial stage in the theoretical analysis of optimization is to derive the optimal conditions of an optimization problem. These conditions play a pivotal role in informing algorithmic design. Let us now focus on our $SL_{0/1}$-SVM model (9).

Definition 4.1 (*P-stationary point of* (9))**.** For a given $C > 0$, we call $\left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}\right)$ a proximal stationary (P-stationary) point of (9) if there is a Lagrangian multiplier $\lambda^{(v)*} \in \mathbb{R}^m$ and a constant $\gamma > 0$ such that

$$\begin{cases} \left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)\mathbf{w}^{(v)*} + \mathbf{M}^{(v)\top}\lambda^{(v)*} = \mathbf{0}, \\ \left\langle \mathbf{y}, \lambda^{(v)*} \right\rangle = 0, \\ \mathbf{t}^{(v)*} + \mathbf{M}^{(v)}\mathbf{w}^{(v)*} + b^{(v)*}\mathbf{y} = \mathbf{1}, \\ \mathrm{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right) = \mathbf{t}^{(v)*}, \end{cases} \tag{11}$$

where

$$\left[\mathrm{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{z}^{(v)*}\right)\right]_i = \begin{cases} 0, & 0 < z_i^{(v)*} \le \sqrt{2\gamma C}, \\ z_i^{(v)*}, & z_i^{(v)*} > \sqrt{2\gamma C} \text{ or } z_i^{(v)*} \le 0, \end{cases} \tag{12}$$

and $\mathbf{z}^{(v)*} := \mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}$. The above Eq. (12) is referred to as the $L_{0/1}$ proximal operator, whose solution has been derived in $L_{0/1}$-SVM [1].

Let us reveal the connection between local/global minimizers and a PSP of $SL_{0/1}$-SVM. To delve further, denote

$$\begin{aligned} \mathbf{E}^{(v)} &:= \begin{bmatrix} \mathbf{I}^{(v)} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix} \in \mathbb{R}^{(n^{(v)}+1)\times(n^{(v)}+1)}, \\ \mathbf{F}^{(v)} &:= \left[\mathbf{M}^{(v)}\mathbf{y}\right] \in \mathbb{R}^{m\times(n^{(v)}+1)}, \\ \mathbf{G}^{(v)} &:= \mathbf{E}^{(v)}\mathbf{F}^{(v)\dagger}, \\ \widetilde{\boldsymbol{\Sigma}}^{(v)} &:= \begin{bmatrix} \boldsymbol{\Sigma}^{(v)} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix} \in \mathbb{R}^{(n^{(v)}+1)\times(n^{(v)}+1)}, \\ \lambda_{\mathbf{G}^{(v)}} &:= \lambda_{\max}\left(\mathbf{G}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{G}^{(v)}\right), \end{aligned} \tag{13}$$

where $\mathbf{I}^{(v)}$ is the identity matrix of order $n^{(v)}$, $\mathbf{F}^{(v)\dagger} \in \mathbb{R}^{(n^{(v)}+1)\times m}$ is the generalized inverse of $\mathbf{F}^{(v)}$, and $\lambda_{\max}\left(\mathbf{G}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{G}^{(v)}\right)$ is the maximum eigenvalue of $\mathbf{G}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{G}^{(v)}$. Thus, we have the following theorem.

Theorem 4.2. (First-order necessary optimality conditions) *Assume that* $\mathbf{F}^{(v)}$ *is full column rank. When* $C > 0$ *is given, if* $\left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}\right)$ *is a globally optimal solution of* (9)*, then it is a P-stationary point with* $0 < \gamma < 1/\lambda_{\mathbf{G}^{(v)}}$.

Proof. Given that $\mathbf{F}^{(v)}$ defined by (13) is full column rank, then we can derive $\mathbf{F}^{(v)\dagger} = \left(\mathbf{F}^{(v)\top}\mathbf{F}^{(v)}\right)^{-1}\mathbf{F}^{(v)\top}$. Therefore, the constraint in (9) is equivalent to $(\mathbf{w}^{(v)}; b^{(v)}) = \mathbf{F}^{(v)\dagger}(\mathbf{1} - \mathbf{t}^{(v)})$ due to $\mathbf{1} - \mathbf{t}^{(v)} = \mathbf{M}^{(v)}\mathbf{w}^{(v)} + b^{(v)}\mathbf{y} = \mathbf{F}^{(v)}(\mathbf{w}^{(v)}; b^{(v)})$. As a result, the problem (9) becomes

$$\min_{\mathbf{t}^{(v)}\in\mathbb{R}^m} \frac{1}{2}\left\|\mathbf{G}^{(v)}(\mathbf{t}^{(v)} - \mathbf{1})\right\|_2^2 + C\left\|\mathbf{t}_+^{(v)}\right\|_0 + \frac{\eta}{2}(\mathbf{t}^{(v)} - \mathbf{1})^\top\mathbf{G}^{(v)\top}\widetilde{\boldsymbol{\Sigma}}^{(v)}\mathbf{G}^{(v)}(\mathbf{t}^{(v)} - \mathbf{1}), \tag{14}$$

where $\mathbf{G}^{(v)}$ is defined in (13). Given that $\mathbf{t}^{(v)*}$ is a global minimizer of (14), and (14) is equivalent to (9), so $\mathbf{t}^{(v)*}$ is also a global minimizer of (9).

To demonstrate (11), we can roughly divide the proof into two parts.

(i) The globally optimal solution $\mathbf{t}^{(v)*}$ satisfies

$$\mathbf{t}^{(v)*} = \mathrm{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)*} - \gamma\mathbf{G}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{G}^{(v)}\left(\mathbf{t}^{(v)*} - \mathbf{1}\right)\right). \tag{15}$$

(ii) The Eq. (15) implies (11).

For simplicity, we define

$$\begin{aligned} g\left(\mathbf{t}^{(v)}\right) &:= \frac{1}{2}\left\|\mathbf{G}^{(v)}(\mathbf{t}^{(v)} - \mathbf{1})\right\|_2^2 + \frac{\eta}{2}(\mathbf{t}^{(v)} - \mathbf{1})^\top\mathbf{G}^{(v)\top}\widetilde{\boldsymbol{\Sigma}}^{(v)}\mathbf{G}^{(v)}(\mathbf{t}^{(v)} - \mathbf{1}) \\ &= \frac{1}{2}\left[\mathbf{G}^{(v)}(\mathbf{t}^{(v)} - \mathbf{1})\right]^\top\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\left[\mathbf{G}^{(v)}(\mathbf{t}^{(v)} - \mathbf{1})\right], \\ \lambda^{(v)*} &:= \mathbf{G}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{G}^{(v)}\left(\mathbf{t}^{(v)*} - \mathbf{1}\right). \end{aligned} \tag{16}$$

To prove (i), it is sufficient to prove $\mathbf{t}^{(v)*} = \mathbf{z}^{(v)}$, where $\mathbf{z}^{(v)} := \mathrm{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right)$. From the definition of the $L_{0/1}$ proximal operator [1], we have

$$\begin{aligned} &C\left\|\mathbf{z}_+^{(v)}\right\|_0 + \frac{1}{2\gamma}\left\|\mathbf{z}^{(v)} - \left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right)\right\|_2^2 \\ \le\ &C\left\|\mathbf{t}_+^{(v)*}\right\|_0 + \frac{1}{2\gamma}\left\|\mathbf{t}^{(v)*} - \left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right)\right\|_2^2 \\ =\ &C\left\|\mathbf{t}_+^{(v)*}\right\|_0 + \frac{\gamma}{2}\left\|\lambda^{(v)*}\right\|_2^2. \end{aligned} \tag{17}$$

Think of the quadratic nature of the function g, we derive that

$$g\left(\mathbf{z}^{(v)}\right) - g\left(\mathbf{t}^{(v)*}\right) \le \left\langle\lambda^{(v)*}, \mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\rangle + \frac{\lambda_{\mathbf{G}^{(v)}}}{2}\left\|\mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\|_2^2. \tag{18}$$

Due to $\mathbf{t}^{(v)*}$ is the global minimizer of (9), we can get

$$C\left\|\mathbf{t}_+^{(v)*}\right\|_0 + g\left(\mathbf{t}^{(v)*}\right) \le C\left\|\mathbf{z}_+^{(v)}\right\|_0 + g\left(\mathbf{z}^{(v)}\right). \tag{19}$$

From the three facts above, we can derive the following chain of inequalities

$$\begin{aligned} 0 &\overset{(19)}{\le} C\left\|\mathbf{z}_+^{(v)}\right\|_0 + g(\mathbf{z}^{(v)}) - \left(C\left\|\mathbf{t}_+^{(v)*}\right\|_0 + g\left(\mathbf{t}^{(v)*}\right)\right) \\ &\overset{(18)}{\le} C\left\|\mathbf{z}_+^{(v)}\right\|_0 + \left\langle\lambda^{(v)*}, \mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\rangle + \frac{\lambda_{\mathbf{G}^{(v)}}}{2}\left\|\mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\|_2^2 - C\left\|\mathbf{t}_+^{(v)*}\right\|_0 \\ &= C\left\|\mathbf{z}_+^{(v)}\right\|_0 + \left\langle\lambda^{(v)*}, \mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\rangle + \frac{1}{2\gamma}\left\|\mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\|_2^2 - C\left\|\mathbf{t}_+^{(v)*}\right\|_0 \\ &\quad - \frac{1}{2\gamma}\left\|\mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\|_2^2 + \frac{\lambda_{\mathbf{G}^{(v)}}}{2}\left\|\mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\|_2^2 \\ &= C\left\|\mathbf{z}_+^{(v)}\right\|_0 + \frac{1}{2\gamma}\left\|\mathbf{z}^{(v)} - \left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right)\right\|_2^2 - \frac{\gamma}{2}\left\|\lambda^{(v)*}\right\|_2^2 \\ &\quad - C\left\|\mathbf{t}_+^{(v)*}\right\|_0 + \frac{\lambda_{\mathbf{G}^{(v)}} - 1/\gamma}{2}\left\|\mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\|_2^2 \\ &\overset{(17)}{\le} \frac{\lambda_{\mathbf{G}^{(v)}} - 1/\gamma}{2}\left\|\mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\|_2^2, \end{aligned}$$

which indicates $\left\|\mathbf{z}^{(v)} - \mathbf{t}^{(v)*}\right\|_2^2 \le 0$ because of $0 < \gamma < 1/\lambda_{\mathbf{G}^{(v)}}$. Therefore, we have $\mathbf{z}^{(v)} = \mathbf{t}^{(v)*}$.

Next, we demonstrate (ii). From (13) and (16), we can get

$$\begin{aligned} -\lambda^{(v)*} &= \mathbf{G}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{G}^{(v)}\left(\mathbf{1} - \mathbf{t}^{(v)*}\right) \\ &= \mathbf{G}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{E}^{(v)}\mathbf{F}^{(v)\dagger}\left(\mathbf{1} - \mathbf{t}^{(v)*}\right) \\ &= \mathbf{G}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{E}^{(v)}\begin{bmatrix} \mathbf{w}^{(v)*} \\ b^{(v)*} \end{bmatrix}, \end{aligned}$$

which leads to

$$\begin{aligned} -\mathbf{F}^{(v)\top}\lambda^{(v)*} &= \mathbf{F}^{(v)\top}\mathbf{G}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{E}^{(v)}\begin{bmatrix} \mathbf{w}^{(v)*} \\ b^{(v)*} \end{bmatrix} \\ &= \mathbf{F}^{(v)\top}\left(\mathbf{F}^{(v)\dagger}\right)^\top\mathbf{E}^{(v)\top}\left(\theta^{(v)}\mathbf{E}^{(v)} + \eta\widetilde{\boldsymbol{\Sigma}}^{(v)}\right)\mathbf{E}^{(v)}\begin{bmatrix} \mathbf{w}^{(v)*} \\ b^{(v)*} \end{bmatrix} \\ &= \begin{bmatrix} \left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)\mathbf{w}^{(v)*} \\ 0 \end{bmatrix}, \end{aligned}$$

where we employed the fact $\mathbf{F}^{(v)\top}\left(\mathbf{F}^{(v)\dagger}\right)^\top = \mathbf{F}^{(v)\top}\mathbf{F}^{(v)}\left(\mathbf{F}^{(v)\top}\mathbf{F}^{(v)}\right)^{-1} = \mathbf{I}^{(v)}$. By the definition (13) that $\mathbf{F}^{(v)} = [\mathbf{M}^{(v)}\mathbf{y}]$, the above equation yields

$$\begin{cases} \left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)\mathbf{w}^{(v)*} + \mathbf{M}^{(v)\top}\lambda^{(v)*} = \mathbf{0}, \\ \left\langle \mathbf{y}, \lambda^{(v)*} \right\rangle = 0. \end{cases}$$

Therefore, the combination of (i) and (ii) completes the proof. □

Theorem 4.3. (First-order sufficient optimality conditions) *When $C > 0$ is given, if $\left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}\right)$ is a P-stationary point with $\gamma > 0$, then it is also a locally optimal solution of (9).*

Proof. Denote $\mathbf{p}^{(v)*} := \left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}\right)$ as a PSP of (9) with $\gamma > 0$ and let $\mathcal{P}^{(v)}$ be the feasible region of (9), i.e.,

$$\mathcal{P}^{(v)} := \{\mathbf{p}^{(v)} := (\mathbf{w}^{(v)}; b^{(v)}; \mathbf{t}^{(v)}) : \mathbf{t}^{(v)} + \mathbf{M}^{(v)}\mathbf{w}^{(v)} + b^{(v)}\mathbf{y} = \mathbf{1}\}. \tag{20}$$

Owing to the function $\left\|\mathbf{t}_+^{(v)}\right\|_0$ is lower semi-continuous at $\mathbf{p}^{(v)*} \in \mathcal{P}^{(v)}$, then according to [42, Proposition 4.3], there exists a neighborhood $U\left(\mathbf{p}^{(v)*}, \delta_1\right)$ of $\mathbf{p}^{(v)*} \in \mathcal{P}^{(v)}$ with $\delta_1 > 0$ such that

$$\left\|\mathbf{t}_+^{(v)}\right\|_0 > \left\|\mathbf{t}_+^{(v)*}\right\|_0 - \frac{1}{2}, \quad \forall\, \mathbf{p} \in \mathcal{P}^{(v)} \cap U\left(\mathbf{p}^{(v)*}, \delta_1\right). \tag{21}$$

Due to the fact that the covariance matrix is positive semi-definite and both $\theta^{(v)}$ and η are positive, we can deduce that $\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}$ is positive definite. Furthermore, owing to the symmetry of the matrices $\mathbf{I}^{(v)}$ and $\boldsymbol{\Sigma}^{(v)}$, $\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}$ is a symmetric positive definite matrix. Thus, we perform the Cholesky decomposition as follows.

$$\mathbf{w}^{(v)\top}\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)\mathbf{w}^{(v)} = \mathbf{w}^{(v)\top}\mathbf{H}^{(v)}\mathbf{H}^{(v)\top}\mathbf{w}^{(v)} = \left\|\mathbf{H}^{(v)\top}\mathbf{w}^{(v)}\right\|_2^2. \tag{22}$$

Additionally, since $\left\|\mathbf{H}^{(v)\top}\mathbf{w}^{(v)}\right\|_2^2$ is locally lipschitz continuous in $\mathbb{R}^{n^{(v)}}$, there exists a neighborhood $U\left(\mathbf{p}^{(v)*}, \delta_2\right)$ of $\mathbf{p}^{(v)*} \in \mathcal{P}^{(v)}$ with $\delta_2 > 0$ such that

$$\left|\left\|\mathbf{H}^{(v)\top}\mathbf{w}^{(v)}\right\|_2^2 - \left\|\mathbf{H}^{(v)\top}\mathbf{w}^{(v)*}\right\|_2^2\right| \le 2C, \quad \forall\, \mathbf{p} \in \mathcal{P}^{(v)} \cap U\left(\mathbf{p}^{(v)*}, \delta_2\right). \tag{23}$$

Denote $\delta := \min\left\{\delta_1, \delta_2\right\}$. Now we demonstrate that $\mathbf{p}^{(v)*}$ is a local minimizer of (9). In other words, there exists a neighborhood $U\left(\mathbf{p}^{(v)*}, \delta\right)$ of $\mathbf{p}^{(v)*} \in \mathcal{P}^{(v)}$ with $\delta > 0$ such that

$$\frac{1}{2}\left\|\mathbf{H}^{(v)\top}\mathbf{w}^{(v)*}\right\|_2^2 + C\left\|\mathbf{t}_+^{(v)*}\right\|_0 \le \frac{1}{2}\left\|\mathbf{H}^{(v)\top}\mathbf{w}^{(v)}\right\|_2^2 + C\left\|\mathbf{t}_+^{(v)}\right\|_0, \quad \forall\, \mathbf{p} \in \mathcal{P}^{(v)} \cap U\left(\mathbf{p}^{(v)*}, \delta\right). \tag{24}$$

Since $\mathbf{p}^{(v)*} := \left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}\right)$ is a PSP of (9) with $\gamma > 0$, there is a $\lambda^{(v)*} \in \mathbb{R}^m$ such that $\left(\mathbf{p}^{(v)*}; \lambda^{(v)*}\right)$ satisfies (11), which indicates that

$$\begin{cases} \left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)\mathbf{w}^{(v)*} + \mathbf{M}^{(v)\top}\lambda^{(v)*} = \mathbf{0}, \\ \left\langle \mathbf{y}, \lambda^{(v)*}\right\rangle = 0, \\ \mathbf{t}^{(v)*} + \mathbf{M}^{(v)}\mathbf{w}^{(v)*} + b^{(v)*}\mathbf{y} = \mathbf{1}, \\ \mathrm{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right) = \mathbf{t}^{(v)*}, \end{cases} \tag{25}$$

where

$$\left[\mathrm{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{z}^{(v)*}\right)\right]_i = \begin{cases} 0, & 0 < z_i^{(v)*} \le \sqrt{2\gamma C}, \\ z_i^{(v)*}, & z_i^{(v)*} > \sqrt{2\gamma C} \text{ or } z_i^{(v)*} \le 0. \end{cases} \tag{26}$$

To accomplish this, define $\mathcal{T}_* := \left\{i : t_i^{(v)*} = 0\right\}$ and $\bar{\mathcal{T}}_* := \mathbb{N}_m \backslash \mathcal{T}_*$. The fourth equation of (25) and (26) implies that

$$\begin{aligned} -\sqrt{2C/\gamma} \le \lambda_i^{(v)*} \le 0,\ t_i^{(v)*} = 0, \quad \forall\, i \in \mathcal{T}_*, \\ \lambda_i^{(v)*} = 0,\ t_i^{(v)*} \ne 0, \quad \forall\, i \in \bar{\mathcal{T}}_*. \end{aligned} \tag{27}$$

Moreover, denote a local region $\mathcal{P}_{\mathcal{T}}^{(v)}$ of $\mathcal{P}^{(v)}$ as

$$\mathcal{P}_{\mathcal{T}}^{(v)} := \mathcal{P}^{(v)} \cap \left\{\mathbf{p} : t_i^{(v)} \le 0,\ i \in \mathcal{T}_*\right\}, \tag{28}$$

then, we can think about the following two cases:

Case (i): $\mathbf{p} \in \mathcal{P}_{\mathcal{T}}^{(v)} \subseteq \mathcal{P}^{(v)}$ and $\mathbf{p} \in U\left(\mathbf{p}^{(v)*}, \delta\right)$. We can check that $\mathbf{p}^{(v)*} \in \mathcal{P}_{\mathcal{T}}^{(v)}$ by (25). Then for any $\mathbf{p} \in \mathcal{P}_{\mathcal{T}}^{(v)}$, we have $\mathbf{t}^{(v)} + \mathbf{M}^{(v)}\mathbf{w}^{(v)} + b^{(v)}\mathbf{y} = \mathbf{1}$, which and (25) suffice to

$$-\mathbf{M}^{(v)}\left(\mathbf{w}^{(v)} - \mathbf{w}^{(v)*}\right) = \mathbf{t}^{(v)} - \mathbf{t}^{(v)*} + \left(b^{(v)} - b^{(v)*}\right)\mathbf{y}. \tag{29}$$

Thus, for any $\mathbf{p} \in \mathcal{P}_{\mathcal{T}}^{(v)}$, the sequence of inequalities below holds:

$$\begin{aligned} &\left\|\mathbf{H}^{(v)\top}\mathbf{w}^{(v)}\right\|_2^2 - \left\|\mathbf{H}^{(v)\top}\mathbf{w}^{(v)*}\right\|_2^2 \\ =\ &\left\|\mathbf{H}^{(v)\top}\left(\mathbf{w}^{(v)} - \mathbf{w}^{(v)*}\right) + \mathbf{H}^{(v)\top}\mathbf{w}^{(v)*}\right\|_2^2 - \left\|\mathbf{H}^{(v)\top}\mathbf{w}^{(v)*}\right\|_2^2 \\ =\ &2\left\langle\mathbf{H}^{(v)\top}\left(\mathbf{w}^{(v)} - \mathbf{w}^{(v)*}\right), \mathbf{H}^{(v)\top}\mathbf{w}^{(v)*}\right\rangle + \left\|\mathbf{H}^{(v)\top}\left(\mathbf{w}^{(v)} - \mathbf{w}^{(v)*}\right)\right\|_2^2 \\ \ge\ &2\left\langle\mathbf{H}^{(v)\top}\left(\mathbf{w}^{(v)} - \mathbf{w}^{(v)*}\right), \mathbf{H}^{(v)\top}\mathbf{w}^{(v)*}\right\rangle \\ \overset{(25)}{=}\ &-2\left\langle\mathbf{H}^{(v)\top}\left(\mathbf{w}^{(v)} - \mathbf{w}^{(v)*}\right), \mathbf{H}^{(v)\top}\left(\mathbf{H}^{(v)}\mathbf{H}^{(v)\top}\right)^{-1}\mathbf{M}^{(v)\top}\lambda^{(v)*}\right\rangle \\ =\ &-2\left\langle\mathbf{M}^{(v)}\left(\mathbf{H}^{(v)}\mathbf{H}^{(v)\top}\right)^{-1}\left(\mathbf{H}^{(v)}\mathbf{H}^{(v)\top}\right)\left(\mathbf{w}^{(v)} - \mathbf{w}^{(v)*}\right), \lambda^{(v)*}\right\rangle \\ \overset{(29)}{=}\ &2\left\langle\mathbf{t}^{(v)} - \mathbf{t}^{(v)*}, \lambda^{(v)*}\right\rangle + 2\left(b^{(v)} - b^{(v)*}\right)\left\langle\mathbf{y}, \lambda^{(v)*}\right\rangle \\ \overset{(25)}{=}\ &2\left\langle\mathbf{t}^{(v)} - \mathbf{t}^{(v)*}, \lambda^{(v)*}\right\rangle \\ =\ &2\left\langle\mathbf{t}_{\mathcal{T}_*}^{(v)} - \mathbf{t}_{\mathcal{T}_*}^{(v)*}, \lambda_{\mathcal{T}_*}^{(v)*}\right\rangle + 2\left\langle\mathbf{t}_{\bar{\mathcal{T}}_*}^{(v)} - \mathbf{t}_{\bar{\mathcal{T}}_*}^{(v)*}, \lambda_{\bar{\mathcal{T}}_*}^{(v)*}\right\rangle \\ \overset{(27)}{=}\ &2\left\langle\mathbf{t}_{\mathcal{T}_*}^{(v)}, \lambda_{\mathcal{T}_*}^{(v)*}\right\rangle \\ \overset{(27),(28)}{\ge}\ &0. \end{aligned} \tag{30}$$

Since $\left\|\mathbf{t}_+^{(v)}\right\|_0$ can only take values from $\{0, 1, \ldots, m\}$, combined with (21), we can conclude that

$$\left\|\mathbf{t}_+^{(v)}\right\|_0 \ge \left\|\mathbf{t}_+^{(v)*}\right\|_0, \ \forall\, \mathbf{p} \in \mathcal{P}^{(v)} \cap U\left(\mathbf{p}^{(v)*}, \delta_1\right). \tag{31}$$

Therefore, for any $\mathbf{p} \in \mathcal{P}_{\mathcal{T}}^{(v)} \cap U\left(\mathbf{p}^{(v)*}, \delta\right) \subseteq \mathcal{P}_{\mathcal{T}}^{(v)} \cap U\left(\mathbf{p}^{(v)*}, \delta_1\right)$, it follows from (30) and (31) that

$$\frac{1}{2}\left\|\mathbf{w}^{(v)*}\right\|_2^2 + C\left\|\mathbf{t}_+^{(v)*}\right\|_0 \le \frac{1}{2}\left\|\mathbf{w}^{(v)}\right\|_2^2 + C\left\|\mathbf{t}_+^{(v)}\right\|_0. \tag{32}$$

Case (ii): $\mathbf{p} \in \left(\mathcal{P}^{(v)} \backslash \mathcal{P}_{\mathcal{T}}^{(v)}\right)$ and $\mathbf{p} \in U\left(\mathbf{p}^{(v)*}, \delta\right)$. Since $\mathbf{p} \in \left(\mathcal{P}^{(v)} \backslash \mathcal{P}_{\mathcal{T}}^{(v)}\right)$, we can derive that there exists $i_0 \in \mathcal{T}_*$ with $t_{i_0}^{(v)*} = 0$ but $t_{i_0}^{(v)} > 0$, implying $\left\|\left(t_{i_0}^{(v)*}\right)_+\right\|_0 = 0$ but $\left\|\left(t_{i_0}^{(v)}\right)_+\right\|_0 = 1$. It can be seen from $\mathbf{p} \in U\left(\mathbf{p}^{(v)*}, \delta\right)$ and (31) that:

$$\left\|\mathbf{t}_+^{(v)}\right\|_0 \ge \left\|\mathbf{t}_+^{(v)*}\right\|_0 + 1. \tag{33}$$

Summarizing (23) and (33), we obtain that for any $\mathbf{p} \in \left(\mathcal{P}^{(v)} \backslash \mathcal{P}_{\mathcal{T}}^{(v)}\right) \cap U\left(\mathbf{p}^*, \delta\right)$,

$$\begin{aligned} \frac{1}{2}\left\|\mathbf{w}^{(v)*}\right\|_2^2 + C\left\|\mathbf{t}_+^{(v)*}\right\|_0 &\le \frac{1}{2}\left\|\mathbf{w}^{(v)*}\right\|_2^2 + C\left\|\mathbf{t}_+^{(v)}\right\|_0 - C \\ &\le \frac{1}{2}\left\|\mathbf{w}^{(v)}\right\|_2^2 + C\left\|\mathbf{t}_+^{(v)}\right\|_0. \end{aligned} \tag{34}$$

Combining the above (32) and (34), we can conclude that $\mathbf{p}^{(v)*}$ is a local minimizer of (9). This completes the proof. □

According to the two theorems above, a PSP has to be a local minimizer in (9). This implies that we can employ the PSP as a termination rule for the iterative points generated by the algorithm proposed in the subsequent section.

5. Solution to Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM

SVs are generally acknowledged to determine the classifier. When designing a solution algorithm, the fewer SVs employed, the faster the computational speed because the classifier is trained on fewer training data samples. Therefore, it is essential to reduce the number of SVs, especially for massive datasets. In the preceding part, we establish the optimality theory, based on which we are going to propose SVs of $SL_{0/1}$-SVM and a working set strategy. Consequently, to solve

the $SL_{0/1}$-SVM (9), we use the widely used ADMM. Furthermore, we provide an analysis of the convergence and time complexity of Mv$L_{0/1}$-SVM and Mv$L_{0/1}$-SVM.

Definition 5.1 (*SVs of* (9))**.** Suppose $\left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}\right)$ is a PSP of (9), hence also a local minimizer of $SL_{0/1}$-SVM. From the fourth equation of (11), we define

$$\mathcal{I}_* := \left\{ i \in \mathbb{N}_m : \lambda_i^{(v)*} \in [-\sqrt{2C/\gamma}, 0) \right\}, \quad \bar{\mathcal{I}}_* := \mathbb{N}_m \backslash \mathcal{I}_*. \tag{35}$$

By (12), we can derive that

$$\begin{bmatrix} \left(\mathrm{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right)\right)_{\mathcal{I}_*} \\ \left(\mathrm{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right)\right)_{\bar{\mathcal{I}}_*} \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{\mathcal{I}_*} \\ \left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right)_{\bar{\mathcal{I}}_*} \end{bmatrix}.$$

This leads to

$$\mathbf{t}^{(v)*} = \mathrm{Prox}_{\gamma C\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)*} - \gamma\lambda^{(v)*}\right) \Longleftrightarrow \begin{bmatrix} \mathbf{t}_{\mathcal{I}_*}^{(v)*} \\ \lambda_{\bar{\mathcal{I}}_*}^{(v)*} \end{bmatrix} = \mathbf{0}. \tag{36}$$

Then $\mathcal{I}_*$ in (35) can be written as

$$\mathcal{I}_* = \left\{ i \in \mathbb{N}_m : \lambda_i^{(v)*} \in [-\sqrt{2C/\gamma}, 0) \right\}.$$

Combined with (36), we can obtain

$$\lambda_i^{(v)*} \begin{cases} \in [-\sqrt{2C/\gamma}, 0), & \text{for } i \in \mathcal{I}_*, \\ = 0, & \text{for } i \in \bar{\mathcal{I}}_*. \end{cases} \tag{37}$$

Taking (37) into (11), we can derive

$$\mathbf{w}^{(v)*} = -\mathbf{M}_{\mathcal{I}_*}^{(v)\top}\lambda_{\mathcal{I}_*}^{(v)*} - \mathbf{M}_{\bar{\mathcal{I}}_*}^{(v)\top}\lambda_{\bar{\mathcal{I}}_*}^{(v)*} = -\mathbf{M}_{\mathcal{I}_*}^{(v)\top}\lambda_{\mathcal{I}_*}^{(v)*} = \sum_{i\in\mathcal{I}_*} -\lambda_i^{(v)*} y_i \mathbf{x}_i^{(v)}.$$

Following the concept of SV of $L_{0/1}$-SVM, $\left\{\mathbf{x}_i^{(v)} : i \in \mathcal{I}_*\right\}$ are the SVs of the $SL_{0/1}$-SVM, here $\mathcal{I}_*$ is the index set of SVs.

From $t_i^{(v)*} = 1 - y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right) = 0$ for $i \in \mathcal{I}_*$, we can obtain $f^{(v)}\left(\mathbf{x}_i^{(v)}\right) = \pm 1$, which means all SVs of $SL_{0/1}$-SVM lie on the support surfaces $f^{(v)}\left(\mathbf{x}_i^{(v)}\right) = -\sum_{i\in\mathcal{I}_*}\lambda_i^{(v)*} y_i \mathbf{x}_i^{(v)} = \pm 1$. This geometric property indicates that the SVs of the $SL_{0/1}$-SVM are systematically arranged and likely sparse. Additionally, this elegant theoretical finding underscores its efficacy in developing an efficient algorithm for the $SL_{0/1}$-SVM.

5.1. Iterative algorithm

In the iterative algorithm, there are two sets of variables that need to be optimized: $\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^{V}$ and θ. Simultaneously optimizing these variables is challenging, so we adopt an alternating optimization approach. First, we fix θ and optimize the set of variables $\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^{V}$. This process is referred to as the inner loop. After updating $\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^{V}$, we then fix $\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^{V}$ and optimize θ in the outer loop. This alternating optimization process continues iteratively until convergence.

5.1.1. Update $\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^{V}$ via SL$_{0/1}$ADMM and working set

In this subsection, we leverage the advantages of ADMM and working set to solve the $SL_{0/1}$-SVM (9). For the problem (9), the augmented Lagrangian function can be obtained by

$$\begin{aligned} &L_\sigma\left(\mathbf{w}^{(v)}; b^{(v)}; \mathbf{t}^{(v)}; \lambda^{(v)}\right) \\ &= \tfrac{1}{2}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|_2^2 + C\left\|\mathbf{t}_+^{(v)}\right\|_0 + \langle\lambda^{(v)}, \mathbf{t}^{(v)} - \mathbf{1} + \mathbf{M}^{(v)}\mathbf{w}^{(v)} + b^{(v)}\mathbf{y}\rangle \\ &\quad + \tfrac{\sigma}{2}\left\|\mathbf{t}^{(v)} - \mathbf{1} + \mathbf{M}^{(v)}\mathbf{w}^{(v)} + b^{(v)}\mathbf{y}\right\|_2^2 + \tfrac{\eta}{2}\mathbf{w}^{(v)\top}\boldsymbol{\Sigma}^{(v)}\mathbf{w}^{(v)}, \end{aligned}$$

where $\lambda^{(v)}$ are the vectors composed of Lagrangian multipliers and $\sigma > 0$ is the nonnegative tradeoff parameter. In the k-th iteration $\left(\mathbf{w}^{(v)k}; b^{(v)k}; \mathbf{t}^{(v)k}; \lambda^{(v)k}\right)$, given that the dual step-size $\tau > 0$ and the proximal term $\left\|\mathbf{w}^{(v)} - \mathbf{w}^{(v)k}\right\|_{\mathbf{D}_k}^2$, the framework for updating each component can be written as follows:

$$\begin{aligned} \mathbf{t}^{(v)k+1} &= \underset{\mathbf{t}^{(v)}\in\mathbb{R}^m}{\mathrm{argmin}}\ L_\sigma\left(\mathbf{w}^{(v)k}, b^{(v)k}, \mathbf{t}^{(v)}, \lambda^{(v)k}\right), \\ \mathbf{w}^{(v)k+1} &= \underset{\mathbf{w}^{(v)}\in\mathbb{R}^{n^{(v)}}}{\mathrm{argmin}}\ L_\sigma\left(\mathbf{w}^{(v)}, b^{(v)k}, \mathbf{t}^{(v)k+1}, \lambda^{(v)k}\right) + \frac{\sigma}{2}\left\|\mathbf{w}^{(v)} - \mathbf{w}^{(v)k}\right\|_{\mathbf{D}_k}^2, \\ b^{(v)k+1} &= \underset{b^{(v)}\in\mathbb{R}}{\mathrm{argmin}}\ L_\sigma\left(\mathbf{w}^{(v)k+1}, b^{(v)}, \mathbf{t}^{(v)k+1}, \lambda^{(v)k}\right), \\ \lambda^{(v)k+1} &= \lambda^{(v)k} + \tau\sigma\left(\mathbf{t}^{(v)k+1} - \mathbf{1} + \mathbf{M}^{(v)}\mathbf{w}^{(v)k+1} + b^{(v)k+1}\mathbf{y}\right). \end{aligned} \tag{38}$$

The fundamental principle guiding the selection of $\mathbf{D}_k$ is to ensure the $\mathbf{w}^{(v)}$-subproblem of (38) to be convex. Here, we choose $\mathbf{D}_k$ to be an indefinite matrix in a flexible manner owing to $L_\sigma\left(\mathbf{w}^{(v)}; b^{(v)k}; \mathbf{t}^{(v)k+1}; \lambda^{(v)k}\right)$ is firmly convex with respect to $\mathbf{w}^{(v)}$.

Denote $\mathbf{z}^{(v)k} := \mathbf{1} - \mathbf{M}^{(v)}\mathbf{w}^{(v)k} - b^{(v)k}\mathbf{y} - \lambda^{(v)k}/\sigma$, and the working set $\mathcal{I}_k$ and its complementarity set with respect to $\mathbb{N}_m$ at the k-th step is

$$\mathcal{I}_k := \left\{ i \in \mathbb{N}_m : z_i^{(v)k} \in (0, \sqrt{2C/\sigma}] \right\}, \quad \bar{\mathcal{I}}_k := \mathbb{N}_m \backslash \mathcal{I}_k. \tag{39}$$

Thus, $\mathbf{D}_k$ can be chosen as

$$\mathbf{D}_k = -\mathbf{M}_{\bar{\mathcal{I}}_k}^{(v)\top}\mathbf{M}_{\bar{\mathcal{I}}_k}^{(v)}. \tag{40}$$

Given a set $\mathcal{I} \subseteq \mathbb{N}_m$, $\mathbf{M}_{\mathcal{I}}^{(v)} \in \mathbb{R}^{|\mathcal{I}|\times n^{(v)}}$ means the submatrix containing rows of $\mathbf{M}^{(v)}$ indexed on $\mathcal{I}$. To be mentioned, the choice of $\mathbf{D}_k$ and the working set $\mathcal{I}_k$ will greatly speed up calculation at each step of the ADMM. To be more precise, we compute each sub-problem in (38) indicated below.

(i) Updating $\mathbf{t}^{(v)k+1}$: We can get $\mathbf{t}^{(v)k+1}$ by the following equivalent problems

$$\begin{aligned} &\mathbf{t}^{(v)k+1} \\ &= \underset{\mathbf{t}^{(v)}\in\mathbb{R}^m}{\mathrm{argmin}}\ C\left\|\mathbf{t}_+^{(v)}\right\|_0 + \left\langle\lambda^{(v)k}, \mathbf{t}^{(v)}\right\rangle + \frac{\sigma}{2}\left\|\mathbf{t}^{(v)} - \mathbf{1} + \mathbf{M}^{(v)}\mathbf{w}^{(v)k} + b^{(v)k}\mathbf{y}\right\|^2 \\ &= \underset{\mathbf{t}^{(v)}\in\mathbb{R}^m}{\mathrm{argmin}}\ C\left\|\mathbf{t}_+^{(v)}\right\|_0 + \frac{\sigma}{2}\left\|\mathbf{t}^{(v)} - \mathbf{z}^{(v)k}\right\|_2^2 \\ &= \mathrm{Prox}_{\frac{C}{\sigma}\|(\cdot)_+\|_0}\left(\mathbf{z}^{(v)k}\right), \end{aligned}$$

where $\gamma = 1/\sigma$ is used in the final equation, which comes from (12). This, along with the working set (12) and (39), is sufficient to

$$\mathbf{t}_{\mathcal{I}_k}^{(v)k+1} = \mathbf{0}, \quad \mathbf{t}_{\bar{\mathcal{I}}_k}^{(v)k+1} = \mathbf{z}_{\bar{\mathcal{I}}_k}^{(v)k}. \tag{41}$$

Therefore, updating $\mathbf{t}^{(v)k+1}$ becomes incredibly simple and fast.

(ii) Updating $\mathbf{w}^{(v)k+1}$. The $\mathbf{w}^{(v)}$-subproblem in (38) is

$$\begin{aligned} \mathbf{w}^{(v)k+1} = \underset{\mathbf{w}^{(v)}\in\mathbb{R}^{n^{(v)}}}{\mathrm{argmin}}\ &\frac{1}{2}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|_2^2 + \frac{\sigma}{2}\left\|\mathbf{w}^{(v)} - \mathbf{w}^{(v)k}\right\|_{-\mathbf{M}_{\bar{\mathcal{I}}_k}^{(v)\top}\mathbf{M}_{\bar{\mathcal{I}}_k}^{(v)}}^2 + \left\langle\lambda^{(v)k}, \mathbf{M}^{(v)}\mathbf{w}^{(v)}\right\rangle \\ &+ \frac{\sigma}{2}\left\|\mathbf{t}^{(v)k+1} - \mathbf{1} + \mathbf{M}^{(v)}\mathbf{w}^{(v)} + b^{(v)k}\mathbf{y}\right\|_2^2 + \frac{\eta}{2}\mathbf{w}^{(v)\top}\boldsymbol{\Sigma}^{(v)}\mathbf{w}^{(v)}. \end{aligned} \tag{42}$$

This quadratic programming problem is convex. We simply need to solve the equations below in order to solve (42).

$$\begin{aligned} \mathbf{0} = &\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)\mathbf{w}^{(v)} - \sigma\mathbf{M}_{\bar{\mathcal{I}}_k}^{(v)\top}\mathbf{M}_{\bar{\mathcal{I}}_k}^{(v)}\left(\mathbf{w}^{(v)} - \mathbf{w}^{(v)k}\right) + \mathbf{M}^{(v)\top}\lambda^{(v)k} \\ &+ \sigma\mathbf{M}^{(v)\top}\left(\mathbf{t}^{(v)k+1} - \mathbf{1} + \mathbf{M}^{(v)}\mathbf{w}^{(v)} + b^{(v)k}\mathbf{y}\right), \end{aligned} \tag{43}$$

which is equivalent to solving the equation

$$\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)} + \sigma\mathbf{M}_{\mathcal{I}_k}^{(v)\top}\mathbf{M}_{\mathcal{I}_k}^{(v)}\right)\mathbf{w}^{(v)} = \sigma\mathbf{M}_{\mathcal{I}_k}^{(v)\top}\mathbf{v}_{\mathcal{I}_k}^{(v)k}, \tag{44}$$

where $\mathbf{v}^{(v)k} := -\left(\mathbf{t}^{(v)k+1} + b^{(v)k}\mathbf{y} - \mathbf{1} + \lambda^{(v)k}/\sigma\right)$. To get (44) from (43), we utilizes two facts that $\mathbf{t}_{\bar{\mathcal{I}}_k}^{(v)k+1} = \mathbf{z}_{\bar{\mathcal{I}}_k}^{(v)k}$ by (41) and $\mathbf{M}_{\mathcal{I}_k}^{(v)\top}\mathbf{M}_{\mathcal{I}_k}^{(v)} = \mathbf{M}^{(v)\top}\mathbf{M}^{(v)} - \mathbf{M}_{\bar{\mathcal{I}}_k}^{(v)\top}\mathbf{M}_{\bar{\mathcal{I}}_k}^{(v)}$.

Consequently, the term $\mathbf{M}_{\bar{I}_k}^{(v)}$ vanishes in (44), indicating that the samples $\left\{\mathbf{x}_j^{(v)}, j \in \bar{I}_k\right\}$ are discarded by the working set I_k. This could greatly speed up the computation if the chosen $|I_k|$ is really small. In practice, the following guidelines can effectively address (44):

If $n^{(v)} \leq |I_k|$, we can solve (44) by

$$\mathbf{w}^{(v)^{k+1}} = \left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)} + \sigma\mathbf{M}_{I_k}^{(v)^\top}\mathbf{M}_{I_k}^{(v)}\right)^{-1}\sigma\mathbf{M}_{I_k}^{(v)^\top}\mathbf{v}_{I_k}^{(v)^k}. \tag{45}$$

If $n^{(v)} > |I_k|$, by employing Sherman–Morrison–Woodbury formula [43], the inverse can be calculated as

$$\begin{aligned}&\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)} + \sigma\mathbf{M}_{I_k}^{(v)^\top}\mathbf{M}_{I_k}^{(v)}\right)^{-1}\\ =\ &\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)^{-1}\\ &-\ \sigma\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)^{-1}\mathbf{M}_{I_k}^{(v)^\top}\left(\mathbf{I}^{(v)} + \sigma\mathbf{M}_{I_k}^{(v)}\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)^{-1}\mathbf{M}_{I_k}^{(v)^\top}\right)^{-1}\mathbf{M}_{I_k}^{(v)}.\end{aligned}$$

Next, we update $\mathbf{w}^{(v)^{k+1}}$ by

$$\begin{aligned}&\mathbf{w}^{(v)^{k+1}}\\ =\ &\sigma\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)^{-1}\mathbf{M}_{I_k}^{(v)^\top}\left(\mathbf{I}^{(v)} + \sigma\mathbf{M}_{I_k}^{(v)}\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)^{-1}\mathbf{M}_{I_k}^{(v)^\top}\right)^{-1}\mathbf{v}_{I_k}^{(v)^k}\\ =\ &\sigma\mathbf{M}_{I_k}^{(v)^\top}\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)} + \sigma\mathbf{M}_{I_k}^{(v)}\mathbf{M}_{I_k}^{(v)^\top}\right)^{-1}\mathbf{v}_{I_k}^{(v)^k}.\end{aligned} \tag{46}$$

(iii) Updating $b^{(v)^{k+1}}$. The b-subproblem in (38) is a convex quadratic programming

$$b^{(v)^{k+1}} = \underset{b^{(v)}\in\mathbb{R}}{\operatorname{argmin}}\left\langle\lambda^{(v)^k}, b^{(v)}\mathbf{y}\right\rangle + \frac{\sigma}{2}\left\|\mathbf{t}^{(v)^{k+1}} - \mathbf{1} + \mathbf{M}^{(v)}\mathbf{w}^{(v)^{k+1}} + b^{(v)}\mathbf{y}\right\|_2^2,$$

which can be solved by

$$b^{(v)^{k+1}} = \left\langle\mathbf{y}, \mathbf{r}^{(v)^k}\right\rangle / \|\mathbf{y}\|^2 = \left\langle\mathbf{y}, \mathbf{r}^{(v)^k}\right\rangle / m, \tag{47}$$

where $\mathbf{r}^{(v)^k} := -\mathbf{M}^{(v)}\mathbf{w}^{(v)^{k+1}} + \mathbf{1} - \mathbf{t}^{(v)^{k+1}} - \lambda^{(v)^k}/\sigma$.

(iv) Updating $\lambda^{(v)^{k+1}}$. We update $\lambda^{(v)^{k+1}}$ in (38) as the following equation

$$\lambda_{I_k}^{(v)^{k+1}} = \lambda_{I_k}^{(v)^k} + \tau\sigma\boldsymbol{\varpi}_{I_k}^{(v)^{k+1}}, \quad \lambda_{\bar{I}_k}^{(v)^{k+1}} = \mathbf{0}, \tag{48}$$

where $\boldsymbol{\varpi}^{(v)^{k+1}} := \mathbf{t}^{(v)^{k+1}} - \mathbf{1} + \mathbf{M}^{(v)}\mathbf{w}^{(v)^{k+1}} + b^{(v)^{k+1}}\mathbf{y}$.

Getting inspiration from Theorem 4.3, we use the PSP as the stopping criterion. In the experiment, we terminate our Mv$SL_{0/1}$-SVM if the point $\left(\mathbf{w}^{(v)^k}; b^{(v)^k}; \mathbf{t}^{(v)^k}; \lambda^{(v)^k}\right)$ closely satisfies the conditions in (11) and ϵ denotes the tolerance level, i.e.,

$$\max\left\{\epsilon_1^k, \epsilon_2^k, \epsilon_3^k, \epsilon_4^k\right\} < \epsilon,$$

where

$$\begin{aligned}\epsilon_1^k &:= \frac{\left\|\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)\mathbf{w}^{(v)^k} + \mathbf{M}_{I_k}^{(v)^\top}\lambda_{I_k}^{(v)^k}\right\|_2}{1 + \left\|\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)\mathbf{w}^{(v)^k}\right\|_2},\\ \epsilon_2^k &:= \frac{\left|\left\langle\mathbf{y}_{I_k}, \lambda_{I_k}^{(v)^k}\right\rangle\right|}{1 + |I_k|},\\ \epsilon_3^k &:= \frac{\left\|\mathbf{t}^{(v)^k} - \mathbf{1} + \mathbf{M}^{(v)}\mathbf{w}^{(v)^k} + b^{(v)^k}\mathbf{y}\right\|_2}{\sqrt{m}},\\ \epsilon_4^k &:= \frac{\left\|\mathbf{t}^{(v)^k} - \operatorname{Prox}_{\frac{C}{\sigma}\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)^k} - \lambda^{(v)^k}/\sigma\right)\right\|_2}{1 + \left\|\mathbf{t}^{(v)^k}\right\|_2}.\end{aligned}$$

Consequently, we obtain **Algorithm 1**, which is used to solve the sub-problem (9) of our Mv$SL_{0/1}$-SVM. We call the algorithm $\mathrm{SL}_{0/1}$ADMM, an abbreviation for $SL_{0/1}$-SVM solved by ADMM.

Algorithm 1: $\mathrm{SL}_{0/1}$ADMM for solving $SL_{0/1}$-SVM.

Input: Training dataset $\{(\mathbf{x}_i^{(v)}, y_i)\}_{i=1}^m$, $\theta^{(v)}$, ϵ, τ, K, C, η, σ.
Output: $\mathbf{w}^{(v)^*}$ and $b^{(v)^*}$.

Initialize: $\mathbf{w}^{(v)^0}$, $b^{(v)^0}$, $\mathbf{t}^{(v)^0}$, $\lambda^{(v)^0}$ and $k = 0$;
While $\max\left\{\epsilon_1^k, \epsilon_2^k, \epsilon_3^k, \epsilon_4^k\right\} < \epsilon$ and $k \leq K$ **do**
  Get $\mathbf{t}^{(v)^{k+1}}$ by (41);
  if $n \leq |I_k|$ holds, **then**
    Get $\mathbf{w}^{(v)^{k+1}}$ by (45);
  else if $n > |I_k|$ holds, **then**
    Get $\mathbf{w}^{(v)^{k+1}}$ by (46);
  end if
  Get $b^{(v)^{k+1}}$ by (47);
  Get $\lambda^{(v)^{k+1}}$ by (48);
  $k = k + 1$;
end while
Return $\left(\mathbf{w}^{(v)^*}, b^{(v)^*}\right)$ to (9).

5.1.2. Update θ

After fixing $\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^V$, we can optimize θ as

$$\begin{aligned}\min_{\boldsymbol{\theta}\in\mathbb{R}^V}\ &\frac{1}{2}\boldsymbol{\theta}^\top\widetilde{\boldsymbol{\pi}} + \frac{\eta}{2}\boldsymbol{\theta}^\top\boldsymbol{\rho} + \frac{\alpha}{2}\|\boldsymbol{\theta}\|_2^2\\ \text{s.t.}\ &\sum_{v=1}^V\theta^{(v)} = 1, \quad \theta^{(v)} > 0, \quad v \in \mathbb{N}_V,\end{aligned} \tag{49}$$

where $\widetilde{\boldsymbol{\pi}} = \left[\left\|\mathbf{w}^{(1)}\right\|_2^2, \left\|\mathbf{w}^{(2)}\right\|_2^2, \ldots, \left\|\mathbf{w}^{(V)}\right\|_2^2\right]^\top$, $\boldsymbol{\rho} \in \mathbb{R}^V$ whose u-th element is $\sum_{v=1}^V\sum_{j=1}^{c_{(u)}^{(v)}}\mathbf{w}^{(v)^\top}\boldsymbol{\Sigma}_{c_{(u)}^{(v)}}^{(v)}\mathbf{w}^{(v)}$. It is evident that (49) is typically a small-scale QP problem. For clarity, we explicitly outline the Mv$SL_{0/1}$-SVM algorithm in **Algorithm 2**.

Algorithm 2: Iterative algorithm based on $\mathrm{SL}_{0/1}$ADMM for solving Mv$SL_{0/1}$-SVM.

Input: Training dataset $S^{(V)}$, ϵ, τ, K, C, η, α, σ, l_m.
Output: θ^*, $\left\{\mathbf{w}^{(v)^*}\right\}_{v=1}^V$ and $\left\{b^{(v)^*}\right\}_{v=1}^V$.

Initialize: θ and $l = 0$;
While θ is not optimal and $l \leq l_m$ **do**
  for $v = 1 : V$ **do**
    Initialize: $\mathbf{w}^{(v)^0}$, $b^{(v)^0}$, $\mathbf{t}^{(v)^0}$, $\lambda^{(v)^0}$ and $k = 0$;
    While $\max\left\{\epsilon_1^k, \epsilon_2^k, \epsilon_3^k, \epsilon_4^k\right\} < \epsilon$ and $k \leq K$ **do**
      Get $\mathbf{t}^{(v)^{k+1}}$ by (41);
      if $n \leq |I_k|$ holds, **then**
        Get $\mathbf{w}^{(v)^{k+1}}$ by (45);
      else if $n > |I_k|$ holds, **then**
        Get $\mathbf{w}^{(v)^{k+1}}$ by (46);
      end if
      Get $b^{(v)^{k+1}}$ by (47);
      Get $\lambda^{(v)^{k+1}}$ by (48);
      $k = k + 1$;
    end while
    Return $\left(\mathbf{w}^{(v)^*}, b^{(v)^*}\right)$ to (9);
  end while
  Get θ by solving equation (49);
  $l = l + 1$;
end while
Return θ^*, $\left\{\mathbf{w}^{(v)^*}\right\}_{v=1}^V$ and $\left\{b^{(v)^*}\right\}_{v=1}^V$.

5.2. Convergence analysis

On the one hand, the sequence produced by $\mathrm{SL}_{0/1}$ADMM must have a limit point in order to be a PSP and a locally optimal solution to (9), as demonstrated by the following theorem. This demonstrates that **Algorithm 1** effectively solves the optimization problem of $SL_{0/1}$-SVM, and thus can efficiently handle the inner loop of the Mv$SL_{0/1}$-SVM.

Theorem 5.1. *Assume that* $\left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}; \lambda^{(v)*}\right)$ *is the limit point of the sequence* $\left\{\left(\mathbf{w}^{(v)k}; b^{(v)k}; \mathbf{t}^{(v)k}; \lambda^{(v)k}\right)\right\}$ *generated by* $\mathrm{SL}_{0/1}$ADMM. *Then* $\left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}\right)$ *is a local minimizer to the problem* (9) *and a P-stationary point with* $\gamma = 1/\sigma$ *as well.*

Proof. Since $\mathcal{I}_k \subseteq \mathbb{N}_m$ has finite many elements, we can always pick up a sufficiently big k, and there exists a subset $\mathcal{J} \subseteq \{1, 2, 3, \ldots\}$ such that

$$\mathcal{I}_j \equiv: \mathcal{I}, \quad \forall j \in \mathcal{J}.$$

For convenience, we denote $\mathbf{s}^{(v)*} := \left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}; \lambda^{(v)*}\right)$ as the limit point of the sequence $\mathbf{s}^{(v)k} := \left(\mathbf{w}^{(v)k}; b^{(v)k}; \mathbf{t}^{(v)k}; \lambda^{(v)k}\right)$, namely $\left\{\mathbf{s}^{(v)k}\right\} \to \mathbf{s}^{(v)*}$. This also indicates $\left\{\mathbf{s}^{(v)j}\right\}_{j\in\mathcal{J}} \to \mathbf{s}^{(v)*}$ and $\left\{\mathbf{s}^{(v)j+1}\right\}_{j\in\mathcal{J}} \to \mathbf{s}^{(v)*}$.

Taking the limit along with $\mathcal{J}$ of (48), to be more precise, $k \in \mathcal{J}, k \to \infty$, we can derive

$$\begin{cases} \lambda_{\mathcal{I}}^{(v)*} = \lambda_{\mathcal{I}}^{(v)*} + \tau\sigma\boldsymbol{\varpi}_{\mathcal{I}}^{(v)*}, \\ \lambda_{\bar{\mathcal{I}}}^{(v)*} = \mathbf{0}, \end{cases} \tag{50}$$

which means $\boldsymbol{\varpi}_{\mathcal{I}}^{(v)*} = \mathbf{0}$. Taking the limit along with $\mathcal{J}$ of (41) and $\mathbf{z}^{(v)k}$ respectively yields

$$\begin{aligned} \mathbf{z}^{(v)*} &= \mathbf{1} - \mathbf{M}^{(v)}\mathbf{w}^{(v)*} - b^{(v)*}\mathbf{y} - \lambda^{(v)*}/\sigma \\ &= \mathbf{1} - \mathbf{M}^{(v)}\mathbf{w}^{(v)*} - b^{(v)*}\mathbf{y} - \mathbf{t}^{(v)*} + \mathbf{t}^{(v)*} - \lambda^{(v)*}/\sigma \\ &= -\boldsymbol{\varpi}^{(v)*} + \mathbf{t}^{(v)*} - \lambda^{(v)*}/\sigma, \end{aligned} \tag{51}$$

and thus

$$\mathbf{t}_{\mathcal{I}}^{(v)*} = \mathbf{0}, \tag{52}$$

$$\mathbf{t}_{\bar{\mathcal{I}}}^{(v)*} = \mathbf{z}_{\bar{\mathcal{I}}}^{(v)*} \overset{(51)}{=} -\boldsymbol{\varpi}_{\bar{\mathcal{I}}}^{(v)*} + \mathbf{t}_{\bar{\mathcal{I}}}^{(v)*} - \lambda_{\bar{\mathcal{I}}}^{(v)*}/\sigma \overset{(50)}{=} -\boldsymbol{\varpi}_{\bar{\mathcal{I}}}^{(v)*} + \mathbf{t}_{\bar{\mathcal{I}}}^{(v)*}. \tag{53}$$

Therefore, we can see $\boldsymbol{\varpi}_{\bar{\mathcal{I}}}^{(v)*} = \mathbf{0}$ and hence $\boldsymbol{\varpi}^{(v)*} = \mathbf{0}$. Again by referring to (51), we obtain $\mathbf{z}^{(v)*} = \mathbf{t}^{(v)*} - \lambda^{(v)*}/\sigma$. Together with (52), (53) and the definition of the $L_{0/1}$ proximal operator, we can demonstrate that

$$\mathbf{t}^{(v)*} = \mathrm{Prox}_{\frac{C}{\sigma}\|(\cdot)_+\|_0}\left(\mathbf{z}^{(v)*}\right) = \mathrm{Prox}_{\frac{C}{\sigma}\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)*} - \lambda^{(v)*}/\sigma\right).$$

Moreover, taking the limit along with $\mathcal{J}$ of (44) shows that

$$\begin{aligned} \left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)} + \sigma\mathbf{M}_{\mathcal{I}}^{(v)\top}\mathbf{M}_{\mathcal{I}}^{(v)}\right)\mathbf{w}^{(v)*} &= \sigma\mathbf{M}_{\mathcal{I}}^{(v)\top}\mathbf{v}_{\mathcal{I}}^{(v)*} \\ &= -\sigma\mathbf{M}_{\mathcal{I}}^{(v)\top}\left(\mathbf{t}_{\mathcal{I}}^{(v)*} + b^{(v)*}\mathbf{y}_{\mathcal{I}} - \mathbf{1} + \lambda_{\mathcal{I}}^{(v)*}/\sigma\right) \\ &= -\sigma\mathbf{M}_{\mathcal{I}}^{(v)\top}\left(\boldsymbol{\varpi}_{\mathcal{I}}^{(v)*} - \mathbf{M}_{\mathcal{I}}^{(v)}\mathbf{w}^{(v)*} + \lambda_{\mathcal{I}}^{(v)*}/\sigma\right) \\ &= -\sigma\mathbf{M}_{\mathcal{I}}^{(v)\top}\left(-\mathbf{M}_{\mathcal{I}}^{(v)}\mathbf{w}^{(v)*} + \lambda_{\mathcal{I}}^{(v)*}/\sigma\right), \end{aligned}$$

where $\mathbf{v}^{(v)*} = -\left(\mathbf{t}^{(v)*} + b^{(v)*}\mathbf{y} - \mathbf{1} + \lambda^{(v)*}/\sigma\right)$ and the last two equations hold because $\boldsymbol{\varpi}^{(v)*} = \mathbf{t}^{(v)*} + \mathbf{M}^{(v)}\mathbf{w}^{(v)*} + b^{(v)*}\mathbf{y} - \mathbf{1} = \mathbf{0}$. From the last equation, we can see that

$$\mathbf{w}^{(v)*} = -\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)^{-1}\mathbf{M}_{\mathcal{I}}^{(v)\top}\lambda_{\mathcal{I}}^{(v)*} \overset{(50)}{=} -\left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)^{-1}\mathbf{M}^{(v)\top}\lambda^{(v)*}.$$

Last but not least, taking the limit along with $\mathcal{J}$ of (47) demonstrates that

$$\begin{aligned} b^{(v)*} = \left\langle \mathbf{y}, \mathbf{r}^{(v)*}\right\rangle/m &= -\left\langle \mathbf{y}, \mathbf{M}^{(v)}\mathbf{w}^{(v)*} - \mathbf{1} + \mathbf{t}^{(v)*} + \lambda^{(v)*}/\sigma\right\rangle/m \\ &= -\left\langle \mathbf{y}, \boldsymbol{\varpi}^{(v)*} - b^{(v)*}\mathbf{y} + \lambda^{(v)*}/\sigma\right\rangle/m \\ &= -\left\langle \mathbf{y}, -b^{(v)*}\mathbf{y} + \lambda^{(v)*}/\sigma\right\rangle/m \\ &= b^{(v)*} - \left\langle \mathbf{y}, \lambda^{(v)*}\right\rangle/(m\sigma), \end{aligned}$$

which means $\left\langle \mathbf{y}, \lambda^{(v)*}\right\rangle = 0$. In summary, we have

$$\begin{cases} \left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)}\right)\mathbf{w}^{(v)*} + \mathbf{M}^{(v)\top}\lambda^{(v)*} = \mathbf{0}, \\ \left\langle \mathbf{y}, \lambda^{(v)*}\right\rangle = 0, \\ \mathbf{t}^{(v)*} + \mathbf{M}^{(v)}\mathbf{w}^{(v)*} + b^{(v)*}\mathbf{y} = \mathbf{1}, \\ \mathrm{Prox}_{\frac{C}{\sigma}\|(\cdot)_+\|_0}\left(\mathbf{t}^{(v)*} - \lambda^{(v)*}/\sigma\right) = \mathbf{t}^{(v)*}. \end{cases}$$

That is to say, $\left(\mathbf{w}^{(v)*}; b^{(v)*}; \mathbf{t}^{(v)*}\right)$ is a PSP of problem (9) where $\gamma = 1/\sigma$. Then, by Theorem 4.3, it is also a locally optimal solution to the problem (9), which completes the proof. □

On the other hand, when $\left\{\mathbf{w}^{(v)}, b^{(v)}\right\}_{v=1}^{V}$ are fixed, we denote $P(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top\widetilde{\boldsymbol{\pi}} + \frac{\eta}{2}\boldsymbol{\theta}^\top\boldsymbol{\rho} + \frac{\alpha}{2}\|\boldsymbol{\theta}\|_2^2$ for the value of the objective function in (49). Let $P_l(\boldsymbol{\theta}_l)$ represent the value of the objective function during the l-th iteration of the outer loop of **Algorithm 2**. It is straightforward to verify that $\boldsymbol{\theta}_{l+1} = \underset{\boldsymbol{\theta}_l}{\mathrm{argmin}}\ P_l\left(\boldsymbol{\theta}_l\right)$, because the minimization problem (49) is convex. Therefore, we have $P_l\left(\boldsymbol{\theta}_{l+1}\right) \le P_l\left(\boldsymbol{\theta}_l\right)$. Consequently, the outer loop of the **Algorithm 2** decreases monotonically with each iteration until convergence.

All in all, we can conclude that **Algorithm 2** effectively addresses the optimization problem of Mv$SL_{0/1}$-SVM.

5.3. Time complexity analysis

Clustering and solving make up the two main parts of the whole process. During clustering, the distance between each pair in the current cluster in each round of aggregation accounts for most of the calculations. Since positive and negative samples are combined separately, the clustering stage's time complexity is $\mathcal{O}\left(n\left(m_+^2 + m_-^2\right)\right)$, where $n = \sum_{v=1}^{V} n^{(v)}$ means total dimension of data, and m_+ and m_- denote the numbers of positive and negative samples, respectively. For the solution stage, we suppose that all variables converge after k iterations.

At the stage of solving, we have the following observations on the computational complexity in each iteration of the suggested algorithm $\mathrm{SL}_{0/1}$ADMM:

- Using (39) to update $\mathcal{I}_k$ requires the complexity $\mathcal{O}(m)$.
- $\mathbf{M}^{(v)}\mathbf{w}^{(v)k}$ is the primary term in the computation of $\mathbf{t}^{(v)k+1}$ by (41), requiring the complexity of $\mathcal{O}(mn^{(v)})$.

  (i) If $n^{(v)} \le \left|\mathcal{I}_k\right|$, we use (45) to update $\mathbf{w}^{(v)k+1}$. When it comes to (45), calculating

  $$\mathbf{M}_{\mathcal{I}_k}^{(v)\top}\mathbf{M}_{\mathcal{I}_k}^{(v)} \quad \text{and} \quad \left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)} + \sigma\mathbf{M}_{\mathcal{I}_k}^{(v)\top}\mathbf{M}_{\mathcal{I}_k}^{(v)}\right)^{-1}$$

  will cost $\mathcal{O}\left(n^{(v)2}\left|\mathcal{I}_k\right|\right)$ and $\mathcal{O}\left(n^{(v)\kappa}\right)$ respectively for their computational complexities, where $\kappa \in (2, 3)$.

  (ii) If $n^{(v)} > \left|\mathcal{I}_k\right|$, we use (46) to update $\mathbf{w}^{(v)k+1}$. Regarding (46), the primary calculations originate from

  $$\mathbf{M}_{\mathcal{I}_k}^{(v)}\mathbf{M}_{\mathcal{I}_k}^{(v)\top} \quad \text{and} \quad \left(\theta^{(v)}\mathbf{I}^{(v)} + \eta\boldsymbol{\Sigma}^{(v)} + \sigma\mathbf{M}_{\mathcal{I}_k}^{(v)}\mathbf{M}_{\mathcal{I}_k}^{(v)\top}\right)^{-1},$$

  which will cost $\mathcal{O}\left(n^{(v)}\left|\mathcal{I}_k\right|^2\right)$ and $\mathcal{O}\left(\left|\mathcal{I}_k\right|^\kappa\right)$ respectively for their computational complexities, where $\kappa \in (2, 3)$.

Consequently, updating $\mathbf{w}^{(v)k+1}$ at each step needs a complexity of

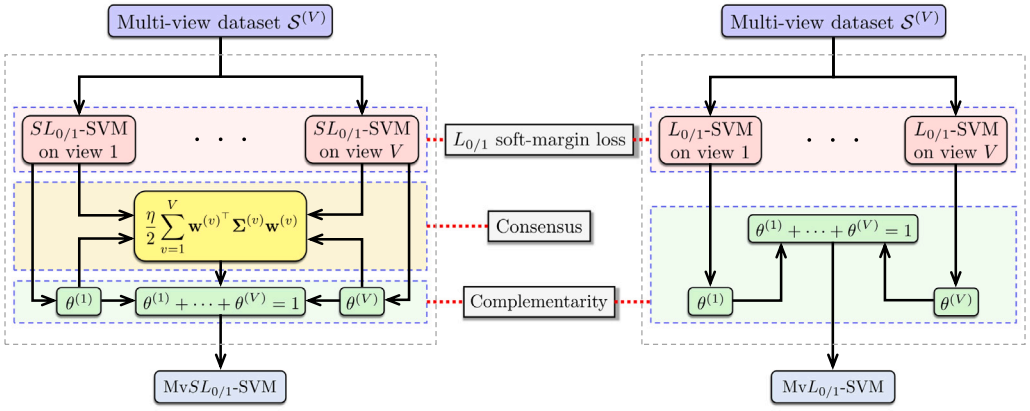


Fig. 4. Comparison between the structures of Mv$SL_{0/1}$-SVM and Mv$L_{0/1}$-SVM.

$$\mathcal{O}\left(\min\left\{n^{(v)^2},|\mathcal{I}_k|^2\right\}\max\left\{n^{(v)},|\mathcal{I}_k|\right\}\right).$$

- In the same way, $\mathbf{M}^{(v)}\mathbf{w}^{(v)^{k+1}}$ is the most expensive computation in (47) to obtain $b^{(v)^{k+1}}$, which has a complexity of $\mathcal{O}(mn^{(v)})$.
- Same as that of updating $b^{(v)^{k+1}}$, the complexity to achieve $\lambda^{(v)^{k+1}}$ via (48) is $\mathcal{O}\left(mn^{(v)}\right)$.

As a total, the computational complexity of $\mathrm{SL}_{0/1}$ADMM in Algorithm 1 at each step is

$$\mathcal{O}\left(mn^{(v)}+\min\left\{n^{(v)^2},|\mathcal{I}_k|^2\right\}\max\left\{n^{(v)},|\mathcal{I}_k|\right\}\right).$$

When it comes to the working sets, if they exhibit small cardinalities $|\mathcal{I}_k|$ or if $n^{(v)}$ is significantly smaller than m (i.e., $n^{(v)} \ll m$), the computational complexity of $\mathrm{SL}_{0/1}$ADMM notably reduces.

Since updating θ may not take much time, it can be negligible. Thus, the overall time complexity of Mv$SL_{0/1}$-SVM is

$$\mathcal{O}\left(n\left(m_+^2+m_-^2\right)+kV\left(mn_{max}+\min\left\{n_{max}^2,|\mathcal{I}_k|^2\right\}\max\left\{n_{max},|\mathcal{I}_k|\right\}\right)\right),$$

where n_{max} denotes the maximum dimensionality of feature space across all views.

In practical datasets, the number of views is typically less than or equal to 10, meaning $V \ll m$ most of the time. Dataset dimensions tend to be small, or features are highly sparse. Therefore, we can generally assume a complexity about $\mathcal{O}\left(3m^3/2\right)$.

5.4. Analysis of Mv$L_{0/1}$-SVM

Compared to Mv$SL_{0/1}$-SVM (5), Mv$L_{0/1}$-SVM (4) omits the structural regularization term. Thus, Mv$L_{0/1}$-SVM is a special case of Mv$SL_{0/1}$-SVM, as it can be obtained by setting $\eta = 0$ in Mv$SL_{0/1}$-SVM. Therefore, the optimality theory established in the previous context applies equally effectively to Mv$L_{0/1}$-SVM.

Moreover, without the structural regularization term, Mv$L_{0/1}$-SVM does not require clustering the samples or solving for their covariance matrices. This implies that Mv$L_{0/1}$-SVM is faster than Mv$SL_{0/1}$-SVM. Fig. 4 shows an intuitive comparison of the structures of Mv$SL_{0/1}$-SVM and Mv$L_{0/1}$-SVM.

Similar to the analysis of the time complexity of Mv$SL_{0/1}$-SVM, we can derive the total time complexity of Mv$L_{0/1}$-SVM is

$$\mathcal{O}\left(kV\left(mn_{max}+\min\left\{n_{max}^2,|\mathcal{I}_k|^2\right\}\max\left\{n_{max},|\mathcal{I}_k|\right\}\right)\right).$$

Generally, we can assume a complexity about $\mathcal{O}\left(m^3\right)$.

6. Numerical experiments

6.1. Algorithm comparison

Here, we contrast our Mv$SL_{0/1}$-SVM with SVM-2K [10], MPWTSVM [28], PSVM-2V [17], MvSVM-2C [22], MvLSSVM [15], MvSLMC [6], PIFMvSVM [31], MVLDM [32] and our Mv$L_{0/1}$-SVM. For simplicity, we assume that $m_+ = m_- = m/2$, which means both the positive and negative classes have equal samples.

6.1.1. Mv$SL_{0/1}$-SVM vs SVM-2K

SVM-2K adheres to the consistency principle of MVL through the use of the KCCA theory but fails to follow the principle of complementarity. It combines two-stage learning and SVM into a single optimization problem. SVM-2K algorithm has a time complexity of $\mathcal{O}\left((4\ m)^3\right)$. Our Mv$SL_{0/1}$-SVM achieves the complementarity principle through a combination of view weights and SI. Furthermore, by maintaining the structural distinctions for every view, the application of structural regularization promotes diversity.

6.1.2. Mv$SL_{0/1}$-SVM vs MPWTSVM

MPWTSVM comprises two convex QPPs, each of which can be worked out within the time complexity of $2\times\mathcal{O}\left((3\ m)^3\right)$. Through the incorporation of coupling terms in the objective function and privileged information, MPWTSVM achieves the consistency and complementarity principles simultaneously. In contrast, our proposed Mv$SL_{0/1}$-SVM achieves the complementarity principle through a combination of view weights and SI. Furthermore, the introduction of a working set significantly decreases the amount of SVs and speeds up the computation, improving overall efficiency.

6.1.3. Mv$SL_{0/1}$-SVM vs PSVM-2V

Leveraging privileged information and tackling with a large-sized QPP, PSVM-2V algorithm has a time complexity of $\mathcal{O}\left((6\ m)^3\right)$. PSVM-2V employs regularization terms to mitigate disparities in prediction results across various views, thereby upholding the consistency principle. Both PSVM-2V and our Mv$SL_{0/1}$-SVM achieve both the 2C principles at the same time. However, unlike PSVM-2V, our model incorporates the concept of weighting, facilitating the preservation of diversity among different views within the data.

6.1.4. Mv$SL_{0/1}$-SVM vs MvSVM-2C

MvSVM-2C incorporates a ϵ-insensitive 1-norm constraint between every two SVMs, allowing it to explore consensus information across diverse views. The combination weight θ maintains balance in relationships and uncovers complementary information among multiple distinct views. Nevertheless, MvSVM-2C utilizes a pairwise approach to

explore the connections among each pair of views, inevitably leading to at least C_V^2 constraints. In contrast, our Mv$SL_{0/1}$-SVM captures and integrates SI through the structural regularization term, therefore adhering to the consensus principle in a non-pairwise manner.

6.1.5. Mv$SL_{0/1}$-SVM vs MvLSSVM

MvLSSVM builds upon LS-SVM classification, introducing a coupling term to facilitate the integration of distinct views. By incorporating information from multiple perspectives, MvLSSVM enhances classification accuracy. The time complexity of the MvLSSVM training process can be considered as $\mathcal{O}\left(m^3\right)$. Yet, MvLSSVM is unable to effectively use the supplementary information between different views, resulting in its inability to satisfy the complementary principle. Conversely, our Mv$SL_{0/1}$-SVM achieves complementarity through a combination of view weights and SI.

6.1.6. Mv$SL_{0/1}$-SVM vs MvSLMC

Through a combination of view weights and SI, MvSLMC achieves 2C principles. The integration of view weights into this regularization process allows for dynamic adjustments throughout the optimization process, ensuring adaptability and improved performance. Structural regularization encourages diversity among different views. By introducing the ADMM algorithm and working set strategy enlightened by $SL_{0/1}$-SVM SVs, our Mv$SL_{0/1}$-SVM significantly accelerates the whole computation, leading to improved efficiency in the overall process.

6.1.7. Mv$SL_{0/1}$-SVM vs PIFMvSVM

Both Mv$SL_{0/1}$-SVM and PIFMvSVM adhere to the 2C principles of MVL, but they differ in their approach to robustness and optimization. Mv$SL_{0/1}$-SVM achieves robustness against outliers through the $L_{0/1}$ soft-margin loss while also accelerating the optimization process with a working set strategy, making it computationally efficient. On the other hand, PIFMvSVM enhances robustness by combining the intuitionistic fuzzy score and the pinball loss function, making it more adept at distinguishing noise and outliers. However, this comes at the cost of increased computational complexity, with PIFMvSVM having a complexity of $\mathcal{O}\left(\left(\left(V^2+V\right)n\right)^3\right)$.

6.1.8. Mv$SL_{0/1}$-SVM vs MVLDM

In MVLDM, a new multi-view margin distribution is constructed for multi-view data to facilitate information complementarity between multiple views while ensuring the consistency principle through similarity constraints across different views. However, MVLDM is limited to handling data from only two views simultaneously. In contrast, Mv$SL_{0/1}$-SVM, while adhering to the 2C principle, can effectively utilize data from three or more views at the same time.

6.1.9. Mv$SL_{0/1}$-SVM vs Mv$L_{0/1}$-SVM

Mv$L_{0/1}$-SVM introduces the ADMM algorithm and working set strategy, which significantly accelerates the computation and improves the efficiency of classification. However, Mv$L_{0/1}$-SVM only achieves the complementarity principle through a combination of view weights, which may result in the classification results from each view cannot be unified. To solve this, Mv$SL_{0/1}$-SVM integrates SI through the structural regularization term on the basis of Mv$L_{0/1}$-SVM, therefore satisfying the consensus principle.

6.2. Experimental settings

Datasets. Through the following datasets, we confirm the performance of our proposed model: 3 synthetic datasets, 8 real-world datasets from UCI machine learning repository,[1] 9 real-world datasets selected from LIBSVM Data[2] and 45 multi-view datasets obtained from CIFAR-10[3].

Table 3
Details of 3 synthetic datasets.

Datasets	#Data m	#Noise ratio r	#View 1 $n^{(1)}$	#View 2 $n^{(2)}$
Synthetic 1	1000	0	2	2
Synthetic 2	1000	0.2	2	2
Synthetic 3	1000	0.4	2	2

Table 4
Details of 8 UCI datasets.

Datasets	#Data m	#View 1 $n^{(1)}$	#View 2 $n^{(2)}$
Balance	576	4	4
Bankruptcy	250	6	5
Heart	270	13	10
Hepatitis	155	19	15
Iris	150	4	2
Pima	768	8	7
Vote	435	16	13
wine	130	13	1

Table 5
Details of 9 LIBSVM datasets.

Datasets	#Training data m	#Testing data m_t	#View 1 $n^{(1)}$	#View 2 $n^{(2)}$
a1a	1605	30 956	123	44
a2a	2265	30 296	123	44
a3a	3185	29 376	123	44
a4a	4781	27 780	123	44
a5a	6414	26 147	123	44
a6a	11 220	21 341	123	44
a7a	16 100	16 461	123	44
satimage	4435	2000	36	6
pendigits	7494	2000	16	9

(1) Synthetic datasets: The synthetic datasets are generated by twin Gaussian normal distributions and twin parallel lines. Then, in each class, r percentage of labels is randomly flipped. Finally, the samples are evenly randomly divided into a testing set and a training set. Table 3 displays the details of the synthetic datasets used in our experiment.

(2) UCI datasets [44]: Table 4 displays the specifics of the UCI datasets used in our experiment. Since these datasets are not inherently multi-view, we consider the raw data as view 1 and the 95% principal components extracted from the raw data as view 2.

(3) LIBSVM datasets [45]: The concise details of the 9 LIBSVM datasets are outlined in Table 5. Since these datasets are not inherently multi-view, we treat them similarly to the UCI datasets by applying the same operations.

(4) CIFAR-10 [46]: CIFAR-10 is a dataset designed for universal object recognition. With a resolution of $32 * 32$ pixels, every image in the collection is described using three different types of features. In our experiments, we extract LBP and HOG features as two separate views. The dataset consists of ten classes (see Fig. 5), and we randomly select 500 samples from each class for classification tasks. To confirm that the model can be used generally, we train every possible pair combination of classes (i.e., 45 binary classifiers).

Benchmark methods. We compare our Mv$SL_{0/1}$-SVM and Mv$L_{0/1}$-SVM models against the following eight related benchmark methods:

(1) SVM-2K (hinge loss): SVM-2K adheres to the consensus principle of MVL through the use of KCCA theory [10].

[1] Available at https://archive.ics.uci.edu/ml/index.php.

[2] Available at https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets.

[3] Available at http://www.cs.toronto.edu/~kriz/cifar.html.

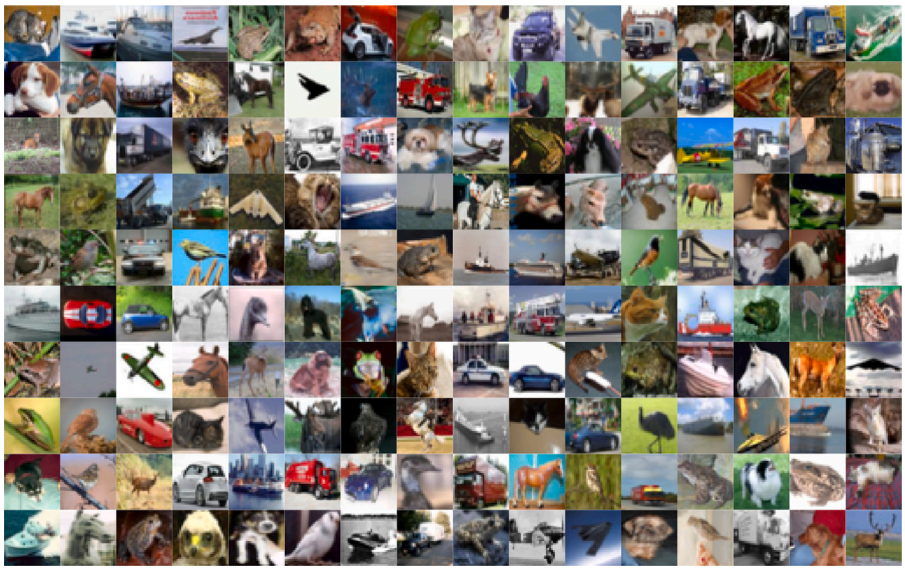


Fig. 5. CIFAR-10 dataset.

(2) MPWTSVM (hinge loss): MPWTSVM achieves the consistency and complementarity principles simultaneously through the coupling terms and privileged information [28].
(3) PSVM-2V (hinge loss): PSVM-2V employs regularization terms and privileged information to mitigate disparities in prediction results across various views, thereby upholding 2C principles at the same time [17].
(4) MvSVM-2C (hinge loss): MvSVM-2C incorporates a similarity constraint between two SVMs to explore consensus information across diverse views. The combination weight θ maintains balance in relationships and uncovers complementary information among multiple distinct views [22].
(5) MvLSSVM (least square loss): MvLSSVM builds upon LS-SVM classification, introducing a coupling term to facilitate the integration of distinct views [15]. However, MvLSSVM only adheres to the complementarity principle.
(6) MvSLMC (hinge loss): Through a combination of view weights and SI, MvSLMC promotes diversity across different views, achieving 2C principles [6].
(7) PIFMvSVM (ε-insensitive pinball loss): PIFMvSVM not only achieves 2C principles of MVL, but also combines the intuitionistic fuzzy score and pinball loss function to determine the impact of samples on decision-making, which is more robust and easier to distinguish noise and outliers [31].
(8) MVLDM (hinge loss): MVLDM constructs a new multiview margin distribution for multiview data to realize information complementarity between multiple views and achieve the consistency principle through the similarity constraints between different views [32].

Measures. All experiments are conducted in MATLAB 2021b on a Windows 7 PC with an Intel(R) Core(TM) i5-7500 processor (3.40 GHz) and 4 GB RAM. To enhance the credibility of the results, we employ the widely used five-fold cross-validation method in UCI and CIFAR-10 datasets.

Parameters. For SVM-2K, PSVM-2V, MvSVM-2C, PIFMvSVM and MVLDM, we fix $\epsilon = 10^{-2}$ and select other parameters over the set $\{2^i | i = -4{:}\ 1{:}\ 4\}$. For MPWTSVM, the neighborhood size l_n is chosen from the set $\{3, 5, 7, 9, 11\}$, while other parameters are kept equal, and they are all searched over the range $\{2^i | i = -4{:}\ 1{:}\ 4\}$. All parameters of MvLSSVM are also tuned over $\{2^i | i = -4{:}\ 1{:}\ 4\}$. In MvSLMC, our Mv$SL_{0/1}$-SVM and Mv$L_{0/1}$-SVM, the number of clusters is determined by the adaptive method, and other parameters are selected from $\{2^i | i = -4{:}\ 1{:}\ 4\}$. Moreover, we set $\tau = 1.618$, $\epsilon = 10^{-2}$, $l_m = 10$ and $K = 10^3$ in our suggested methods. In LIBSVM datasets and CIFAR-10, due to the large sample sizes, we appropriately reduced the number of parameter combinations.

Evaluation metrics. To quantitatively evaluate the models, we use accuracy as the primary performance metric, defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + TN + FP},$$

where TP, FN, TN and FP represent the number of true positive, false negative, true negative and false positive, respectively. In addition, we assess the performance of the proposed models by using the Precision, Recall and F1-score for a comprehensive evaluation.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}},$$

where Precision and Recall are computed as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}.$$

6.3. Experimental results

In this section, we compare Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM against other benchmark methods to assess their effectiveness. The best results are highlighted in **boldface**.

6.3.1. Results on synthetic datasets

Based on Table 6, both Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM deliver strong accuracy, precision, recall, and F1-scores. Mv$SL_{0/1}$-SVM distinguishes itself by maintaining high performance across key metrics, especially in noisy environments. Its optimized working set strategy further enhances its computational efficiency, making it well-suited for complex tasks involving multiple data views. However, while Mv$SL_{0/1}$-SVM shows slightly superior accuracy and F1-score, Mv$L_{0/1}$-SVM consistently excels in computation time, making it the faster of the two. This speed advantage could be crucial in time-sensitive applications.

To gain deeper insights into model behavior, we provide a visual comparison of classification results on the Synthetic 2 dataset, as shown in Figs. 6–15. The comparative analysis of different classification result images reveals that the hyperplanes of Mv$SL_{0/1}$-SVM and Mv$L_{0/1}$-SVM are less sensitive to noisy points compared to other models, resulting in better classification outcomes.

Table 6
Results of 10 models on 3 Synthetic datasets.

Metrics	Noise ratio	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
Accuracy	0	99.00	99.20	99.00	98.20	99.20	**99.60**	99.40	99.40	99.40	**99.60**
Accuracy	0.2	80.00	**80.60**	**80.60**	**80.60**	79.80	80.00	80.00	80.40	**80.60**	**80.60**
Accuracy	0.4	60.40	61.40	61.40	59.80	60.40	61.60	61.60	61.80	**62.00**	**62.00**
Accuracy	Avg.	79.80	80.40	80.33	79.53	79.80	80.40	80.33	80.53	80.67	**80.73**
Accuracy	Avg.Rank.	8.33	5.33	6.00	7.67	8.33	4.67	5.50	4.33	2.83	**2.00**
Precision	0	**99.60**	98.81	**99.60**	97.25	**99.60**	99.21	99.20	99.20	99.20	**99.60**
Precision	0.2	80.00	80.48	80.24	80.24	80.16	80.49	80.00	**80.65**	80.48	80.48
Precision	0.4	60.32	61.83	61.54	59.53	60.24	61.79	61.69	61.94	**62.40**	**62.40**
Precision	Avg.	79.97	80.37	80.46	79.01	80.00	80.50	80.30	80.60	80.69	**80.83**
Precision	Avg.Rank.	6.67	5.67	5.33	8.83	6.50	4.00	7.50	3.67	4.17	**2.67**
Recall	0	98.40	99.60	98.40	99.20	98.80	**100.00**	99.60	99.60	99.60	99.60
Recall	0.2	80.00	80.80	**81.20**	**81.20**	79.20	79.20	80.00	80.00	80.80	80.80
Recall	0.4	60.80	59.60	60.80	**61.20**	**61.20**	60.80	**61.20**	**61.20**	60.40	60.40
Recall	Avg.	79.73	80.00	80.13	**80.53**	79.73	80.00	80.27	80.27	80.27	80.27
Recall	Avg.Rank.	7.50	6.00	5.67	**3.67**	6.67	5.50	4.50	4.50	5.50	5.50
F1-score	0	98.99	99.20	98.99	98.22	99.20	**99.60**	99.40	99.40	99.40	**99.60**
F1-score	0.2	80.00	80.64	**80.72**	**80.72**	79.68	79.84	80.00	80.32	80.64	80.64
F1-score	0.4	60.56	60.69	61.17	60.36	60.71	61.29	61.45	**61.57**	61.38	61.38
F1-score	Avg.	79.85	80.18	80.29	79.77	79.86	80.24	80.28	80.43	80.47	**80.54**
F1-score	Avg.Rank.	8.33	6.00	5.33	7.17	8.00	5.17	4.50	3.67	3.83	**3.00**
Time	0	2.5102	0.6675	12.9879	2.4520	0.1666	4.5153	17.9049	2.0605	**0.1036**	4.8747
Time	0.2	2.3455	2.5339	18.6099	2.6504	0.1448	5.3697	13.7763	2.0948	**0.0125**	0.9484
Time	0.4	2.9011	0.6715	17.0649	2.2568	0.1586	0.5754	11.0111	2.0569	**0.1352**	1.0582
Time	Avg.	2.5856	1.2910	16.2209	2.4531	0.1567	3.4868	14.2308	2.0707	**0.0838**	2.2938
Time	Avg.Rank.	6.33	4.33	9.67	6.33	2.00	6.00	9.33	4.67	**1.00**	5.33

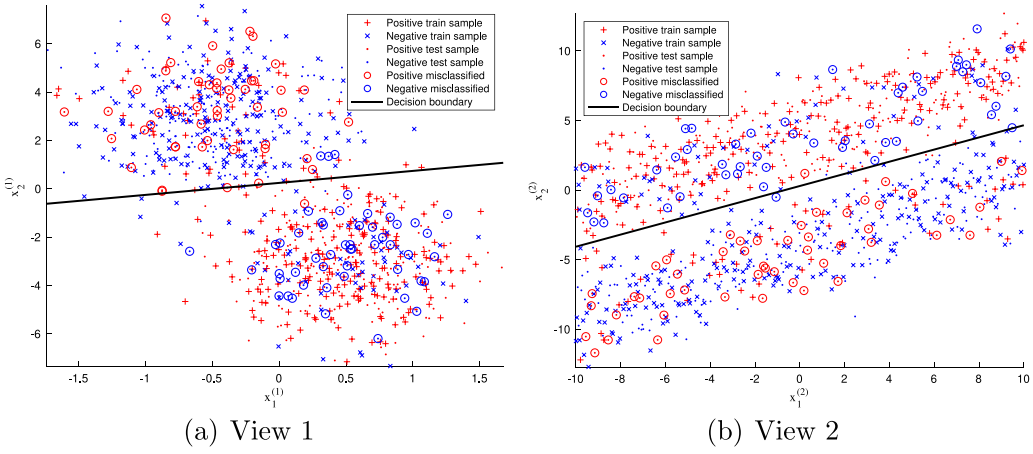


Fig. 6. Classification results of SVM-2K on Synthetic 2 dataset ($r = 0.2$).

Mv$SL_{0/1}$-SVM employs the $L_{0/1}$ soft-margin loss and incorporates SI into the model. This approach provides superior classification performance compared to other state-of-the-art models that use hinge loss (MVLDM) or ε-insensitive pinball loss (PIFMvSVM).

6.3.2. Results on UCI datasets

Tables 7–11 illustrate that our proposed Mv$SL_{0/1}$-SVM and Mv$L_{0/1}$-SVM demonstrate outstanding classification performance, achieving optimal results in seven out of eight datasets in terms of accuracy. Additionally, while Mv$SL_{0/1}$-SVM may not be the top performer on every dataset for precision, recall, or F1-score, it consistently ranks first on average across these metrics, highlighting its overall effectiveness. In terms of training time, MvLSSVM demonstrated the fastest average processing time among all models tested, with our algorithm showing comparable efficiency. Although our models do not match the speed of MvLSSVM due to the relatively small sample sizes of the datasets, they still exhibit significant efficiency and outperform several other models in terms of processing time.

6.3.3. Results on LIBSVM datasets

The data size of the LIBSVM datasets we selected for training ranges from 1605 to 16,100, indicating a large sample size. In Tables 12 and 16, we provide a list of the training time and accuracy for LIBSVM datasets, where "–" represents that the memory required exceeds the capacity of our PC.

In Tables 12–16, Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM are optimal in most cases. While Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM perform a little worse in certain situations, overall, they are nearly as well as the best outcome. With regard to training time, Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM outperform other models by a significant margin. Furthermore, the speed superiority of Mv$L_{0/1}$-SVM becomes even more noticeable as the training dataset size increases.

Overall, both Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM have quick training speeds and are generally more stable than their competitors. It is because Mv$SL_{0/1}$-SVM uses the covariance matrix to fuse the SI of each view beforehand. Consequently, in Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM, the size of every subproblem is similar to the original single-view SVM.

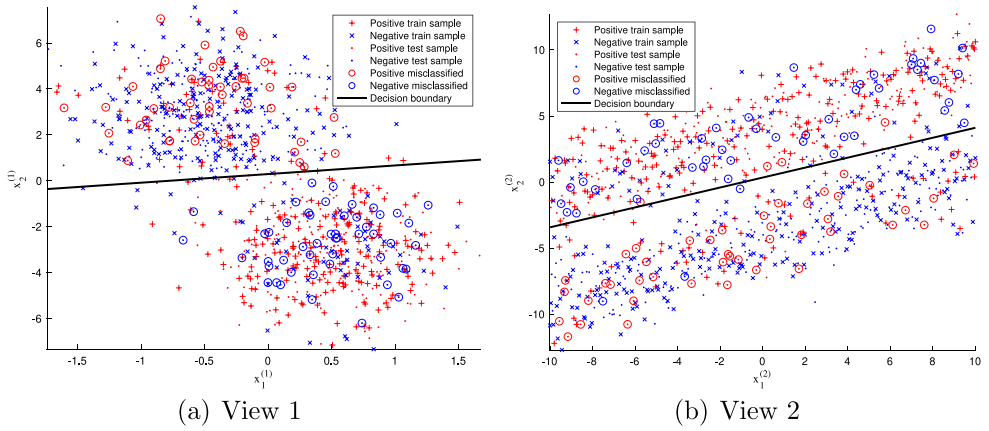


Fig. 7. Classification results of MPWTSVM on Synthetic 2 dataset ($r = 0.2$).

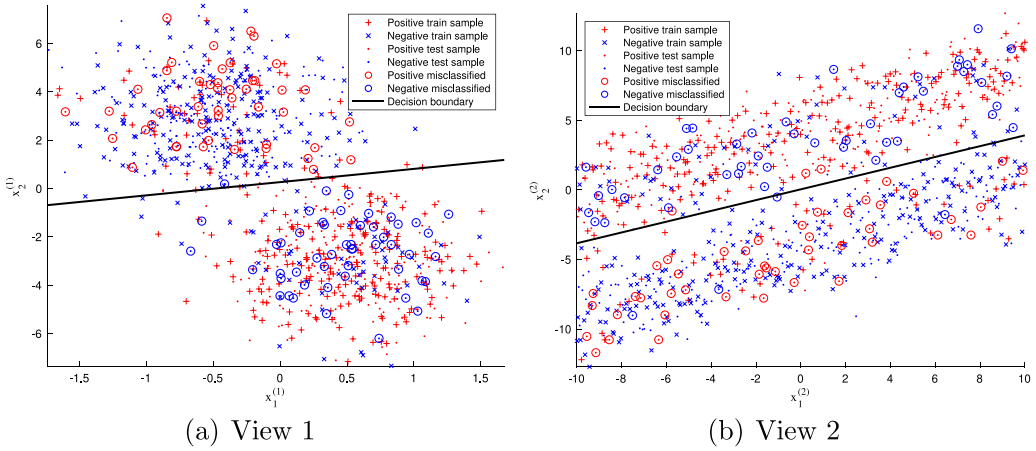


Fig. 8. Classification results of PSVM-2V on Synthetic 2 dataset ($r = 0.2$).

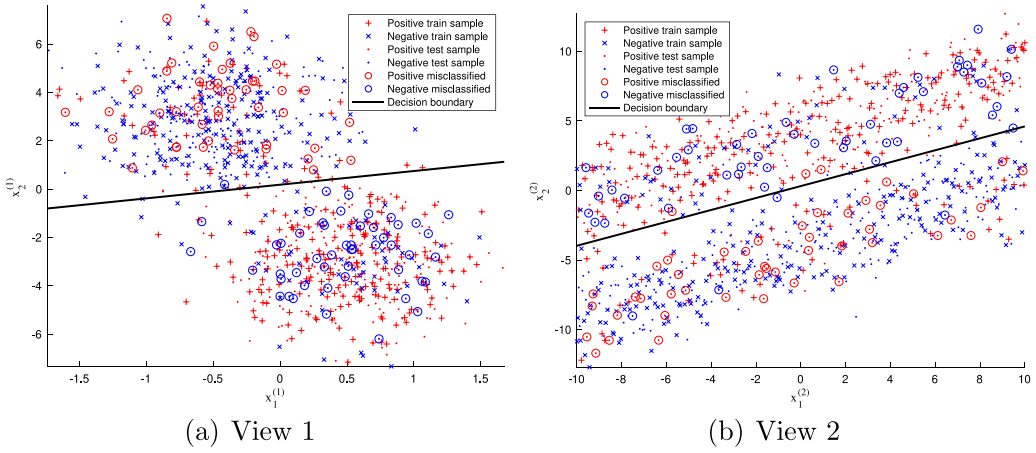


Fig. 9. Classification results of MvSVM-2C on Synthetic 2 dataset ($r = 0.2$).

Moreover, the strategy of the working set significantly accelerates the computation process.

Additionally, SI is advantageous for harnessing the potential of Mv$SL_{0/1}$-SVM, enabling it to fulfill 2C principles, thereby enhancing classification performance.

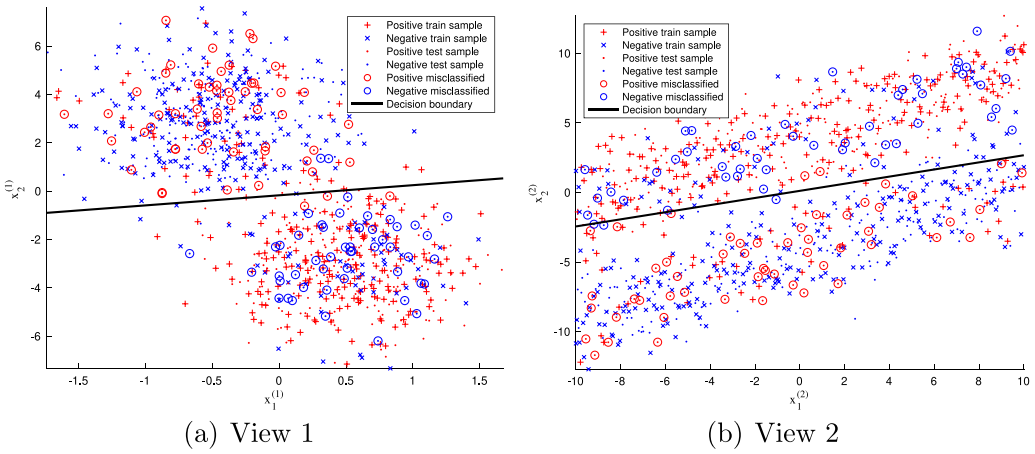


Fig. 10. Classification results of MvLSSVM on Synthetic 2 dataset ($r = 0.2$).

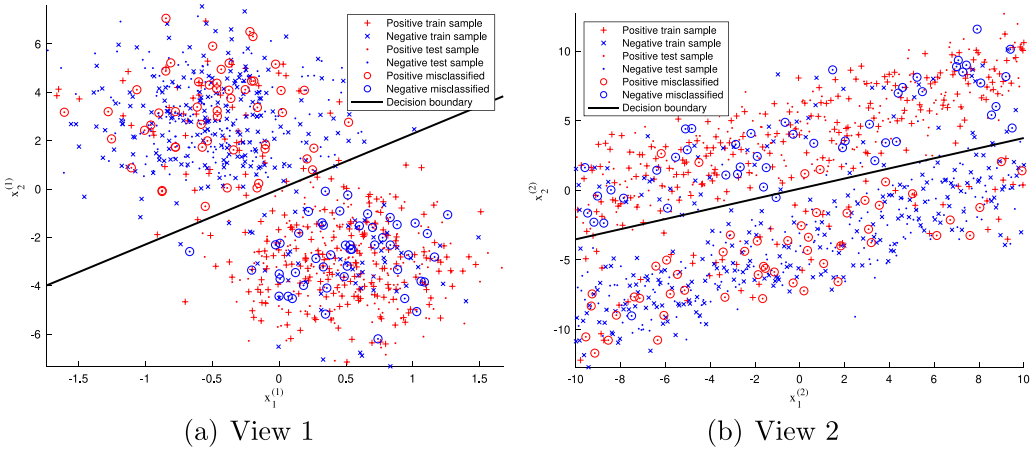


Fig. 11. Classification results of MvSLMC on Synthetic 2 dataset ($r = 0.2$).

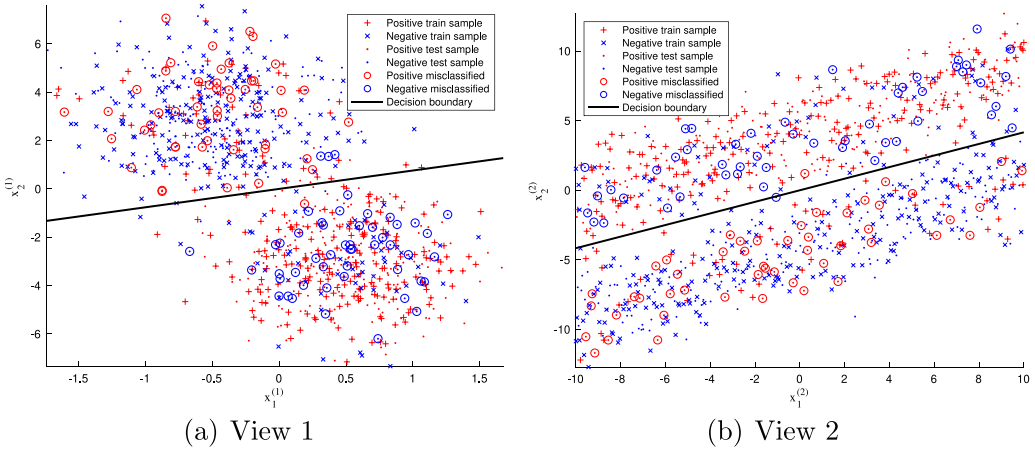


Fig. 12. Classification results of PIFMvSVM on Synthetic 2 dataset ($r = 0.2$).

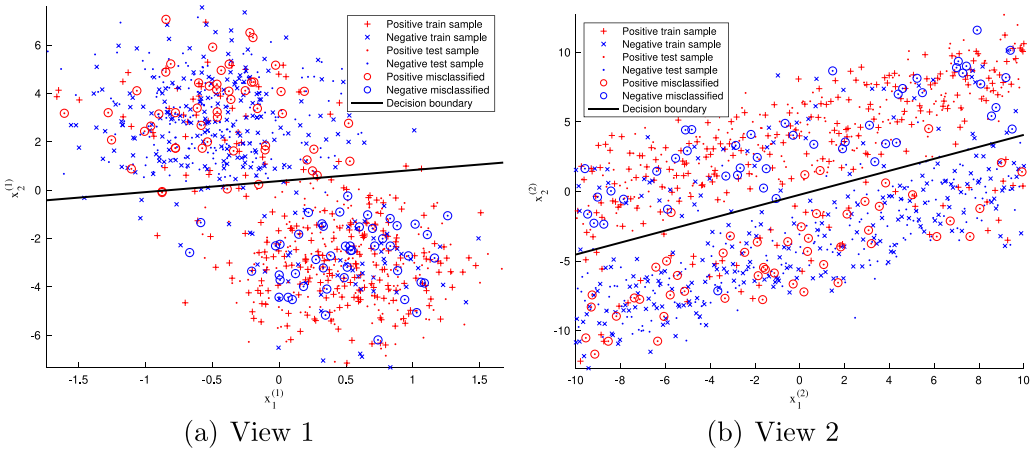


Fig. 13. Classification results of MVLDM on Synthetic 2 dataset ($r = 0.2$).

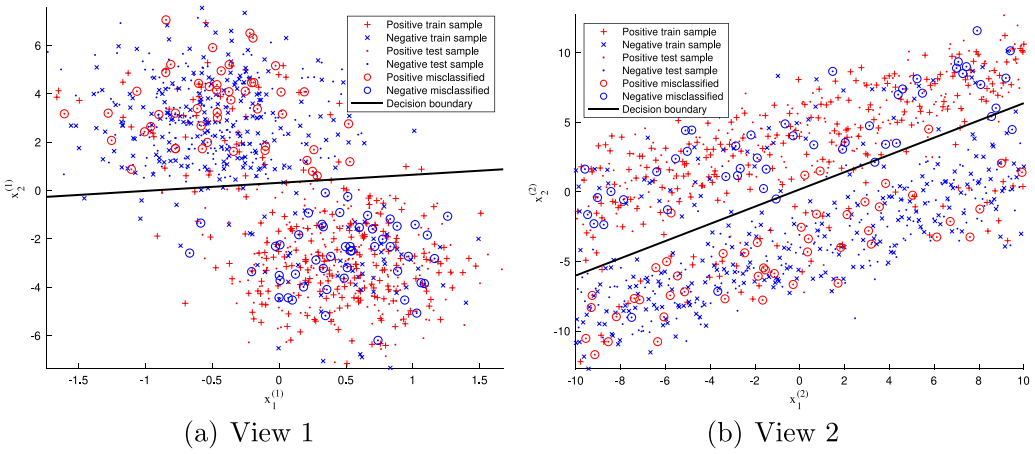


Fig. 14. Classification results of Mv$L_{0/1}$-SVM on Synthetic 2 dataset ($r = 0.2$).

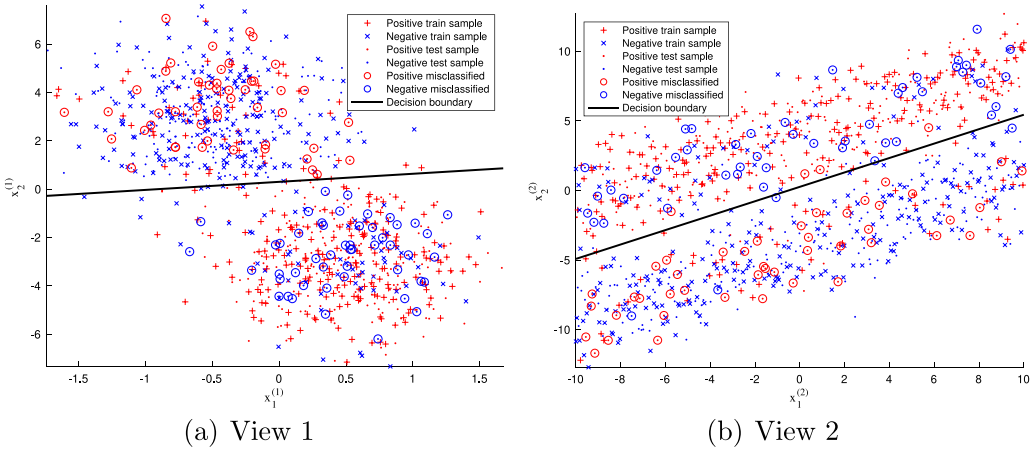


Fig. 15. Classification results of Mv$SL_{0/1}$-SVM on Synthetic 2 dataset ($r = 0.2$).

6.3.4. Results on CIFAR-10 dataset

Table 17 summarizes the experimental results of Mv$L_{0/1}$-SVM, Mv$SL_{0/1}$-SVM, and eight benchmark methods on 45 multi-view datasets. Out of all the eight methods, Mv$SL_{0/1}$-SVM has the greatest average accuracy of 78.94% across 45 datasets, followed by Mv$L_{0/1}$-SVM (77.75%), PIFMvSVM (77.27%), MVLDM (74.90%), MvSLMC (73.49%), MPWTSVM (69.80%), PSVM-2V (69.48%), MvLSSVM (66.78%), MvSVM-2C (58.55%) and SVM-2K (57.57%).

It is notable to mention that our model outperforms PIFMvSVM, MVLDM, MvSLMC, MPWTSVM, PSVM-2V, and MvSVM-2C in terms of generalization performance. While SVM-2K only satisfies the consensus principle, MvLSSVM only achieves the complementary principle.

Table 7
Accuracy results of 10 models on 8 UCI datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
Balance	83.08 ± 6.32	95.34 ± 3.23	90.27 ± 10.80	83.61 ± 15.44	90.88 ± 4.95	94.11 ± 2.52	93.06 ± 2.53	85.64 ± 12.44	97.39 ± 2.57	**97.56 ± 0.76**
Bankruptcy	57.20 ± 0.12	99.59 ± 0.91	98.78 ± 2.74	57.20 ± 0.12	99.18 ± 1.83	95.92 ± 8.03	**100.00 ± 0.00**	93.88 ± 6.61	**100.00 ± 0.00**	**100.00 ± 0.00**
Heart	55.93 ± 0.83	87.04 ± 3.93	84.07 ± 3.84	59.26 ± 8.28	85.56 ± 5.14	85.93 ± 2.81	85.93 ± 4.26	85.19 ± 2.62	87.78 ± 3.84	**88.52 ± 4.22**
Hepatitis	92.19 ± 3.91	97.52 ± 2.46	92.29 ± 5.49	90.95 ± 3.64	94.19 ± 1.44	94.67 ± 5.06	95.33 ± 2.98	94.10 ± 1.70	96.00 ± 2.79	**98.00 ± 2.98**
Iris	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Pima	65.89 ± 1.15	77.99 ± 2.23	74.46 ± 3.56	66.02 ± 1.02	76.42 ± 3.77	77.86 ± 1.38	77.98 ± 2.18	74.35 ± 2.41	78.24 ± 3.46	**79.28 ± 2.46**
Vote	93.18 ± 4.05	**96.85 ± 3.20**	91.59 ± 5.49	90.24 ± 6.43	95.68 ± 2.55	96.15 ± 2.81	95.68 ± 2.55	95.67 ± 3.88	96.37 ± 2.60	96.60 ± 2.54
Wine	92.80 ± 5.22	**100.00 ± 0.00**	96.00 ± 2.83	96.00 ± 2.83	94.40 ± 6.69	95.20 ± 3.35	98.40 ± 2.19	85.73 ± 12.34	99.20 ± 1.79	**100.00 ± 0.00**
Avg.Accuracy.	80.03	94.29	90.93	80.41	92.04	92.48	93.30	89.32	94.37	**95.00**
Avg.Rank.	8.88	2.88	7.00	8.44	6.00	5.25	4.31	7.56	2.81	**1.88**

Table 8
Precision results of 10 models on 8 UCI datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
Balance	75.25 ± 7.09	93.74 ± 5.79	92.09 ± 14.70	79.85 ± 19.12	90.48 ± 9.66	94.41 ± 6.21	94.19 ± 4.74	81.47 ± 15.13	**97.55 ± 0.98**	97.30 ± 1.89
Bankruptcy	0.00 ± 0.00	**100.00 ± 0.00**	98.18 ± 4.07	0.00 ± 0.00	99.05 ± 2.13	93.09 ± 13.06	**100.00 ± 0.00**	90.09 ± 11.78	**100.00 ± 0.00**	**100.00 ± 0.00**
Heart	55.77 ± 0.47	87.58 ± 2.57	82.48 ± 3.44	61.11 ± 12.42	85.41 ± 4.51	**87.88 ± 2.63**	86.43 ± 3.80	85.80 ± 3.13	87.31 ± 3.65	86.97 ± 3.24
Hepatitis	**97.50 ± 5.59**	96.00 ± 8.94	87.58 ± 19.82	95.00 ± 11.18	86.00 ± 7.96	96.67 ± 7.45	90.48 ± 8.75	90.00 ± 9.13	93.33 ± 9.13	96.67 ± 7.45
Iris	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Pima	0.00 ± 0.00	**71.85 ± 3.31**	68.85 ± 6.27	0.00 ± 0.00	70.77 ± 7.28	68.57 ± 2.80	68.71 ± 4.03	66.16 ± 3.73	70.45 ± 5.53	71.78 ± 3.74
Vote	91.00 ± 9.77	94.87 ± 6.58	87.40 ± 10.32	**98.67 ± 1.83**	92.59 ± 5.62	95.39 ± 4.19	92.10 ± 5.56	93.05 ± 7.34	94.93 ± 4.46	94.70 ± 5.62
Wine	93.18 ± 7.03	**100.00 ± 0.00**	96.52 ± 4.78	98.33 ± 3.73	93.72 ± 10.06	95.10 ± 7.06	98.33 ± 3.73	79.30 ± 17.22	**100.00 ± 0.00**	**100.00 ± 0.00**
Avg.Precision.	64.09	93.01	89.14	66.62	89.75	91.39	91.28	85.73	92.95	**93.43**
Avg.Rank.	7.94	3.25	6.94	6.63	6.56	4.38	5.31	7.44	3.38	**3.19**

Table 9
Recall results of 10 models on 8 UCI datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
Balance	**100.00 ± 0.00**	97.54 ± 2.35	91.93 ± 12.99	98.60 ± 1.92	92.98 ± 5.55	94.39 ± 5.32	92.28 ± 8.00	97.89 ± 3.80	97.19 ± 4.57	97.89 ± 1.92
Bankruptcy	0.00 ± 0.00	99.05 ± 2.13	99.05 ± 2.13	0.00 ± 0.00	99.05 ± 2.13	**100.00 ± 0.00**	**100.00 ± 0.00**	98.10 ± 2.61	**100.00 ± 0.00**	**100.00 ± 0.00**
Heart	**100.00 ± 0.00**	89.33 ± 5.96	90.67 ± 4.35	93.33 ± 14.91	89.33 ± 5.96	86.67 ± 4.08	88.67 ± 5.58	88.00 ± 2.98	91.33 ± 3.80	93.33 ± 5.27
Hepatitis	64.17 ± 22.75	**93.33 ± 9.13**	82.50 ± 13.31	59.17 ± 14.55	86.67 ± 7.45	76.67 ± 25.28	86.67 ± 13.94	80.83 ± 8.12	86.67 ± 7.45	**93.33 ± 9.13**
Iris	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Pima	2.26 ± 2.07	60.74 ± 9.07	47.99 ± 9.22	2.62 ± 2.16	55.16 ± 5.95	67.49 ± 2.70	**67.85 ± 3.38**	54.07 ± 7.90	65.23 ± 10.93	66.70 ± 5.08
Vote	92.83 ± 3.51	**97.63 ± 1.33**	92.98 ± 5.50	76.11 ± 17.05	97.02 ± 2.15	94.75 ± 3.53	**97.63 ± 2.52**	96.52 ± 3.01	95.91 ± 3.20	97.02 ± 2.15
Wine	90.91 ± 11.13	**100.00 ± 0.00**	94.55 ± 4.98	92.73 ± 7.61	94.55 ± 8.13	94.55 ± 4.98	98.18 ± 4.07	98.18 ± 4.07	98.18 ± 4.07	**100.00 ± 0.00**
Avg.Recall.	68.77	92.20	87.46	65.32	89.35	89.32	91.41	89.20	91.81	**93.53**
Avg.Rank.	6.88	4.06	6.94	7.19	5.81	6.13	4.44	6.13	4.50	**2.94**

Table 10
F1-score results of 10 models on 8 UCI datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
Balance	85.73 ± 4.61	95.51 ± 3.02	90.76 ± 9.43	87.07 ± 11.15	91.26 ± 4.02	94.15 ± 2.44	92.91 ± 2.86	88.08 ± 8.50	97.34 ± 2.68	**97.57 ± 0.75**
Bankruptcy	0.00 ± 0.00	99.51 ± 1.09	98.60 ± 3.12	0.00 ± 0.00	99.05 ± 2.13	96.01 ± 7.70	**100.00 ± 0.00**	93.58 ± 6.70	**100.00 ± 0.00**	**100.00 ± 0.00**
Heart	71.60 ± 0.38	88.39 ± 3.74	86.34 ± 3.26	71.96 ± 1.18	87.27 ± 4.56	87.23 ± 2.71	87.46 ± 3.95	86.85 ± 2.29	89.25 ± 3.33	**90.00 ± 3.75**
Hepatitis	75.02 ± 16.80	94.14 ± 5.41	82.53 ± 8.88	72.23 ± 12.07	85.96 ± 3.67	83.03 ± 19.39	87.79 ± 8.21	84.67 ± 5.06	89.70 ± 6.89	**94.85 ± 7.55**
Iris	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Pima	4.36 ± 3.98	65.52 ± 5.78	56.41 ± 8.33	5.04 ± 4.08	61.98 ± 6.42	68.00 ± 2.30	68.24 ± 3.28	59.31 ± 5.50	67.29 ± 7.26	**69.12 ± 4.32**
Vote	91.57 ± 4.29	**96.13 ± 3.65**	89.79 ± 6.11	84.89 ± 11.90	94.65 ± 2.82	95.05 ± 3.50	94.68 ± 2.83	94.65 ± 4.64	95.38 ± 3.19	95.76 ± 2.91
Wine	91.55 ± 6.40	**100.00 ± 0.00**	95.41 ± 3.22	95.23 ± 3.55	93.80 ± 7.15	94.61 ± 3.61	98.18 ± 2.50	86.91 ± 10.87	99.05 ± 2.13	**100.00 ± 0.00**
Avg.F1-score.	64.98	92.40	87.48	64.55	89.25	89.76	91.16	86.76	92.25	**93.41**
Avg.Rank.	8.88	3.13	7.06	8.50	5.88	5.44	3.94	7.25	3.06	**1.88**

Table 11
Time results of 10 models on 8 UCI datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
Balance	1.6818	6.1164	13.6579	3.1466	0.1276	3.6395	5.2154	2.1507	**0.0658**	1.0355
Bankruptcy	0.2504	0.1219	0.9756	0.2532	**0.0169**	0.6524	0.6401	0.2910	0.0245	0.4823
Heart	0.4236	0.1515	1.7990	0.5161	**0.0194**	1.8681	1.0572	0.5051	0.1917	0.6194
Hepatitis	0.1146	0.0595	0.3452	0.1378	**0.0052**	0.4845	0.2403	0.1246	0.0960	0.3879
Iris	0.0692	0.0465	0.2720	0.0938	**0.0051**	0.1363	0.1962	0.0526	0.0279	0.2518
Pima	12.0416	47.3203	37.5668	12.4497	0.2696	10.3526	22.8540	8.0417	**0.1206**	5.7984
Vote	1.4133	2.5432	7.3627	1.8809	**0.0572**	3.6513	3.4243	1.4247	0.1824	0.8136
Wine	0.0560	0.0293	0.2103	0.0845	**0.0037**	0.0968	0.1290	0.1162	0.0514	0.3212
Avg.Time.	2.0063	7.0486	7.7737	2.3203	**0.0631**	2.6102	4.2196	1.5883	0.0950	1.2138
Avg.Rank.	4.38	4.75	9.38	5.88	**1.25**	7.88	7.88	5.13	2.13	6.38

Therefore, compared to other algorithms that simultaneously satisfy the complementary and consensus principle, their generalization ability is slightly inadequate.

With an average time of 0.264 s, Mv$L_{0/1}$-SVM is the fastest compared to MvLSSVM (0.538 s), Mv$SL_{0/1}$-SVM (2.746 s), MvSLMC (5.019 s), MVLDM (8.773 s), MvSVM-2C (15.385 s), SVM-2K (25.390 s), PIFMvSVM (88.931 s), PSVM-2V (97.412 s), and MPWTSVM (133.348

Table 12
Accuracy results of 10 models on 9 LIBSVM datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
a1a	83.60	83.42	77.91	83.82	75.95	81.87	62.13	**84.02**	82.91	83.30
a2a	83.44	83.66	–	83.62	76.01	**83.89**	–	83.74	82.92	83.68
a3a	–	–	–	–	76.94	81.77	–	83.33	83.43	**83.63**
a4a	–	–	–	–	76.06	80.93	–	–	**84.33**	84.08
a5a	–	–	–	–	78.48	81.78	–	–	**84.21**	84.15
a6a	–	–	–	–	–	84.54	–	–	84.46	**84.95**
a7a	–	–	–	–	–	–	–	–	**84.78**	–
satimage	–	–	–	–	87.30	85.35	–	–	92.00	92.25
pendigits	–	–	–	–	54.63	68.10	–	–	79.47	**79.53**

Table 13
Precision results of 10 models on 9 LIBSVM datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
a1a	70.72	69.58	61.85	70.87	0.00	63.29	33.80	**71.89**	65.21	68.46
a2a	70.86	69.15	–	**71.98**	0.00	67.07	–	69.78	65.26	67.40
a3a	–	–	–	–	51.23	62.20	–	68.04	67.78	**69.05**
a4a	–	–	–	–	**80.00**	58.70	–	–	69.64	68.50
a5a	–	–	–	–	53.24	62.72	–	–	**69.81**	69.12
a6a	–	–	–	–	–	70.06	–	–	**72.46**	72.43
a7a	–	–	–	–	–	–	–	–	**70.68**	–
satimage	–	–	–	–	87.64	90.88	–	–	91.16	**92.41**
pendigits	–	–	–	–	81.88	**86.02**	–	–	79.17	78.51

Table 14
Recall results of 10 models on 9 LIBSVM datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
a1a	54.31	55.25	21.34	55.57	0.00	58.62	59.88	55.09	**62.10**	56.69
a2a	52.59	57.61	–	51.97	0.00	**64.58**	–	56.87	61.63	61.95
a3a	–	–	–	–	**87.05**	61.80	–	57.95	59.34	57.92
a4a	–	–	–	–	0.06	**68.69**	–	–	61.30	62.05
a5a	–	–	–	–	**84.60**	59.26	–	–	60.20	61.30
a6a	–	–	–	–	–	**61.44**	–	–	56.22	59.59
a7a	–	–	–	–	–	–	–	–	**59.63**	–
satimage	–	–	–	–	89.09	81.05	–	–	**94.36**	93.35
pendigits	–	–	–	–	14.63	45.00	–	–	81.30	**82.64**

Table 15
F1-score results of 10 models on 9 LIBSVM datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
a1a	61.44	61.59	31.73	62.30	0.00	60.87	43.21	62.38	**63.62**	62.02
a2a	60.38	62.86	–	60.36	0.00	**65.80**	–	62.67	63.39	64.56
a3a	–	–	–	–	**64.50**	62.00	–	62.59	63.28	63.00
a4a	–	–	–	–	0.12	63.30	–	–	**65.20**	65.12
a5a	–	–	–	–	**65.35**	60.94	–	–	64.65	64.98
a6a	–	–	–	–	–	**65.47**	–	–	63.31	65.38
a7a	–	–	–	–	–	–	–	–	**64.69**	–
satimage	–	–	–	–	88.36	85.69	–	–	92.73	**92.87**
pendigits	–	–	–	–	24.82	59.09	–	–	80.22	**80.52**

Table 16
Time results of 10 models on 9 LIBSVM datasets.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
a1a	71.70	73.69	197.81	113.08	**2.89**	11.91	11460.80	39.86	4.17	6.51
a2a	180.91	88.26	–	320.87	7.33	16.97	–	78.52	**2.30**	11.81
a3a	–	–	–	–	18.09	22.77	–	5582.72	**2.60**	21.81
a4a	–	–	–	–	55.14	217.67	–	–	**2.90**	51.80
a5a	–	–	–	–	439.46	489.84	–	–	**3.47**	124.27
a6a	–	–	–	–	–	3826.41	–	–	**6.26**	392.30
a7a	–	–	–	–	–	–	–	–	**6.50**	–
satimage	–	–	–	–	46.81	85.53	–	–	**1.39**	25.52
pendigits	–	–	–	–	1070.27	2032.03	–	–	**1.09**	77.84

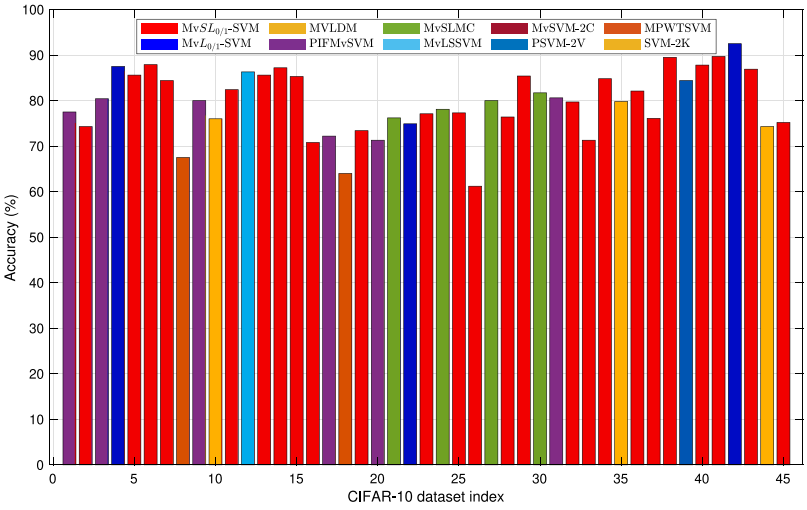


Fig. 16. Accuracy of winning methods on CIFAR-10 dataset.

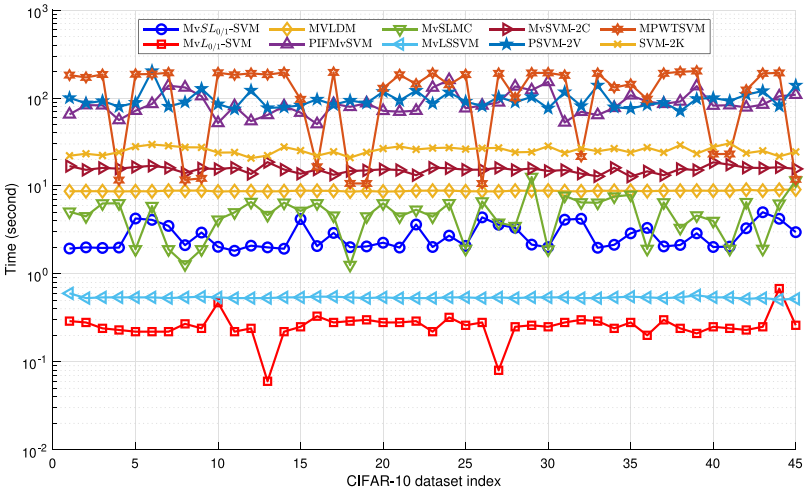


Fig. 17. Training time of algorithms on CIFAR-10 dataset.

s). The reason is that our proposed Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM adopt a fast ADMM with a working set strategy integrated into each step. However, the utilization of SI by Mv$SL_{0/1}$-SVM may result in a loss of competitiveness in terms of speed.

Fig. 16 illustrates the accuracy of the winning algorithm. The generalization performance of the algorithm is indicated by the length of each bar, with distinct colors denoting different algorithms. Among these models, it is evident that Mv$SL_{0/1}$-SVM has superior accuracy.

Fig. 17 describes the training time of all methods. The vertical axis displays the training time for each method, and the horizontal axis displays the number of datasets. The reduction in running time and enhancement in computational efficiency is attributed to the elimination of redundant samples using working sets.

These results confirm the idea that Mv$L_{0/1}$-SVM shows a significant advantage in terms of time, indicating its strong generalization ability. Additionally, Mv$SL_{0/1}$-SVM effectively leverages SI to achieve improved classification performance. This demonstrates the superior efficiency of Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM.

6.4. Parameter sensitivity analysis

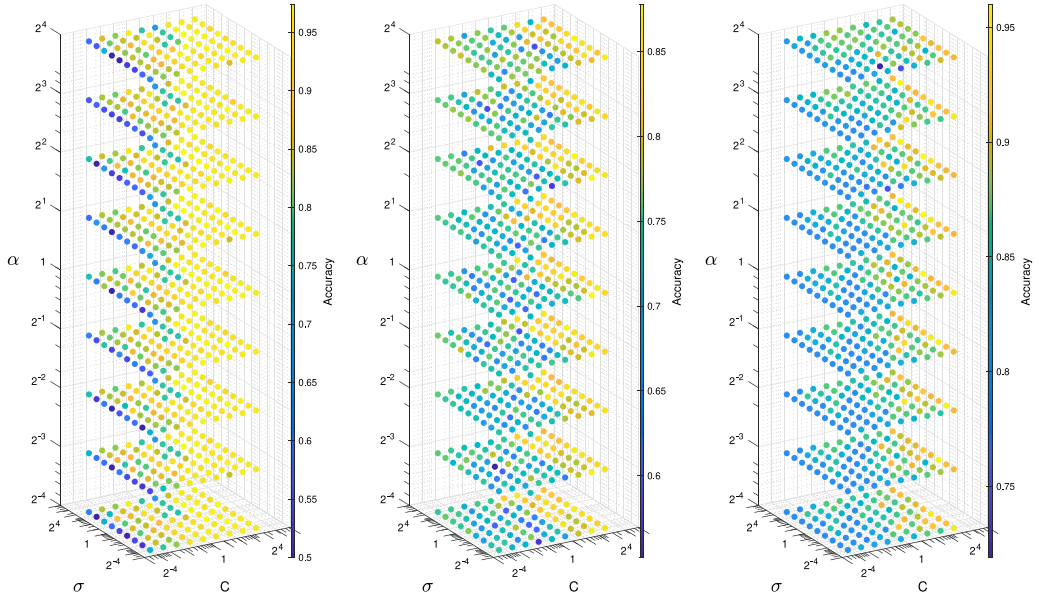
Here, we use the UCI datasets to examine the parameter sensitivity of Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM in order to analyze the impact of parameters. The accuracy of Mv$L_{0/1}$-SVM on Balance, Heart, and Hepatitis is shown in Fig. 18, where the color of the dot indicates the accuracy for the relevant parameter combination. The results of Mv$SL_{0/1}$-SVM on Wine are shown in Figs. 19–22. We fixed four parameters separately and explored combinations of varying parameters. These graphs show that the accuracy of Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM have something to do with their respective parameters on all datasets and are sensitive to them all. The number of SVs in our algorithms is mostly controlled by the values of C and σ, as illustrated in Eq. (39). This emphasizes the importance of carefully adjusting these parameters for optimal performance.

6.5. Discussion on SVs

To investigate the impact of reducing SVs on computational efficiency, we compared our proposed Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM

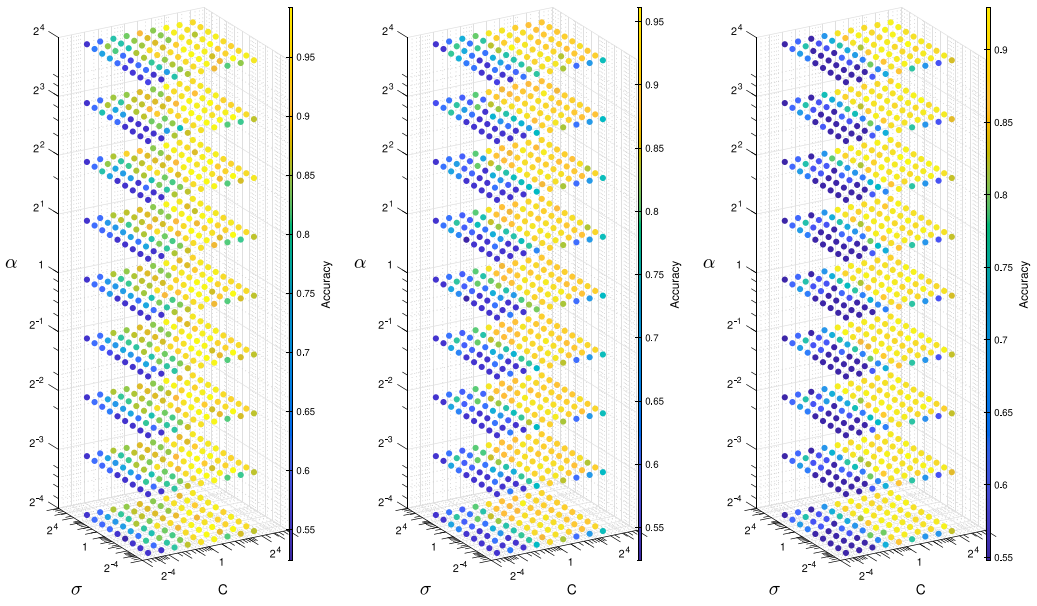
Table 17
Experimental results of 10 models on CIFAR-10 dataset.

Datasets	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
air. vs auto.	51.70 ± 1.04	71.90 ± 4.67	62.30 ± 4.98	51.60 ± 0.96	67.10 ± 15.77	71.10 ± 2.90	**77.50 ± 3.45**	74.50 ± 2.09	71.80 ± 4.62	73.60 ± 2.95
air. vs bird.	51.00 ± 0.79	64.60 ± 11.52	69.00 ± 3.14	51.00 ± 0.79	65.80 ± 14.49	69.50 ± 2.76	71.10 ± 3.60	70.90 ± 2.77	73.20 ± 4.64	**74.30 ± 3.29**
air. vs cat.	52.50 ± 1.37	66.70 ± 12.35	75.60 ± 2.70	52.50 ± 1.37	63.40 ± 18.36	78.40 ± 3.29	**80.40 ± 1.39**	77.30 ± 2.14	79.00 ± 2.21	79.80 ± 2.11
air. vs deer.	51.20 ± 0.91	77.40 ± 8.29	81.50 ± 2.76	51.00 ± 0.94	79.30 ± 16.50	83.80 ± 2.41	83.80 ± 2.49	79.50 ± 3.92	**87.50 ± 1.58**	86.30 ± 2.73
air. vs dog.	53.90 ± 0.65	62.80 ± 14.93	78.20 ± 3.13	53.90 ± 0.96	64.30 ± 19.60	80.10 ± 2.43	82.20 ± 3.49	82.60 ± 2.10	84.30 ± 2.08	**85.60 ± 1.85**
air. vs frog.	50.40 ± 0.55	78.30 ± 17.62	83.00 ± 2.45	50.60 ± 0.65	57.40 ± 16.55	84.20 ± 9.23	85.40 ± 2.82	82.60 ± 2.63	87.30 ± 1.35	**87.90 ± 2.30**
air. vs horse.	51.30 ± 0.76	76.60 ± 14.45	82.20 ± 1.64	51.10 ± 0.82	72.30 ± 20.37	81.20 ± 1.86	83.50 ± 2.35	83.70 ± 1.20	84.20 ± 1.64	**84.40 ± 1.98**
air. vs ship.	53.00 ± 1.50	**67.50 ± 1.62**	50.00 ± 0.00	52.40 ± 0.96	63.20 ± 12.16	66.20 ± 0.84	64.80 ± 4.40	58.20 ± 3.77	64.10 ± 1.98	66.20 ± 2.51
air. vs truck.	51.40 ± 0.89	75.30 ± 5.08	75.30 ± 1.82	51.10 ± 0.82	77.50 ± 15.49	71.80 ± 3.33	**80.00 ± 2.98**	75.90 ± 2.46	76.40 ± 4.83	78.60 ± 1.64
auto. vs bird.	52.90 ± 0.89	65.00 ± 14.93	71.70 ± 3.49	53.10 ± 1.08	61.50 ± 15.87	73.30 ± 6.14	74.00 ± 3.82	**76.00 ± 4.08**	75.90 ± 5.49	75.90 ± 5.56
auto. vs cat.	64.10 ± 2.46	76.10 ± 14.33	79.40 ± 2.95	64.40 ± 2.48	77.50 ± 15.40	81.40 ± 1.43	80.70 ± 3.29	81.80 ± 1.89	81.10 ± 0.42	**82.40 ± 1.52**
auto. vs deer.	67.90 ± 5.20	75.90 ± 15.47	83.90 ± 2.33	73.20 ± 4.45	**86.30 ± 3.05**	78.30 ± 10.14	83.50 ± 2.06	85.60 ± 2.38	82.50 ± 1.90	83.70 ± 1.60
auto. vs dog.	58.00 ± 3.41	73.00 ± 16.72	81.50 ± 2.69	58.00 ± 3.67	57.70 ± 17.22	82.20 ± 5.71	84.30 ± 1.92	84.80 ± 2.20	84.30 ± 2.80	**85.60 ± 1.92**
auto. vs frog.	70.50 ± 9.19	80.10 ± 15.19	85.90 ± 1.56	78.70 ± 3.49	65.00 ± 20.65	81.90 ± 5.95	85.00 ± 1.90	86.70 ± 2.17	86.80 ± 2.64	**87.20 ± 1.15**
auto. vs horse.	72.10 ± 5.82	71.30 ± 9.83	83.40 ± 1.29	73.50 ± 2.42	65.20 ± 20.82	81.40 ± 2.27	82.60 ± 3.76	85.00 ± 1.73	83.20 ± 3.73	**85.30 ± 3.95**
auto. vs ship.	52.40 ± 1.78	69.70 ± 1.92	59.00 ± 1.27	52.40 ± 1.92	57.90 ± 32.48	61.70 ± 8.61	68.80 ± 2.51	66.90 ± 2.10	69.10 ± 3.23	**70.80 ± 1.68**
auto. vs truck.	51.20 ± 2.49	66.10 ± 9.39	64.90 ± 1.92	52.10 ± 2.95	70.00 ± 11.91	61.70 ± 4.91	**72.20 ± 1.60**	71.50 ± 5.09	70.60 ± 3.05	71.00 ± 2.24
bird. vs cat.	58.90 ± 2.43	**64.00 ± 3.16**	50.00 ± 0.00	57.10 ± 2.43	55.60 ± 7.68	62.80 ± 2.71	63.40 ± 3.93	54.20 ± 1.15	62.90 ± 4.88	62.70 ± 4.75
bird. vs deer.	50.70 ± 0.76	70.30 ± 2.77	50.00 ± 0.00	50.90 ± 0.65	50.00 ± 0.00	73.30 ± 2.75	71.40 ± 4.02	60.80 ± 3.33	69.90 ± 5.72	**73.40 ± 1.64**
bird. vs dog.	66.40 ± 5.00	55.80 ± 8.25	53.00 ± 1.17	65.90 ± 4.92	64.00 ± 13.06	70.10 ± 4.41	**71.30 ± 3.55**	60.20 ± 3.83	70.00 ± 5.40	69.60 ± 5.31
bird. vs frog.	51.40 ± 0.96	75.20 ± 2.97	50.00 ± 0.00	52.90 ± 2.36	66.50 ± 15.07	**76.20 ± 4.45**	73.10 ± 2.33	59.90 ± 3.73	74.80 ± 3.09	74.90 ± 2.82
bird. vs horse.	50.30 ± 0.27	61.40 ± 13.97	50.10 ± 0.22	50.40 ± 0.65	66.30 ± 15.06	70.60 ± 9.49	72.20 ± 4.12	61.10 ± 4.13	**74.90 ± 2.82**	71.00 ± 6.39
bird. vs ship.	66.10 ± 2.63	68.90 ± 14.34	73.60 ± 2.36	68.30 ± 5.07	60.20 ± 14.08	67.10 ± 1.24	74.50 ± 3.77	73.50 ± 2.89	76.30 ± 2.80	**77.10 ± 1.82**
bird. vs truck.	50.30 ± 0.45	68.80 ± 7.26	55.00 ± 4.72	50.40 ± 0.42	68.00 ± 16.53	**78.10 ± 1.78**	73.50 ± 2.92	69.10 ± 1.85	76.30 ± 8.71	76.90 ± 7.92
cat. vs deer.	50.10 ± 0.22	67.90 ± 13.09	50.50 ± 0.50	50.10 ± 0.22	71.80 ± 12.29	75.60 ± 1.85	75.40 ± 1.39	70.30 ± 4.64	73.90 ± 4.29	**77.30 ± 2.17**
cat. vs dog.	55.10 ± 4.02	59.80 ± 3.35	50.00 ± 0.00	56.10 ± 3.21	52.50 ± 5.59	57.90 ± 5.89	57.40 ± 3.61	51.40 ± 1.98	58.20 ± 5.46	**61.20 ± 4.66**
cat. vs frog.	50.80 ± 0.57	73.10 ± 13.73	51.20 ± 0.91	51.30 ± 1.04	58.00 ± 35.05	**80.00 ± 2.76**	76.80 ± 3.35	76.50 ± 2.42	76.70 ± 3.27	79.50 ± 4.50
cat. vs horse.	51.40 ± 1.39	64.00 ± 15.66	50.10 ± 0.22	51.90 ± 1.56	61.90 ± 16.32	69.20 ± 7.29	74.90 ± 3.11	66.50 ± 2.87	75.40 ± 3.54	**76.40 ± 1.56**
cat. vs ship.	74.00 ± 2.69	81.60 ± 14.60	78.50 ± 1.17	74.50 ± 2.69	71.70 ± 19.83	71.30 ± 18.10	81.40 ± 3.81	80.20 ± 2.41	83.90 ± 3.58	**85.40 ± 3.25**
cat. vs truck.	52.00 ± 1.27	68.50 ± 11.89	80.60 ± 4.13	52.30 ± 1.79	70.30 ± 18.60	**81.70 ± 3.35**	78.50 ± 3.02	76.60 ± 4.49	79.10 ± 6.28	81.60 ± 3.07
deer. vs dog.	50.70 ± 0.57	63.90 ± 13.72	75.20 ± 2.31	50.60 ± 0.55	62.30 ± 16.89	75.20 ± 6.32	**80.60 ± 2.48**	78.50 ± 3.43	77.10 ± 2.43	76.90 ± 2.88
deer. vs frog.	73.20 ± 1.48	67.90 ± 6.11	60.50 ± 5.57	72.80 ± 0.97	69.20 ± 17.54	62.80 ± 14.83	78.50 ± 2.52	76.10 ± 3.45	79.30 ± 2.02	**79.70 ± 1.75**
deer. vs horse.	53.10 ± 3.80	70.60 ± 3.96	50.20 ± 0.45	53.10 ± 3.54	65.40 ± 14.38	66.80 ± 5.17	70.20 ± 4.04	66.60 ± 3.58	69.20 ± 4.89	**71.30 ± 2.44**
deer. vs ship.	52.80 ± 2.25	75.30 ± 12.32	82.30 ± 1.60	54.50 ± 2.83	71.60 ± 19.73	79.60 ± 9.17	83.10 ± 2.16	84.20 ± 2.99	84.20 ± 1.82	**84.80 ± 1.48**
deer. vs truck.	70.80 ± 3.21	61.00 ± 15.12	75.80 ± 2.66	70.50 ± 4.91	67.40 ± 37.76	69.00 ± 13.99	76.40 ± 2.30	**79.80 ± 2.36**	76.70 ± 4.92	78.40 ± 4.10
dog. vs frog.	53.70 ± 1.57	76.30 ± 8.08	81.30 ± 2.93	53.90 ± 1.56	78.30 ± 15.98	81.70 ± 2.20	80.40 ± 3.91	81.80 ± 2.66	79.70 ± 6.18	**82.10 ± 2.10**
dog. vs horse.	53.00 ± 1.62	65.80 ± 10.45	51.00 ± 0.61	53.40 ± 1.56	71.10 ± 11.83	69.50 ± 9.63	73.40 ± 3.11	72.60 ± 4.67	73.00 ± 6.64	**76.10 ± 0.65**
dog. vs ship.	76.10 ± 2.38	77.60 ± 16.30	82.50 ± 2.74	82.10 ± 1.71	66.30 ± 22.44	68.70 ± 4.52	86.60 ± 2.63	86.10 ± 3.11	88.30 ± 1.35	**89.50 ± 2.69**
dog. vs truck.	51.30 ± 1.20	72.90 ± 15.73	**84.40 ± 2.84**	52.10 ± 2.10	64.50 ± 19.86	74.70 ± 7.75	81.60 ± 2.33	82.50 ± 2.67	82.80 ± 2.80	82.40 ± 4.83
frog. vs horse.	73.70 ± 4.19	69.70 ± 7.01	85.20 ± 2.59	73.50 ± 4.27	80.70 ± 17.25	76.00 ± 4.47	83.30 ± 3.75	82.10 ± 2.58	81.40 ± 11.47	**87.80 ± 4.15**
frog. vs ship.	61.90 ± 4.94	82.80 ± 4.40	85.70 ± 0.84	66.20 ± 6.13	73.60 ± 21.55	82.20 ± 1.89	87.80 ± 1.82	84.30 ± 1.72	88.60 ± 3.27	**89.70 ± 1.92**
frog. vs truck.	81.80 ± 3.93	60.40 ± 5.13	84.10 ± 2.16	81.80 ± 3.80	76.00 ± 23.74	62.80 ± 9.18	84.20 ± 1.79	81.50 ± 6.82	**92.50 ± 0.94**	92.40 ± 1.39
horse. vs ship.	51.00 ± 0.35	64.00 ± 15.99	82.10 ± 3.71	52.00 ± 1.54	50.00 ± 0.00	80.70 ± 2.59	84.10 ± 2.22	84.90 ± 4.49	84.40 ± 1.88	**86.90 ± 2.01**
horse. vs truck.	54.40 ± 3.27	69.70 ± 10.47	70.60 ± 3.86	65.40 ± 5.10	67.80 ± 16.35	69.80 ± 9.02	71.00 ± 4.00	**74.30 ± 5.13**	73.40 ± 5.19	73.70 ± 5.40
ship. vs truck.	50.00 ± 0.00	65.50 ± 9.33	66.50 ± 2.09	50.10 ± 0.22	72.70 ± 12.97	65.40 ± 7.07	72.20 ± 3.21	71.40 ± 2.95	74.40 ± 1.92	**75.20 ± 1.48**
Avg.Accuracy.	57.57	69.80	69.48	58.55	66.78	73.49	77.27	74.90	77.75	**78.94**
Avg.Time.	25.390	133.348	97.412	15.385	0.538	5.019	88.931	8.773	**0.264**	2.746
Avg.Rank.	8.78	6.37	6.41	8.38	7.28	5.21	3.36	4.34	3.08	**1.80**



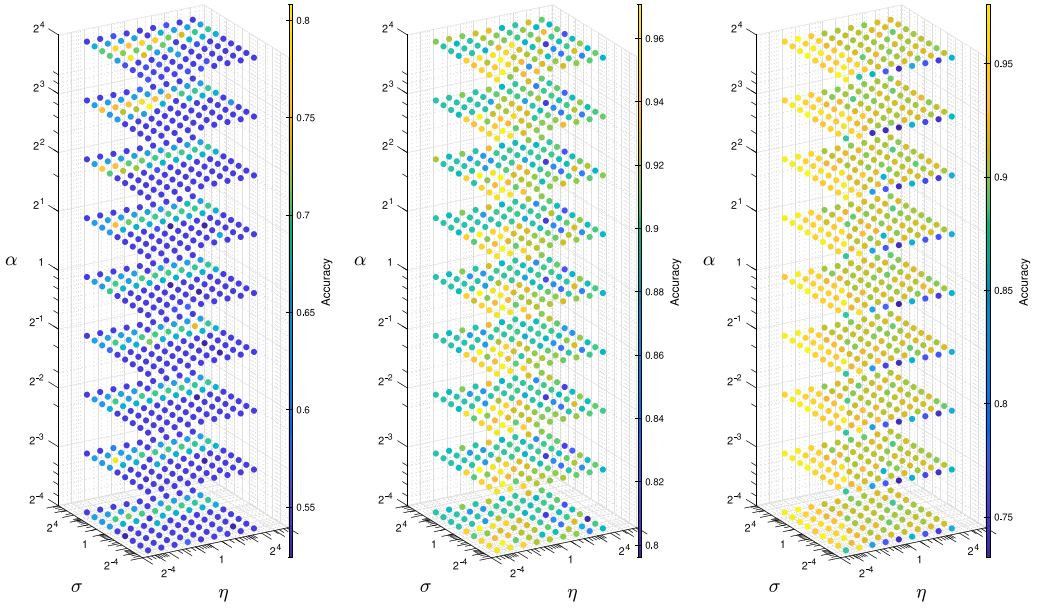
(a) Balance (b) Heart (c) Hepatitis

Fig. 18. Accuracies of Mv$L_{0/1}$-SVM under different parameter combinations of C, σ and α on the Balance, Heart and Hepatitis datasets.



(a) $\eta = 2^{-4}$ (b) $\eta = 1$ (c) $\eta = 2^{4}$

Fig. 19. Accuracies of Mv$SL_{0/1}$-SVM under different parameter combinations of C, σ and α with fixed η on the Wine dataset.



(a) $C = 2^{-4}$ (b) $C = 1$ (c) $C = 2^{4}$

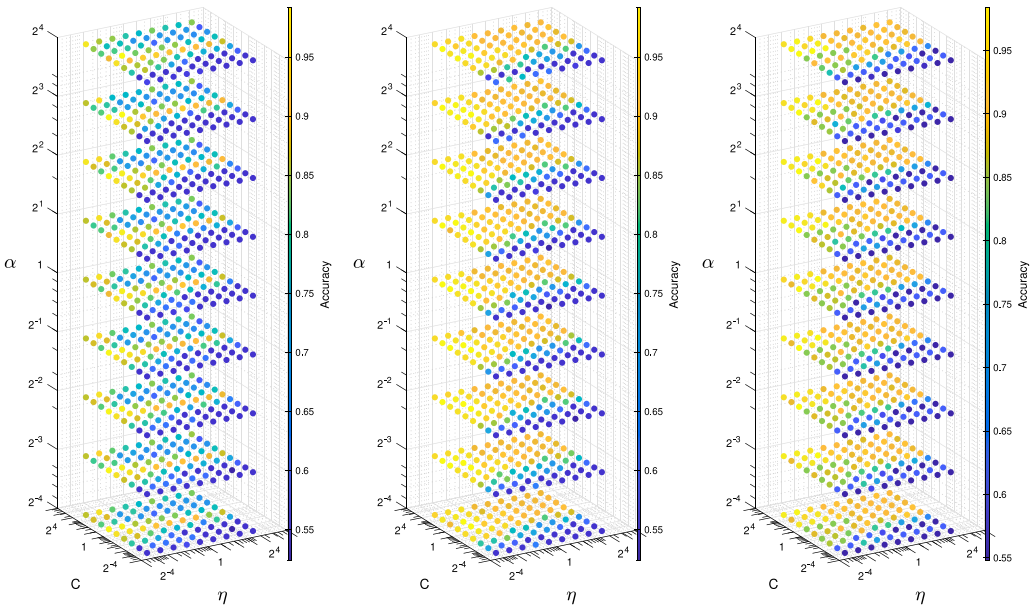
Fig. 20. Accuracies of Mv$SL_{0/1}$-SVM under different parameter combinations of η, σ and α with fixed C on the Wine dataset.

with the classic multi-view model SVM-2K across the Pima, Vote, and a1a datasets. We recorded the number of SVs for each view and the training time for each model. As shown in Fig. 23, Mv$L_{0/1}$-SVM consistently had the fewest SVs for each view across all three datasets and demonstrated faster training times compared to the other models.

To further explore the relationship between SVs and time for the Mv$L_{0/1}$-SVM model, we conducted experiments on the a7a dataset. The results, illustrated in Figs. 24 and 25, show the number of SVs, time, and iteration error under outer loop iterations $l = 0$ and $l = 1$. These figures reveal that the number of SVs and time follow similar decreasing trends, with SVs beginning to decrease early in the iterations while iteration error decreases and converges over time. Combining these observations, it is clear that the number of SVs is closely related to computational efficiency.

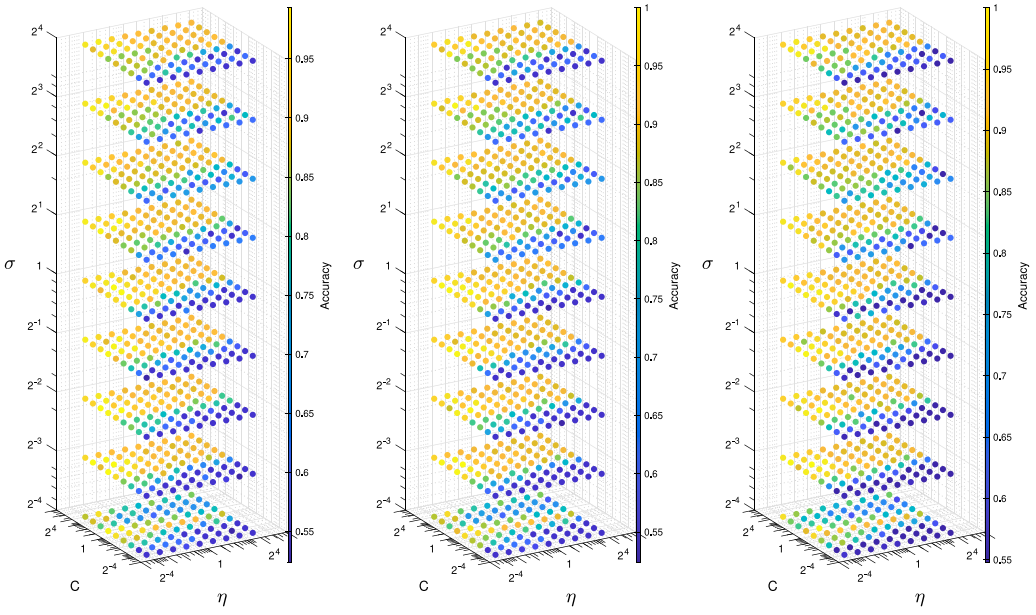
6.6. Friedman test

Since the outcomes of the proposed Mv$SL_{0/1}$-SVM and Mv$L_{0/1}$-SVM are not consistently optimal in the numerical experiments conducted, the Friedman test [47], a common statistical method in machine learning, is employed to assess whether there are significant differences among these algorithms. The null hypothesis H_0 posits that there are no significant differences between the ten algorithms. Conversely, the



(a) $\sigma = 2^{-4}$ (b) $\sigma = 1$ (c) $\sigma = 2^{4}$

Fig. 21. Accuracies of Mv$SL_{0/1}$-SVM under different parameter combinations of η, C and α with fixed σ on the Wine dataset.



(a) $\alpha = 2^{-4}$ (b) $\alpha = 1$ (c) $\alpha = 2^{4}$

Fig. 22. Accuracies of Mv$SL_{0/1}$-SVM under different parameter combinations of η, C and σ with fixed α on the Wine dataset.

alternative hypothesis H_1 suggests that significant differences do exist among the ten algorithms.

By ranking the ten algorithms based on their accuracy for each dataset – where the algorithm with the highest accuracy is assigned rank 1, the second highest rank 2, and so forth – the results are summarized in Tables 6–11 and 17. The average rank for each method is computed across these datasets, with the results displayed in the final line of these tables. Overall, the proposed Mv$SL_{0/1}$-SVM achieves the lowest average rank, indicating the best performance, while Mv$L_{0/1}$-SVM ranks second in terms of average rank.

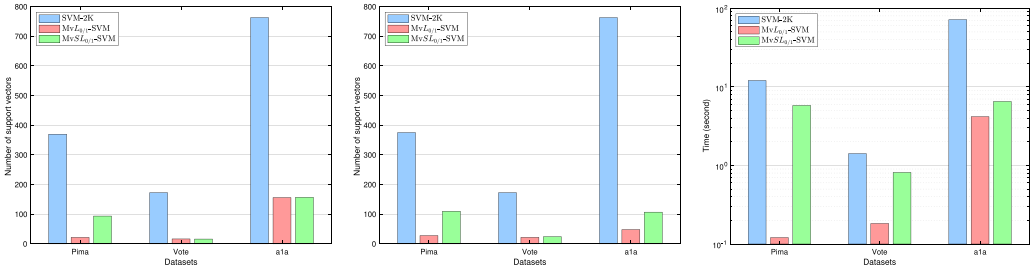
The formula for calculating the Friedman test statistic is given by:

$$\mathcal{X}^2 = \frac{12N}{p(p+1)}\left[\sum_j R_j^2 - \frac{p(p+1)^2}{4}\right], \tag{54}$$

where N represents the number of datasets, p represents the number of algorithms, R_j represents the average ranking of the j-algorithm on all datasets.

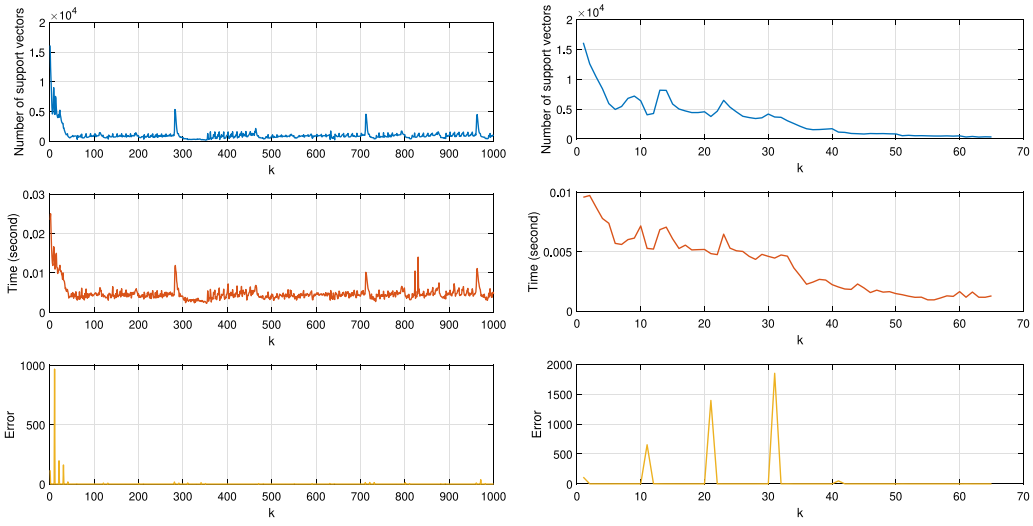
The distribution of F with degrees of freedom of $p-1$ and $(p-1)(N-1)$:

$$F_F = \frac{(N-1)\mathcal{X}^2}{N(p-1)-\mathcal{X}^2}. \tag{55}$$



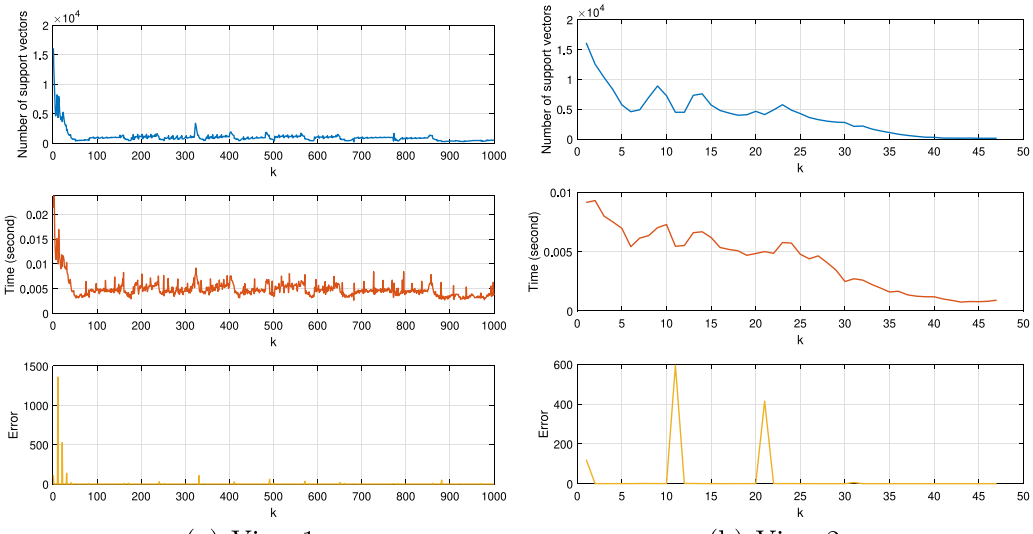
(a) View 1 (b) View 2 (c) Time

Fig. 23. Number of SVs and time of SVM-2K and our models on Pima, Vote and a1a datasets.



(a) View 1 (b) View 2

Fig. 24. Iterative performance of Mv$L_{0/1}$-SVM on the a7a dataset ($l = 0$).



(a) View 1 (b) View 2

Fig. 25. Iterative performance of Mv$L_{0/1}$-SVM on the a7a dataset ($l = 1$).

Table 18
Results of 10 models on WebKB dataset.

Metrics	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
Accuracy	89.05	95.71	92.38	94.76	96.67	98.10	**98.57**	97.62	98.10	**98.57**
Precision	96.00	89.36	**100.00**	85.71	**100.00**	95.65	**100.00**	**100.00**	92.00	95.74
Recall	52.17	91.30	65.22	91.30	84.78	95.65	93.48	89.13	**100.00**	97.83
F1-score	67.61	90.32	78.95	88.42	91.76	95.65	96.63	94.25	95.83	**96.77**
Time	18.7187	494.6883	91.2294	28.0523	**0.6475**	10.5884	36.5153	9.7710	3.0729	9.6632
Avg.Rank.	8.20	7.70	7.70	7.70	4.70	4.50	3.60	4.70	3.50	**2.70**

Table 19
Results of 10 models on Adult dataset.

Metrics	SVM-2K	MPWTSVM	PSVM-2V	MvSVM-2C	MvLSSVM	MvSLMC	PIFMvSVM	MVLDM	Mv$L_{0/1}$-SVM	Mv$SL_{0/1}$-SVM
Accuracy	50.00	57.10	60.00	50.00	54.80	53.40	59.60	50.00	60.60	**63.10**
Precision	50.00	55.59	55.77	50.00	53.61	51.80	55.47	50.00	56.24	**59.29**
Recall	**100.00**	70.60	96.60	**100.00**	71.20	98.00	97.40	**100.00**	95.60	83.60
F1-score	66.67	62.20	70.72	66.67	61.17	67.77	70.68	66.67	**70.81**	69.38
Time	11.8849	527.5921	423.1599	50.1347	0.8992	16.8556	79.3891	11.0119	**0.0873**	12.3724
Avg.Rank.	6.20	7.60	4.60	6.80	6.60	5.80	5.00	6.00	**2.60**	3.80

In our experiment, we perform the Friedman test on the UCI and CIFAR-10 datasets. For the UCI datasets, with $N = 8, p = 10$, according to (54) and (55), we obtain $\mathcal{X}^2 = 48.5281$ and $F_F = 14.4725$ when the metric is accuracy; $\mathcal{X}^2 = 26.7009$ and $F_F = 4.1261$ when the metric is precision; $\mathcal{X}^2 = 16.3155$ and $F_F = 2.0510$ when the metric is recall; $\mathcal{X}^2 = 46.6010$ and $F_F = 12.8433$ when the metric is f1-score; $\mathcal{X}^2 = 51.5914$ and $F_F = 17.6955$ when the metric is time. For the significance level $\alpha = 0.05$, the critical value of $F(7, 63)$ is 2.0322. The F_F values for all metrics in the UCI datasets are greater than the critical value 2.0322. For the CIFAR-10 datasets, with $N = 45, p = 10$, according to (54) and (55), we obtain $\mathcal{X}^2 = 242.8621$ and $F_F = 65.9064$. For the significance level $\alpha = 0.05$, the critical value of $F(9, 396)$ is 1.9035. The F_F value for the CIFAR-10 datasets is greater than the critical value of 1.9035. Therefore, the null hypothesis H_0 is rejected, and the alternative hypothesis H_1 is accepted. Namely, there are significant differences between the ten algorithms.

6.7. Application

In order to verify the performance of Mv$SL_{0/1}$-SVM and Mv$L_{0/1}$-SVM in real applications, we also conduct experiments on the WebKB and Adult datasets for real-world classification applications.

6.7.1. WebKB

The WebKB dataset[4] utilized in [48] comprises web pages collected from the websites of computer science departments at various universities. This dataset features two types of descriptions: (1) $fulltext$—the text content on the web pages, and (2) $inlinks$—the hyperlinks pointing to the page. These two descriptions can be naturally viewed as two distinct perspectives: the $fulltext$ view and the $inlinks$ view. The dataset includes a total of 1051 pages, which are manually categorized into two classes: course (230) and non-course (821). The original $fulltext$ and $inlinks$ documents are represented as 3000-dimensional and 1840-dimensional vectors, respectively. For our experiments, we apply PCA to reduce the dimensionality of both the $fulltext$ and $inlinks$ vectors to 100, in accordance with [48]. Additionally, we randomly allocate 80% of each class for training purposes and reserve the remaining 20% for testing.

Here, SVM-2K, MPWTSVM, PSVM-2V, MvSVM-2C, MvLSSVM, MvSLMC, PIFMvSVM, MVLDM, and our proposed models are applied to the WebKB dataset. As shown in Table 18, among the five evaluation criteria, Mv$SL_{0/1}$-SVM achieved the best performance in two of them. Additionally, our proposed models ranked first and second in average ranks.

6.7.2. Adult

The UCI Adult dataset [49], used for predicting whether a household's income exceeds 50,000 annually, contains 48,842 samples and 14 features. These features are age, work class, weight, education, education number, marital status, occupation, relationship, race, sex, capital gain, capital loss, hours worked per week, and native country.

Following a method similar to that in [12], we generated multi-view datasets from the UCI Adult dataset by using two different subsets of attributes. The first view comprises five features: age, education number, marital status, occupation, and sex. The second view includes the following five features: age, marital status, occupation, sex, and hours worked per week. Furthermore, 500 samples were randomly selected for the training set and another 500 samples for the test set.

Examining the results in Table 19, our proposed models, Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM, each achieved the highest performance in two of the evaluation metrics. Additionally, Mv$L_{0/1}$-SVM demonstrated a significant reduction in computation time compared to other models. Consistent with their performance on the WebKB dataset, our proposed models also achieved the top two average ranks on the Adult dataset.

7. Conclusion and future directions

In this paper, on the basis of $L_{0/1}$ soft-margin loss, we propose two novel models Mv$L_{0/1}$-SVM and Mv$SL_{0/1}$-SVM for multi-view classification. By fusing structural information and adding view weights, Mv$SL_{0/1}$-SVM simultaneously satisfies the consensus and complementarity principles. Despite the fact that Mv$SL_{0/1}$-SVM is an NP-hard problem, we show the existence of globally optimal solutions and also establish their optimality condition. Drawing inspiration from the P-stationary point optimality condition, we design and integrate a working set strategy into the proximal ADMM. This integration significantly accelerates the overall computation and diminishes the number of SVs. Consequently, numerical experiments show that our suggested models perform exceptionally well and have faster computational speed. The advantage becomes more apparent with larger dataset sizes.

There are numerous opportunities for future research. For instance, our method encounters challenges when dealing with high-dimensional

[4] Available at http://www.cs.cmu.edu/afs/cs/project/theo-11/www/wwkb/.

data. In this context, [50] proposed a novel hybrid feature weighting-selection algorithm that effectively reduces dimensionality while preserving the most critical features. Integrating this approach into our MVL framework could provide valuable insights. Furthermore, the field of multi-task unsupervised learning is continually advancing, as demonstrated by [51]. Exploring how to extend our models to handle classification tasks within the realm of multi-task unsupervised learning would be an intriguing and promising direction for future research.

CRediT authorship contribution statement

Chen Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Qianfei Liu:** Methodology, Data curation. **Renpeng Xu:** Software, Formal analysis. **Ying Zhang:** Writing – original draft, Validation, Formal analysis. **Huiru Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization, Methodology, Resources. **Qingmin Yu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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